

K-Bound: When Is Label-Free Adaptation Knowable?

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Abstract

Test-time adaptation can improve or degrade a model, leaving a separate decision: when does label-free evidence justify replacing the frozen predictor? K-Bound characterizes this decision in a binary population model under an externally declared calibration-residual bound β . Holding label-free observations fixed, we derive the exact feasible interval for paired benefit: a strict decision is justified precisely when $|M| > \beta$ on disagreement. An evidence-fibre audit shows why the same observations cannot justify a smaller uniform residual bound without additional structure. A separate finite-sample multiclass construction bounds paired benefit under an explicit conditional-transport budget; synthetic examples illustrate its value and failure under assumption violations. Knowability-Guided Adaptation (KGA) is an empirical companion, not an implementation of the population frontier. Using labeled historical outcomes, it predicts observed evaluation-cell benefit and chooses ADAPT, FREEZE, or ABSTAIN through residual intervals. Controlled studies show candidate-dependent routing gains, while SAR favors always-adapt. A shared-predictor comparison finds lower observed loss for CIFAR Tent at a prespecified utility-harm trade-off; this advantage is smaller at matched exposure and depends on the corruption mix. Simpler rules often win. Nominal 90% constant-cell inclusion is only 50.51–60.69% on held-out CIFAR corruptions. CCT-20 has no ADAPT decisions, and So2Sat stops before target scoring. These findings do not establish interval coverage or population-risk protection on unseen natural shifts; useful prospective routing remains open.

Keywords: test-time adaptation; abstention; distribution shift; label-free risk estimation; conformal calibration; selective decisions.

1 Introduction

An update that improves a model on one deployment distribution can degrade it on another. Test-time adaptation proposes updates using unlabeled

data through entropy minimization, filtering, regularization, and related methods [1, 4, 5, 6]. Choosing whether to serve the resulting candidate is a separate decision. The challenge is to justify that choice when the target labels needed to measure its benefit are unavailable.

Our primary research question is: *When can evidence available without new target labels justify replacing a frozen predictor with a fixed adapted candidate?* Let f_0 be the frozen predictor and f_a the candidate. Their population benefit is

$$\Delta = R_T(f_0) - R_T(f_a). \quad (1)$$

A positive benefit favors the update; a negative benefit favors the frozen model. We distinguish three conclusions: ADAPT supports positive benefit, FREEZE supports negative benefit, and ABSTAIN records insufficient evidence for either strict direction. Both non-adapt actions retain f_0 , but only freeze asserts that the candidate is worse.

K-Bound studies this decision as a partial-identification problem. In a binary model, we hold the input law, predictors, and label-free evidence fixed and vary the target labels within a declared class. The resulting feasible benefit set gives an exact boundary for a uniformly justified strict decision. The bound on the hidden calibration residual is an assumption about that class, not a quantity that an unlabeled batch supplies for free. A second result makes this information limit explicit: an audit using the same evidence must still cover the unresolved residual range.

The theory asks what a fixed label-free evidence law and a declared residual bound can identify. Knowability-Guided Adaptation (KGA) studies the same update decision using additional information: labeled historical shift outcomes. It fits an empirical benefit predictor and calibrates a three-way decision rule. The connection is the decision problem; the empirical gate does not implement the population frontier. Section 6.1 gives the full relationship between their targets and assumptions. Together, these analyses separate two possible obstacles to a justified update: the permitted evidence may leave its benefit unidentified, or an empirical interval may fail to

transfer. The first requires additional assumptions or information; the second requires evidence for the calibration procedure at the intended decision unit.

1.1 Three Scientific Contributions

1. **An exact benefit frontier on a declared class.** We construct target label laws with identical label-free evidence, derive their feasible benefit interval, and characterize its strict positive, strict negative, and unresolved regions, including the zero-benefit boundary.
2. **An evidence-fibre audit limit.** We identify the least residual bound that the same observable law can support uniformly, explaining why historical labels help only through a stated link to the deployment domain.
3. **An empirical paired-benefit gate.** We give a reproducible three-way interval procedure and evaluate its commitment behavior under controlled and natural shifts. The standard coverage-to-action implication states exactly what follows if the interval covers its declared target.

The empirical studies then ask whether an interval improves the update decision and whether its calibration transfers beyond the fitting conditions. Controlled CIFAR-10-C results are candidate-dependent: Tent and EATA have lower point regret than either fixed policy, whereas SAR favors always-adapt. A comparison using a shared point predictor isolates the contribution of calibration from the predictor itself (Section 8.7). It finds a benefit for CIFAR Tent at one harm-weighted operating point, but simple rules win elsewhere and nominal interval inclusion deteriorates on held-out corruptions.

Natural-shift studies expose a further distinction between retaining utility and selecting useful updates. On CCT-20, KGA retained the frozen model in all 45 cells, avoiding the measured loss of always-adapt while making no ADAPT decisions. The So2Sat studies stopped before target scoring. These findings leave useful prospective routing unresolved. A separate paired-transport construction shows how a declared conditional-shift assumption can support a finite-sample multiclass decision, with synthetic examples illustrating its value and its failure when that assumption is violated. Additional adverse and incomplete studies are reported in the supplement.

2 Related Work

2.1 Candidate Adaptation and Update Selection

Tent, EATA, SAR, CoTTA, MEMO, SHOT, and test-time training improve candidate construction through entropy objectives, sample filtering, regularization, temporal stabilization, augmentation, or auxiliary tasks [1, 4, 5, 6, 7, 2, 3]. Selective methods also choose where or when to update. TTSA selects modality-specific adapters [10]; GALA selects layers and filters samples using gradient alignment [11]; Informed Adaptation learns topological features associated with useful updates [9]. KGA evaluates a fixed proposed candidate against its frozen baseline. Learned routing and rollback are established ideas; their existence is not our novelty claim. MORPHEUS [58], presented at the ICLR 2026 Test-Time Updates workshop, learns historical regressors from source-model entropy and embedding geometry to predict post-adaptation accuracy and choose a method before executing it. This directly overlaps the practical motivation of avoiding harmful updates. KGA’s proposed distinction is the calibrated paired-benefit operating point for one executed shadow candidate; whether that calibration improves useful decisions over a shared point predictor is an empirical question evaluated separately.

2.2 Label-Free Performance and Strategy Estimation

ATC, agreement-on-the-line, AETTA, and prediction-matrix statistics estimate performance from unlabeled outputs and suitable development information [15, 17, 16, 18, 35, 34]. The closest empirical comparison is Kim et al.’s NeurIPS 2024 work, *Test-Time Adaptation Induces Stronger Accuracy and Agreement-on-the-Line* [51]. It uses agreement-based ALine estimators after adaptation, supports hyperparameter selection, and compares adaptation strategies without labeled OOD data.

KGA instead fits the paired observed benefit of one frozen/candidate pair and uses an interval to make a three-way decision. This is a difference in the fitted target and decision procedure, not a claim that accuracy estimates cannot be differenced to compare models. In Section 8.3 (Table 9), we compare current KGA with fixed policies, scalar evidence rules and a point gate sharing its exact predictions. These saved-feature comparisons isolate decision rules; they do not establish equivalence to complete published TTA/ALine, AETTA, or POEM systems. Method-

KGA decision flow

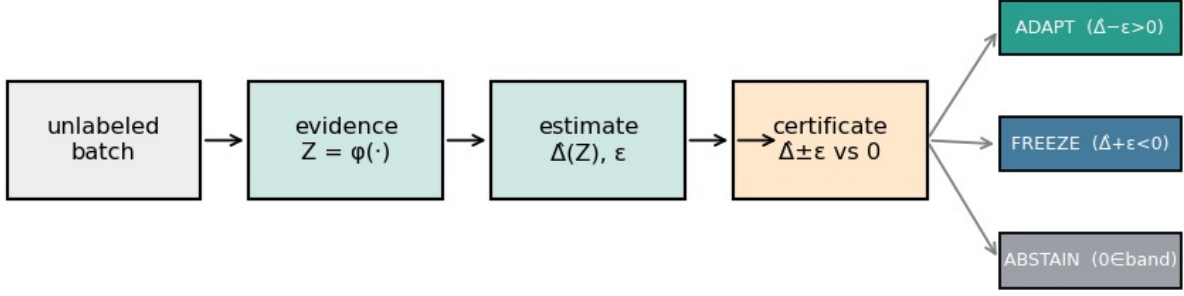


Figure 1: KGA’s empirical workflow. A shadow candidate proposes an update; a learned benefit prediction and residual interval determine whether to accept it. Historical labels support development and calibration, while the new decision uses no evaluation labels.

specific implementation and calibration qualifications are given in Appendix B.3.4.

2.3 Monitoring and Adaptation Controllers

POEM protects adaptation through entropy matching and online betting [14]. Schirmer et al. monitor the running risk of an adapting model relative to a source-risk reference, under a proxy-informativeness condition [8]; the supervised precursor uses a labeled test stream [13]. Drift-to-Action Controllers use delayed labels and online certificates to gate adaptation, abstention, rollback, and retraining [50]. These methods differ in the risk target, label access, and sequential guarantee. KGA concerns one paired benefit and does not establish an anytime-valid risk guarantee.

2.4 Partial Identification and Robust Decisions

An identified set describes the values compatible with the observations and assumptions. Decision theory under partial identification studies how to act when that set leaves payoffs ambiguous; Christensen et al. combine worst-case payoff uncertainty with statistical learning of identified parameters [54]. K-Bound addresses the narrower question of a uniformly correct strict benefit sign. It does not solve their general optimal-decision problem or claim that interval thresholding itself is new.

Indistinguishable adaptation problems are already central to domain-adaptation impossibility [20, 49]. K-Bound’s specific contribution is a disagreement-

conditional label-kernel construction, its clipped benefit set, and the associated residual-audit floor. The proof does not introduce label-free impossibility. The study in [56] independently examines when an unlabeled observation channel identifies the best TTA action. That September 2026 preprint includes a finite-batch Gaussian keep/recenter boundary and benchmark examples where stream structure changes the preferred action. Thus action identification being weaker than mechanism identification is not a unique claim of this paper. Our population result fixes a realized predictor pair and a disagreement-conditional residual class, deriving its clipped paired-benefit set and strict-sign boundary. The distinction lies in that explicit compatible-law construction and its audit requirements, not in the general fibre-consistency principle. Risk-estimation assumptions based on conditional independence [19], label shift [44, 45], or disagreement discrepancy [21] restrict the compatible laws. Likewise, MMD-credal risk intervals impose RKHS and covariate-shift structure [12]. Such restrictions can change the identified set; they are not contradicted by a converse theorem for a richer class. The construction in [57] obtains finite-sample label-shift confidence regions through matrix constraints and linear programming. Consequently, optimizing a compatible probability table is established machinery. The separate paired-transport extension evaluated here targets the benefit contrast directly, retains joint prediction-pair information, and reports sensitivity to an externally specified conditional-transport violation. It does not estimate that violation from unlabeled data or claim a new general-purpose LP confidence

method.

2.5 Marginal and Conditional Coverage

Selective classification and reject-option learning abstain on individual predictions [22, 46, 47]. KGA abstains on an update. Its coverage-to-action proposition applies standard interval logic [36, 37, 40, 41]. The main statistical work is to justify coverage, especially after selection or under shift.

Jin and Ren’s JOMI procedure targets coverage conditional on a unit being selected, under exchangeability and a compatible selection rule [52]. Gibbs et al. provide conditional guarantees over specified classes of shifts using a different calibration construction [53]. KGA’s ordinary residual interval implements neither procedure. Its marginal false-adapt bound therefore must be distinguished from error conditional on ADAPT and from coverage within each target environment. Learn Then Test separately addresses configuration selection and multiplicity [41].

3 Setup, Notation, and Claim Levels

3.1 Model Pair, Evidence, and Evaluation Unit

Let P_S and P_T be the source and target laws and $R_T(f) = \mathbb{E}_{(X,Y) \sim P_T}[\ell(f(X), Y)]$. Population risks integrate a fresh draw from P_T , holding the realized model pair fixed. An adapter $\mathcal{A}$ proposes f_a from f_0 and unlabeled inputs $X_{1:m}$. The decision can read

$$Z = \phi(X_{1:m}, f_0, f_a),$$

including predictions, logits, and update statistics, but not the new evaluation labels.

For evaluation unit j , with records $\mathcal{E}_j$ and a fixed higher-is-better score S , define

$$\Delta_j^{\text{cell}} = S(f_a; \mathcal{E}_j) - S(f_0; \mathcal{E}_j).$$

The two models are scored on the same records. This measured quantity is not population risk. For accuracy it is a sample average of paired correctness differences; for macro-F1 it is a difference of dataset-level scores, not an average of per-example losses. Metric scales are reported separately.

3.2 Actions and Their Operational Meaning

Definition 1 (Decision semantics). ADAPT commits f_a when the interval supports positive benefit.

FREEZE rejects the candidate when it supports negative benefit. ABSTAIN retains f_0 when neither strict direction is supported or the assessment is unavailable.

Freeze is a supported rejection. Abstention is an unresolved assessment. An operator can attach monitoring, label acquisition, or human review to the abstention record; the experiments here evaluate only the frozen fallback and uncertainty log, not those additional interventions. Retaining f_0 after missing evidence must be logged as ABSTAIN, not as a certified FREEZE.

The observed-cell oracle chooses the larger score. Regret is the policy’s score shortfall from that oracle, averaged over the named units. Always-freeze regret is $\max(\Delta_j^{\text{cell}}, 0)$; always-adapt regret is $\max(-\Delta_j^{\text{cell}}, 0)$. KGA uses the candidate score only after ADAPT. Lower regret is better, and a zero-benefit tie carries zero regret for either model even though it supports neither strict sign.

3.3 Commitment, Interval Coverage, and Error

These are three different measurements:

$$\text{Commit} = \mathbb{P}(g \in \{\text{ADAPT}, \text{FREEZE}\}), \quad (2)$$

$$\text{Cov}_{\text{cell}} = \mathbb{P}(|\hat{\Delta} - \Delta^{\text{cell}}| \leq \varepsilon), \quad (3)$$

$$\text{FA}_u = \mathbb{P}(g = \text{ADAPT}, \Delta^{\text{cell}} \leq 0). \quad (4)$$

Conditional false adaptation is $\text{FA}_c = \mathbb{P}(\Delta^{\text{cell}} \leq 0 \mid g = \text{ADAPT})$, defined only when $\mathbb{P}(g = \text{ADAPT}) > 0$. The operational proposition below bounds the marginal error under interval coverage; it does not bound FA_c . A gate that never adapts has zero false adaptations mechanically, so every error report includes action exposure. Analogously, a false FREEZE is a FREEZE decision with $\Delta^{\text{cell}} \geq 0$; its conditional frequency divides by FREEZE decisions and is undefined when that action is absent.

The population theorem uses binary labels and 0/1 loss. The empirical gate can instead target any predeclared scalar score difference, including multi-class metrics. This interface does not extend the binary theorem. In general multiclass classification, $\Delta = \mathbb{P}_T(D)(p_a - p_0)$, with $p_a = \mathbb{P}_T(f_a = Y \mid D)$ and $p_0 = \mathbb{P}_T(f_0 = Y \mid D)$; only the binary setting forces $p_0 = 1 - p_a$.

Table 1: Closest roles, targets, and remaining distinctions. Literature comparisons are conceptual here; a historical AETTA comparison is reported in Appendix B.3.4, as a descriptive comparison whose rank-clipped calibration does not establish nominal coverage, while native TTA/ALine and full POEM comparisons remain incomplete.

Approach	Main target	Relation to this paper
TTA/ALine; AETTA	adapted-model performance and selection	closest empirical estimators; their scores can inform paired comparisons
POEM; risk monitors	protected adaptation or running risk	different monitored target and sequential assumptions
Partial-identification decisions	uncertain payoffs across compatible laws	context for the population benefit set and robust sign decision
Selection/shift-conditional conformal	coverage after selection or within a shift class	stronger, specialized targets not supplied by KGA’s marginal interval
K-Bound / KGA	feasible population benefit / predicted cell benefit	shared update question, distinct uncertainty objects and guarantees

Table 2: Notation and probability targets. Population and measured-cell quantities are kept separate.

Quantity	Level	Meaning
f_0, f_a	both	frozen predictor and realized shadow candidate
Δ	population	expected risk reduction $R_T(f_0) - R_T(f_a)$
$D, \mu_T(D)$	population	disagreement region and its target probability
M	population	label-free mean correctness-score margin on D
γ, β	population	hidden calibration residual and its externally declared absolute bound
Δ^{cell}	empirical	observed score difference on one predeclared evaluation unit
$\hat{\Delta}, \varepsilon$	empirical	predicted cell benefit and residual-based interval radius
FA_u	empirical	probability of ADAPT and a nonpositive cell benefit
FA_c	empirical	probability of nonpositive cell benefit conditional on ADAPT
Commitment rate	empirical	fraction of ADAPT or FREEZE decisions; not interval coverage

4 Population Partial-Identification Frontier

The result describes which benefit signs are compatible with a fixed label-free evidence law and a declared bound on the hidden residual. The population analysis uses binary classification and 0/1 loss, where disagreement implies that exactly one predictor is correct. This permits the exact benefit reduction and strict-commitment frontier derived below. The impossibility result applies within the declared label-kernel class; it does not rule out certification under additional structural restrictions or establish the same frontier for general multiclass problems. KGA separately evaluates an empirical scalar-benefit gate, including on multiclass benchmarks.

4.1 Disagreement-Region Model

Assumption 1 (Deployment setup). Let $Y \in \{0, 1\}$, use 0/1 loss, fix measurable binary predictors f_0, f_a , and assume $\mu_T(D) > 0$ for $D = \{x : f_0(x) \neq f_a(x)\}$. Let $s : \mathcal{X} \rightarrow [0, 1]$ be a fixed measurable candidate-correctness score, fitted using source information, and

let the target law admit the measurable pointwise correctness kernel $\eta_a(x) = \mathbb{P}_T(f_a(X) = Y \mid X = x)$. On D , set

$$M = \mathbb{E}[s(X) \mid D] - \frac{1}{2}, \quad \gamma = \mathbb{E}[\eta_a(X) - s(X) \mid D]. \quad (5)$$

The declared population class is $\mathcal{C}_\beta = \{P_T : |\gamma(P_T)| \leq \beta\}$ for an externally supplied $\beta \geq 0$. Because $s \in [0, 1]$, the feasible margin satisfies $M \in [-\frac{1}{2}, \frac{1}{2}]$. The identities and frontier below do not require a source-calibration property of s .

Remark 1 (Residual identity). The definition gives

$$\begin{aligned} M + \gamma &= (\mathbb{E}[s(X) \mid D] - \frac{1}{2}) + \mathbb{E}[\eta_a(X) - s(X) \mid D] \\ &= \mathbb{E}[\eta_a(X) \mid D] - \frac{1}{2}. \end{aligned}$$

This is an identity, not empirical evidence that calibration transfers. We call γ a target disagreement-conditional calibration residual, not automatically source-to-target drift. Ordinary source calibration need not hold after conditioning on D , so γ can be nonzero even when $P_T = P_S$. A literal label-kernel shift interpretation would require an additional reference, for example the pointwise source correctness

Table 3: Three claim levels and their validity obligations. This table collects the distinctions used throughout the paper.

Claim level	Required premise	Conclusion and boundary
Population frontier	binary 0/1 loss, fixed augmented evidence, declared residual class; richness for necessity	exact strict-sign frontier within that class, not a learned residual bound
Operational proposition	marginal interval coverage for a named scalar target and unit	marginal false commitment for that target, not automatic conditional or anytime control
Empirical evaluation	fixed recorded candidates, splits, metric and action rule	measured cell regret, interval inclusion and action exposure; population transfer needs a separate argument

$s(x) = \mathbb{P}_S(f_a(X) = Y \mid X = x)$. The no-shift counterexample is given in Appendix L. Calibration notions also depend on the conditioning information: top-label calibration, for example, additionally conditions on the predicted class [43]. Neither score nor top-label calibration automatically supplies calibration conditional on D .

Definition 2 (Risk alignment). Evidence Z is risk-aligned on a class $\mathcal{P}$ if no two laws in $\mathcal{P}$ with opposite nonzero benefit signs induce the same law of Z . This is weaker than uniform strict-sign identification: a fibre containing zero and positive benefit is risk-aligned by this definition, but does not support a uniformly positive claim. Population frontier claims take M to be fixed on each augmented evidence fibre, equivalently by using $\tilde{Z} = (Z, M)$.

Definition 3 (Strict directional soundness). At a fixed augmented-evidence law, ADAPT is uniformly directionally sound when $\Delta(P) > 0$ for every compatible $P \in \mathcal{C}_\beta$; FREEZE is sound when $\Delta(P) < 0$ for every such law. A pointwise maximal three-way rule commits whenever one strict action is sound and abstains otherwise. The strict convention treats $\Delta = 0$ as insufficient for either directional claim, although the actions then have equal task risk.

4.2 Label-Kernel Freedom on the Disagreement Region

Lemma 1 (Label-kernel freedom on disagreement). Fix $(\mu_T, P_S, f_0, f_a, s)$, hence D and M , and fix the conditional label law outside D . For every measurable $\eta : D \rightarrow [0, 1]$ there is a target law with input marginal μ_T and $\mathbb{P}(f_a(X) = Y \mid X = x) = \eta(x)$ on D . Every label-free observable formed from the input batch, the fixed models, and their outputs has the same law for all such η , while

$$\gamma = \mathbb{E}[\eta(X) \mid D] - \frac{1}{2} - M.$$

Proof. On D , binary predictors disagree, so their outputs are 0 and 1. Assign conditional mass $\eta(x)$ to $Y = f_a(x)$ and $1 - \eta(x)$ to $Y = f_0(x)$, retaining the fixed kernel outside D . This changes no input marginal or model output and therefore changes no label-free evidence law. The expression for γ follows from Eq. (5). $\square$

4.3 Benefit Reduction and Matched-Evidence Impossibility

Lemma 2 (Disagreement-region reduction). Under Assumption 1, with $\bar{a} := \mathbb{P}_T(f_a = Y \mid D)$,

$$\begin{aligned} \Delta &= 2\mu_T(D)(\bar{a} - \tfrac{1}{2}), \\ \text{sign } \Delta &= \text{sign}(\bar{a} - \tfrac{1}{2}) = \text{sign}(M + \gamma). \end{aligned}$$

The identity $\text{sign } \Delta = \text{sign}(M + \gamma)$ holds for every target law; it does not assume $|\gamma| \leq \beta$.

Proof. On D exactly one of f_0, f_a is correct under 0/1 loss, so the benefit equals $2\mu_T(D)(\bar{a} - \frac{1}{2})$. By the definitions of M and γ ,

$$\begin{aligned} M + \gamma &= \left(\mathbb{E}_{\mu_T|D}[s] - \tfrac{1}{2} \right) + \mathbb{E}_{\mu_T|D}[\eta_a - s] \\ &= \mathbb{E}_{\mu_T|D}[\eta_a] - \tfrac{1}{2} = \bar{a} - \tfrac{1}{2}, \end{aligned}$$

so $\text{sign } \Delta = \text{sign}(\bar{a} - \frac{1}{2}) = \text{sign}(M + \gamma)$. $\square$

Theorem 1 (Interior matched-evidence impossibility). Fix $\beta > 0$. For any feasible margin $M \in [-\frac{1}{2}, \frac{1}{2}]$ with $|M| < \beta$ there exist $P_T^1, P_T^2 \in \mathcal{C}_\beta$ such that $\text{Law}(\tilde{Z} \mid P_T^1) = \text{Law}(\tilde{Z} \mid P_T^2)$ for $\tilde{Z} = (Z, M)$, $M(P_T^1) = M(P_T^2) = M$, and $\text{sign } \Delta_1 = -\text{sign } \Delta_2 \neq 0$. (For $\beta = 0$ the class $\mathcal{C}_0$ forces $\gamma = 0$, so $\text{sign } \Delta = \text{sign } M$ is determined and no such pair exists: matched-evidence ambiguity requires a positive declared residual bound.)

Proof. This is Lemma 1 read at one radius. Choose $0 < \delta < \min\{\beta - |M|, \frac{1}{2}\}$ and apply the lemma with the two constant kernels $\eta \equiv \frac{1}{2} + \theta\delta$, $\theta \in \{+1, -1\}$,

holding the off- D kernel K° common to both worlds. The lemma supplies the two laws P_T^θ , and: *class membership*, $\bar{a}_\theta = \frac{1}{2} + \theta\delta$ so $\gamma_\theta = \theta\delta - M$ with $|\gamma_\theta| \leq \delta + |M| < \beta$, hence both worlds lie in $\mathcal{C}_\beta$; *matched evidence*, $\text{Law}(W \mid P_T^\eta)$ does not depend on η by the lemma and $\tilde{Z} = (Z, M)$ is a function of W , so $\text{Law}(\tilde{Z} \mid P_T^+) = \text{Law}(\tilde{Z} \mid P_T^-)$; *opposite signs*, Lemma 2 gives $\Delta_\theta = 2\mu_T(D)\theta\delta \neq 0$ with $\text{sign } \Delta_\theta = \theta$. $\square$

The construction changes only the target label kernel on D , leaving the label-free evidence laws exactly identical. The result is therefore an exact non-identifiability statement rather than an estimation-rate bound.

Corollary 1 (Matched-evidence abstention lower bound). *Fix $\alpha \in (0, \frac{1}{2})$. On the shared evidence law constructed in Theorem 1, any possibly randomized label-free rule g satisfying $\mathbb{P}[g = \text{ADAPT}, \Delta \leq 0] \leq \alpha$ in every admissible world and $\mathbb{P}[g = \text{FREEZE}, \Delta \geq 0] \leq \alpha$ in every admissible world must satisfy $\mathbb{P}[g = \text{ABSTAIN}] \geq 1 - 2\alpha$.*

Proof. The rule has the same action probabilities in the two evidence-identical worlds. Its adapt probability is at most α in the negative world and its freeze probability is at most α in the positive world. The three action probabilities sum to one. $\square$

4.4 Boundary Case

Proposition 1 (Boundary case and closed-band abstention). *Fix $\alpha \in (0, \frac{1}{2})$. Under Definition 3, abstention is the pointwise maximal sound three-way action throughout the closed band $|M| \leq \beta$, and any label-free rule controlling both marginal errors at level α abstains there with probability at least $1 - 2\alpha$. Opposite nonzero signs are asserted only in the interior $|M| < \beta$.*

Proof. For $|M| < \beta$ this is Theorem 1 together with Corollary 1. For $|M| = \beta > 0$, the declared class contains an augmented-evidence-identical zero-benefit world (take $\gamma = -M$, admissible since $|\gamma| = \beta$) and a strict-benefit world of sign M (take $\gamma = 0$). In the zero-benefit world $\Delta = 0$, so both directional error events $\{\text{ADAPT}, \Delta \leq 0\}$ and $\{\text{FREEZE}, \Delta \geq 0\}$ occur whenever the rule commits; since the rule's action probabilities are identical across the two evidence-identical worlds, dual control at level α gives $\mathbb{P}[\text{ADAPT}] \leq \alpha$ and $\mathbb{P}[\text{FREEZE}] \leq \alpha$, hence $\mathbb{P}[\text{ABSTAIN}] \geq 1 - 2\alpha$, and neither strict action is uniformly directionally sound. For $M = \beta = 0$ the class

$\mathcal{C}_0$ forces $\gamma = 0$, so $\Delta \equiv 0$ and the same conclusion holds directly. $\square$

This directional convention is stricter than task regret: at $\Delta = 0$, adapt and freeze are risk-equivalent but neither certifies a strict sign.

4.5 Exact Strict-Commitment Frontier

Theorem 2 (Exact strict-commitment frontier). *Under Assumption 1, fix the observables $(\mu_T, P_S, f_0, f_a, s)$ so M is determined, and let $\beta \geq 0$. The necessity and maximality clauses use the full class $\mathcal{C}_\beta$ and therefore its richness: it contains the evidence-identical label kernels constructed in Theorem 1 and the zero-benefit boundary law. For a narrower declared subclass not closed under these constructions, only the sufficiency clause follows without an additional richness assumption.*

- (i) **Reduction.** *For every P_T , $\text{sign } \Delta = \text{sign}(M + \gamma)$ (Lemma 2).*
- (ii) **Sufficiency.** *If $|M| > \beta$ and $P_T \in \mathcal{C}_\beta$, then $\text{sign } \Delta = \text{sign } M$.*
- (iii) **Necessity.** *If $|M| < \beta$ (which requires $\beta > 0$), there exist $P_T^1, P_T^2 \in \mathcal{C}_\beta$ with the same augmented-evidence law and opposite nonzero benefit signs; at $|M| = \beta > 0$ the admissible residual $\gamma = -\text{sign}(M)\beta$ yields $\Delta = 0$, and in the degenerate case $M = \beta = 0$ the class $\mathcal{C}_0$ forces $\Delta \equiv 0$. Hence no strict benefit direction is uniformly supported by $\text{Law}(\tilde{Z})$ over $\mathcal{C}_\beta$ whenever $|M| \leq \beta$.*
- (iv) **Strict-commitment iff.** *At the fixed augmented-evidence law, a strict action in $\{\text{ADAPT}, \text{FREEZE}\}$ is uniformly directionally sound in the sense of Definition 3 if and only if $|M| > \beta$; on the closed band $|M| \leq \beta$ (including $M = \beta = 0$, where $\Delta \equiv 0$) abstention is the pointwise maximal sound action. The population rule $M > \beta \Rightarrow \text{ADAPT}$, $M < -\beta \Rightarrow \text{FREEZE}$, $|M| \leq \beta \Rightarrow \text{ABSTAIN}$ is the pointwise maximal sound certificate on $\mathcal{C}_\beta$.*

Proof. Part (i) is Lemma 2. For (ii), if $M > \beta$ and $|\gamma| \leq \beta$, then $M + \gamma > 0$, so $\text{sign } \Delta = \text{sign}(M + \gamma) = +1$; the negative case is symmetric. For (iii), Theorem 1 supplies two laws in the same augmented-evidence fibre with opposite nonzero signs whenever $|M| < \beta$. At $|M| = \beta$ the boundary law with $\gamma = -\text{sign}(M)\beta$ gives $\Delta = 0$, still precluding a determined nonzero sign. Part (iv) combines (ii), (iii), and Proposition 1. Pointwise maximality follows because (ii) certifies the unique strict direction outside the band, while Theorem 1 and Proposition 1 rule out both strict directions inside and on the boundary. $\square$

The feasible range can also be read directly from the identified benefit set. Write $d = \mu_T(D) > 0$. Over the full declared class with the observables fixed,

$$\Delta(\mathcal{C}_\beta) = 2d \left[\max\{-\tfrac{1}{2}, M - \beta\}, \min\{\tfrac{1}{2}, M + \beta\} \right].$$

Indeed, $\bar{a} - \frac{1}{2} \in [-\frac{1}{2}, \frac{1}{2}]$ and $|\bar{a} - \frac{1}{2} - M| \leq \beta$; every value in their intersection is attained by a constant correctness kernel on D . A strict direction is available exactly when this whole interval lies above or below zero.

Region	Meaning	Action
$M > \beta$	strictly positive benefit	ADAPT
$M < -\beta$	strictly negative benefit	FREEZE
$ M < \beta$	both strict signs are feasible	ABSTAIN
$ M = \beta > 0$	zero benefit is feasible	ABSTAIN
$M = \beta = 0$	benefit is identically zero	ABSTAIN

4.6 Interpretation and Limits of the Frontier

The theorem depends on a declared class. Choosing $\beta = 0$ assumes a zero target disagreement-conditional residual; it is not a conservative default. Without a credible upper bound on $|\gamma|$, the population frontier cannot support a robust strict action. More unlabeled samples do not solve this problem because the missing difference lies in the target labels, not in the input batch.

Empirical abstention is not automatically evidence of structural unknowability. A wide finite-sample interval may instead reflect limited calibration data, estimator error, or transfer failure. The frontier therefore leaves one central question: can its required residual bound be learned from the same label-free evidence?

5 What Label-Free Evidence Cannot Learn

5.1 Audit Setup and Evidence Fibres

Let W collect everything a deployment-time audit may read: the unlabeled target batch, the fixed models and score function, label-free statistics derived from them, and any source or calibration sample whose law is specified independently of the target label kernel. An audit is a nonnegative random quantity $\hat{\beta} = g(W, U)$, where U is independent randomization.

For a declared class $\mathcal{C}$ and an observable law $z = \text{Law}(W)$, define the evidence fibre

$$\mathcal{C}(z) = \{P \in \mathcal{C} : \text{Law}(W | P) = z\}$$

and its unresolved residual radius

$$\Gamma_z(\mathcal{C}) = \sup_{P \in \mathcal{C}(z)} |\gamma(P)|. \quad (6)$$

The fibre collects target laws that the audit cannot distinguish. Its radius is therefore the smallest uniform residual bound that those observations could support.

5.2 Exact Fibre-Wise Audit Floor

Theorem 3 (Exact label-free audit floor). *Fix a nonempty fibre $\mathcal{C}(z)$ and $\delta \in [0, 1]$. If $\hat{\beta}$ satisfies $\mathbb{P}_P(|\gamma(P)| \leq \hat{\beta}) \geq 1 - \delta$ for every $P \in \mathcal{C}(z)$, then*

$$\mathbb{P}(\hat{\beta} \geq \Gamma_z(\mathcal{C})) \geq 1 - \delta.$$

The constant audit $\hat{\beta} \equiv \Gamma_z(\mathcal{C})$ is valid with $\delta = 0$. Consequently, on the high-probability event above, an audited frontier commits only where the frontier using the declared radius $\Gamma_z(\mathcal{C})$ already commits.

Proof. Every law in the fibre induces the same law for W , and hence the same law for $\hat{\beta}$. Validity for each fixed P requires that common law to exceed $|\gamma(P)|$ with probability at least $1 - \delta$. Taking an increasing sequence of attainable absolute residuals approaching the supremum gives the lower bound. The constant audit is valid by the definition of the supremum. If $\hat{\beta} \geq \Gamma_z$, then $|M| > \hat{\beta}$ implies $|M| > \Gamma_z$. $\square$

This result does not say that useful budgets never exist. It says that the same label-free fibre cannot create information about the target label kernel that the fibre definition removes. A valid audit must cover the unresolved radius; reading more functions of the same W does not shrink it. The constant $\Gamma_z(\mathcal{C})$ is an oracle benchmark for the fixed evidence-law fibre. The theorem does not provide an algorithm that learns the unknown law z or its radius from a finite observed batch. We therefore describe this as a fibre-wise audit floor rather than an unspecified minimax estimation result.

5.3 Declared Budget and Abstention Radius

For the full declared class $\mathcal{C}_\beta = \{P : |\gamma(P)| \leq \beta\}$ with $0 \leq \beta \leq \frac{1}{2}$, the label-kernel construction in Lemma 1 gives

$$\Gamma_z(\mathcal{C}_\beta) = \beta. \quad (7)$$

Thus for any $\beta \in [0, 1/2]$, the least fibre-wide budget a label-free audit can certify is the budget already

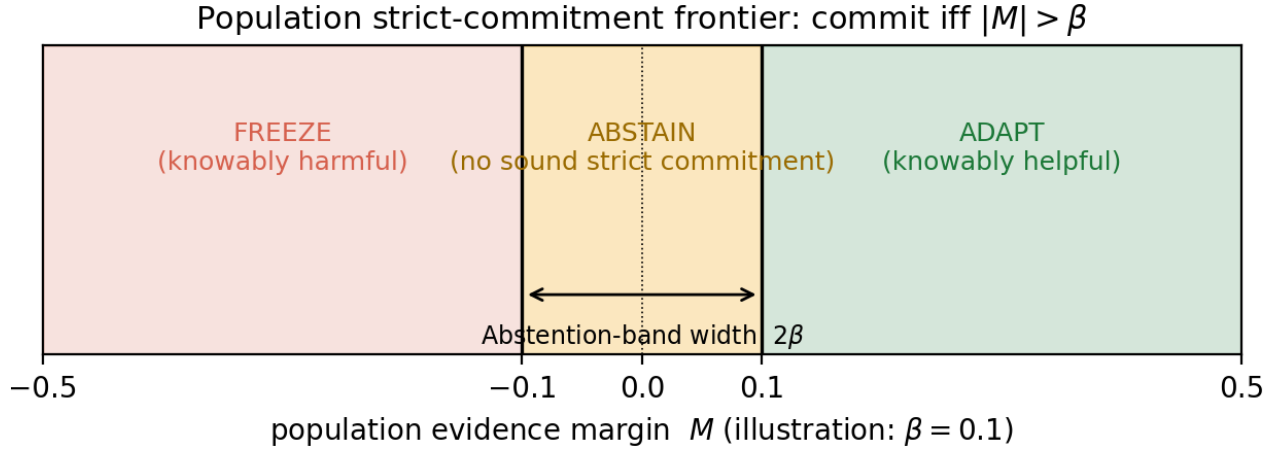


Figure 2: Population strict-commitment frontier. $M > \beta$ supports adapt; $M < -\beta$ supports freeze; and $|M| \leq \beta$ supports no uniform strict commitment. Interior ambiguity uses opposite nonzero benefit signs, whereas the boundary uses zero-benefit ambiguity. This population rule is not the numerical KGA rule used on the real-data panels.

declared by the class. The same number is also the population abstention radius:

least valid fibre-wide residual bound
 = population abstention radius = β ($0 \leq \beta \leq \frac{1}{2}$)

This equality connects the audit problem to the strict-commitment frontier. For the full declared class, Theorem 3 rules out a uniformly tighter residual budget from the same evidence. We therefore treat β as an *externally justified sensitivity parameter*, by analogy to Rosenbaum’s sensitivity parameter Γ for unobserved confounding in observational studies [55] or the ambiguity set radius in distributionally robust optimization (an analogy, not a mathematical equivalence). Extended exploratory formulations of break-even sensitivity frontiers and continuous selection odds-ratio curves are preserved in the supplementary research archive; the primary manuscript retains the exact population frontier $\Delta(\mathcal{C}_\beta)$ established in Theorem 2.

5.4 Historical Labels Require a Transfer Assumption

Suppose the audit also observes a fully labeled calibration domain, but the declared class contains no coupling between that domain’s label kernel and the deployment label kernel. The construction in Lemma 1 changes only deployment labels on D and leaves the calibration sample unchanged. The augmented evidence law is therefore identical across the same target fibre, so Theorem 3 applies without a smaller radius.

Labels help only when they come from the deployment domain or when a stated assumption links the labeled and deployment domains.

5.5 What Additional Structure Can Buy

The missing information can be supplied, but each route changes the problem. Deployment labels can estimate target correctness directly. Historical labeled deployments can support an episode-marginal budget under an exchangeability model. A structural coupling can bound the change between calibration and target domains. Domain knowledge can declare a physically meaningful limit.

Multiclass extensions under entropy-constrained support conditions require separate assumptions and proofs; they are preserved as unproved exploratory research notes outside the maintained claims of this manuscript.

5.6 Implications for K-Bound and KGA

The evidence-fibre result motivates a practical question: what additional information can support an update decision? KGA uses labeled historical shifts to predict and calibrate measured benefit. Section 6.1 states its inferential target and the extra assumptions required for population interpretation.

Table 4: Correspondence between population theory (K-Bound) and the empirical decision rule (KGA) addressing the same paired deployment choice. Distinct inferential targets do not make the quantities equal.

Quantity	Role in Population Frontier (K-Bound)	Status in Empirical Rule (KGA)
M	Observable margin on disagreement region	Not explicitly estimated
γ	Hidden calibration residual ($M + \gamma = \bar{a} - \frac{1}{2}$)	Unobserved / unestimated
β	Externally declared hidden-residual bound	Not a numerical KGA input
Δ	Population benefit $2\mu_T(D)(M + \gamma)$	Target estimand under population transfer
Δ^{cell}	Not present in pure population model	Measured evaluation-cell benefit $S(f_a; \mathcal{E}) - S(f_0; \mathcal{E})$
$\hat{\Delta}$	Not present in pure population model	Fitted prediction $h_\theta(Z)$ of empirical target Δ^{cell}
ε	Not present in pure population model	Empirical residual-calibration radius
$b(m, \delta)$	Not present in pure population model	Separately justified sampling allowance

6 KGA: An Empirical Benefit-Interval Gate

6.1 From Population Identification to Empirical Decisions

K-Bound characterizes population benefit over a declared class of target laws. KGA predicts measured evaluation-cell benefit using labeled historical outcomes and a separately calibrated residual interval. It does not numerically apply $|M| > \beta$, and its empirical radius ε is not an estimate of the population residual bound β . When a credible population budget is supplied externally, a direct finite-sample frontier rule is available (Appendix A.6). The empirical route instead depends on support for its calibration assumptions.

The binary population model gives

$$\Delta = 2\mu_T(D)(M + \gamma), \quad (8)$$

while the observed cell outcome and fitted empirical prediction are

$$\Delta^{\text{cell}} = S(f_a; \mathcal{E}) - S(f_0; \mathcal{E}), \quad \hat{\Delta} = h_\theta(Z). \quad (9)$$

Table 4 summarizes this architectural correspondence.

A population interpretation requires both cell-level coverage and a separate sampling bound:

$$\begin{aligned} \mathbb{P}(|\hat{\Delta} - \Delta^{\text{cell}}| \leq \varepsilon) &\geq 1 - \alpha_{\text{cell}}, \\ \mathbb{P}(|\Delta^{\text{cell}} - \Delta| \leq b(m, \delta)) &\geq 1 - \delta. \end{aligned}$$

The cell-coverage and fresh-sample concentration

premises imply, by a union bound,

$$\begin{aligned} \mathbb{P}(|\hat{\Delta} - \Delta| \leq \varepsilon + b(m, \delta)) \\ \geq 1 - \alpha_{\text{cell}} - \delta. \end{aligned}$$

The enlarged interval therefore covers population benefit only when both premises hold. Proposition 3 states this implication; its sampling assumptions are not established for the benchmark panels. Table 4 keeps the population and empirical quantities distinct.

KGA is label-free at deployment, not during development and calibration. Historical labeled cell outcomes fit h_θ and supply residuals. At the new decision, KGA reads only Z and the frozen fitted artifacts. The interval premise for the empirical target is

$$\mathbb{P}(|\hat{\Delta} - \Delta^{\text{cell}}| \leq \varepsilon) \geq 1 - \alpha. \quad (10)$$

The static interval requires its stated calibration assumptions. Sequential monitoring is a separate extension requiring its own null model and validity conditions. The empirical results reported here do not establish an anytime guarantee or a guaranteed response to every harmful shift.

6.2 The Coverage-to-Action Proposition

The following standard implication makes the decision semantics explicit. It is not a new calibration theorem: coverage is the premise, and establishing it is the statistical burden.

Proposition 2 (Coverage-to-action implication). *Fix the scalar benefit target and its evaluation unit before calibration. Let B and $\hat{B}$ be finite real random variables, and let $\varepsilon \in [0, \infty]$ be a measurable random radius on a common probability space containing all calibration, benefit-model fitting, candidate-construction, and evaluation-unit randomness. The prediction $\hat{B}$ and radius ε must be available without reading the new unit's evaluation labels. For $\alpha \in (0, 1)$, suppose the interval satisfies marginal coverage*

$$\mathbb{P}[|\hat{B} - B| \leq \varepsilon] \geq 1 - \alpha.$$

Define

$$g = \begin{cases} \text{ADAPT} & \hat{B} - \varepsilon > 0, \\ \text{FREEZE} & \hat{B} + \varepsilon < 0, \\ \text{ABSTAIN} & \text{otherwise.} \end{cases}$$

Then, marginally over that common probability space,

$$\begin{aligned} \mathbb{P}[g = \text{ADAPT}, B \leq 0] &\leq \alpha, \\ \mathbb{P}[g = \text{FREEZE}, B \geq 0] &\leq \alpha. \end{aligned}$$

Proof. On the event $\{|\hat{B} - B| \leq \varepsilon\}$, the condition $\hat{B} - \varepsilon > 0$ implies $B > 0$, so a false adapt requires the complement of this event, which has probability at most α . The freeze case is symmetric. $\square$

Remark 2 (Unit scaling: empirical radius versus theoretical budget). The empirical radius ε and theoretical residual bound β live on fundamentally different scales. The population identity is $\Delta = 2d(M + \gamma)$ where $d = P_T(D) > 0$ is the disagreement probability and $2d$ is twice that probability. Thus, $|\Delta - 2dM| = 2d|\gamma| \leq 2d\beta$. The unclipped population envelope centered at $2dM$ has radius $2d\beta$ on the benefit scale, not β ; the exact population benefit set $\Delta(\mathcal{C}_\beta)$ in Theorem 2 includes clipping to $[-d, d]$: $\Delta(\mathcal{C}_\beta) = 2d[\max(-\frac{1}{2}, M - \beta), \min(\frac{1}{2}, M + \beta)]$. The quantities have different definitions. The declared bound β limits the absolute target disagreement-conditional residual, whereas ε is calibrated from prediction errors for the empirical target. They may coincide numerically, but that does not make them interchangeable.

Proposition 3 (Cell-to-population transfer via compound bound). *Let $m \geq 1$ be an integer and let $\Delta^{\text{cell}} = \frac{1}{m} \sum_{j=1}^m W_j$ be the measured cell benefit on m fresh, conditionally i.i.d. evaluation draws, where $W_j = \mathbf{1}\{f_a(X_j) = Y_j\} - \mathbf{1}\{f_0(X_j) = Y_j\} \in [-1, 1]$, with conditional mean Δ . Suppose an independently fitted benefit predictor h and split-conformal radius ε achieve marginal cell coverage $\mathbb{P}(|\hat{\Delta} - \Delta^{\text{cell}}| \leq \varepsilon) \geq 1 - \alpha_{\text{cell}}$ over exchangeable deployment episodes. For a sampling budget $\delta \in (0, 1)$, define $b(m, \delta) = \min\{2, \sqrt{2\log(2/\delta)/m}\}$. Then the compound interval $I_{\text{population}} = [\hat{\Delta} - (\varepsilon + b), \hat{\Delta} + (\varepsilon + b)]$ satisfies*

$$\mathbb{P}(\Delta \in I_{\text{population}}) \geq 1 - \alpha_{\text{cell}} - \delta.$$

The corresponding population decision rule g_{pop} using radius $\varepsilon + b$ satisfies

$$\mathbb{P}(\{g_{\text{pop}} = \text{ADAPT}, \Delta \leq 0\} \cup \{g_{\text{pop}} = \text{FREEZE}, \Delta \geq 0\}) \leq \alpha_{\text{cell}} + \delta.$$

Proof. By Hoeffding’s inequality for bounded variables in $[-1, 1]$ of range 2, $\mathbb{P}(|\Delta^{\text{cell}} - \Delta| \leq b) \geq 1 - \delta$. On the intersection of cell coverage and concentration, the triangle inequality yields $|\hat{\Delta} - \Delta| \leq |\hat{\Delta} - \Delta^{\text{cell}}| + |\Delta^{\text{cell}} - \Delta| \leq \varepsilon + b$. A union bound proves coverage at level $1 - \alpha_{\text{cell}} - \delta$. Event containment (as in Proposition 2) immediately bounds the probability of an incorrect strict direction by $\alpha_{\text{cell}} + \delta$. $\square$

Remark 3 (Direct empirical sampling of the population frontier). An alternative route directly operationalizes the population frontier $|M| > \beta$. Fix the predictor pair and score independently of a fresh i.i.d. sample of unlabeled target inputs, and let $n_D > 0$ of those inputs fall in D . The empirical disagreement margin $\hat{M} = \frac{1}{n_D} \sum_{X_i \in D} s(X_i) - \frac{1}{2}$ concentrates with radius $r_M = \sqrt{\log(2/\alpha)/(2n_D)}$. Under the externally declared bound $|\gamma| \leq \beta$, the rule $\hat{M} - r_M > \beta \Rightarrow \text{ADAPT}$, $\hat{M} + r_M < -\beta \Rightarrow \text{FREEZE}$, and ABSTAIN otherwise, guarantees directional error at most α . This couples sampling uncertainty r_M additively with structural uncertainty β .

Remark 4 (Marginal vs. conditional error). Proposition 2 bounds the *marginal, unconditional* false-commit probabilities for the declared target B . Its false-adapt rate is $\text{FA}_u = \mathbb{P}[g = \text{ADAPT}, B \leq 0]$. It does *not* bound $\text{FA}_c = \mathbb{P}[B \leq 0 \mid g = \text{ADAPT}]$, defined when $\mathbb{P}[g = \text{ADAPT}] > 0$ and reported descriptively for $B = \Delta^{\text{cell}}$. Selection-conditional coverage requires a separate construction, such as that of Jin and Ren [52]; conditional guarantees over specified shift classes also require their own calibration procedure [53]. Neither is supplied by the marginal premise above. Simultaneous protection across candidates and time-uniform protection likewise require additional guarantees. Risk alignment (Definition 2) concerns population sign identifiability; coverage for B alone yields the *interval* implication above.

Remark 5 (Fallback when assumptions are unsupported). If interval coverage for the declared target and evaluation unit—or the assumptions supporting it—cannot be justified, Proposition 2 supplies no level- α protection for either strict action. A separately declared operational fallback may retain the frozen model and record abstention. Serving the frozen model without a valid negative interval is not a certified FREEZE decision. This policy recommendation does not establish that the retained model is safe or optimal. If *risk alignment* (Definition 2) cannot be justified for an intended population sign claim, that interpretation remains unsupported. The interval implication of Proposition 2 still holds when coverage for the declared B is established (cf. Theorem 1).

Proposition 3 formalizes the conditional bridge to population risk Δ : under independent fresh evaluation draws, adding the Hoeffding concentration radius $b(m, \delta)$ to the empirical radius ε yields a compound population interval with radius $\varepsilon + b$ and error budget $\alpha_{\text{cell}} + \delta$. The full protocol, counterexample, and proof are detailed in Appendix A. No such sampling-

Table 5: 108-cell ImageNet-C population replay: cell gate ($\hat{\Delta} \pm \varepsilon$) vs. compound population interval ($\hat{\Delta} \pm (\varepsilon + b)$) under 5-fold cross-fitting ($\alpha_{\text{cell}} = \delta = 0.05$, $\alpha_{\text{pop}} = 0.10$, $m = 2,000$, $b = 0.0607$). Full 12-column breakdown with fold details and radii ranges in Table 16.

Candidate	N	Regret (Cell / Pop)	Cell Gate (A / F / U)	Compound Pop (A / F / U)	FA_u
Tent	36	0.0162 / 0.0468	18 / 0 / 18	1 / 0 / 35	0.000
EATA	36	0.0095 / 0.0386	20 / 0 / 16	0 / 0 / 36	0.000
SAR	36	0.0245 / 0.0267	2 / 11 / 23	0 / 10 / 26	0.000
Total / Panel	108	0.0167 / 0.0374	40 / 11 / 57	1 / 10 / 97	0.000

error bound is established for the reported panels; in particular, when a candidate adapts to evaluation inputs, transductive normalization couples the candidate across batches.

The formalized action rule proves that a strict decision has the correct sign whenever the compound interval contains population benefit. This implication does not establish the sampling or calibration premises for the benchmarks; the scope and theorem mapping are given in Appendix M.

ImageNet-C 108-cell population replay. To audit this procedure on existing benchmark data without changing historical estimands, we replayed the outcome-blind compound interval $[\hat{\Delta} \pm (\varepsilon + b)]$ across 108 ImageNet-C development cells (36 cells each for Tent, EATA, and SAR; five-fold cell-outcome-disjoint cross-fitting; sensitivity parameter $m = 2,000$, yielding uniform concentration radius $b = 0.0607$; Table 5). Full fold breakdowns and radius distributions appear in Table 16 (Appendix A.7). Finite radii were obtained on 108/108 cells. Across the panel, the cell-level gate ($\hat{\Delta} \pm \varepsilon$) issued 40 ADAPT, 11 FREEZE, and 57 ABSTAIN decisions with zero false adaptations ($\text{FA}_u = 0.0000$) and mean oracle regret 0.0167. Incorporating the sampling radius ($b = 0.0607$, with mean compound half-width $\bar{\varepsilon} + b = 0.1272$) shifted compound decisions to 1 ADAPT, 10 FREEZE, and 97 ABSTAIN, preserving zero observed false adaptations ($\text{FA}_u = 0.0000$) while shifting regret to 0.0374. This concretely demonstrates the sensitivity to the added concentration radius $b = 0.0607$: because most empirical gains on this benchmark are smaller than the universal Hoeffding penalty for $m = 2,000$, the compound rule withholds action on nearly all adaptations, illustrating the additional abstention induced by this sampling allowance rather than establishing prospective population truth. As clarified in Appendix A.7, $m = 2,000$ is an evaluation sensitivity parameter, not fresh sampling; population truth remains unestablished on historical development records.

Non-oracle population diagnostic. While the synthetic diagnostic in Table 15 tested exact Bayes prediction ($\Delta(Z) = 0.6Z$), evaluating an imperfect predictor subject to estimation noise requires an extended non-oracle study. We conducted a diagnostic across 1,000 independent trials per configuration over 16 (N_{cal}, m) pairs, fitting an imperfect linear predictor on finite development episodes ($m = 32$; Table 14, Appendix A.7). An unhedged point rule ($\text{sign}(\hat{\Delta})$) incurs a 30.9%–35.6% directional error rate ($\{g = \text{ADAPT} \wedge \Delta \leq 0\} \cup \{g = \text{FREEZE} \wedge \Delta \geq 0\}$) with 0% abstention (where FA_u is defined strictly for false ADAPT; here directional errors combine false ADAPT and false FREEZE on null/tie states); crucially, 100% of these errors occur on null/tie states ($Z = 0, \Delta = 0$) due to estimation noise around zero, with zero errors on strictly harmful states ($\Delta = -0.6$). In the finite-radius configurations, the calibrated cell interval recorded 93.9%–96.5% empirical cell inclusion against a nominal 95% level and zero observed wrong-direction decisions. Observed population inclusion was approximately 100% for the cell interval in this synthetic model. The compound interval also recorded 100% empirical population inclusion and zero observed wrong-direction decisions. Its abstention rate was 99.9%–100% at $m = 32$ and approximately one third for the finite-radius configurations with $m \geq 128$. At $N_{\text{cal}} = 18$, the required calibration rank is unavailable, so the infinite-radius rule abstains throughout. These are finite-simulation observations, not a proof of universal coverage.

6.3 Evidence, Calibration, and the Action Rule

The controlled implementation stores 11 label-free features, with algebraic rank at most 9 because two features are deterministic differences. They summarize confidence, class balance, model disagreement, and update size. The current principal CIFAR replay fits a separate **GradientBoostingRegressor** [33] for each adapter, stream seed and score fold, using squared-error loss, 250 trees, depth 2, learning rate 0.05, subsample 0.8, and model seed 0. The later shared-predictor comparison has a separately specified estimator configuration (Appendix N.1). Natural protocols retain their own fitted model and feature schema; for example, CCT-20 uses a ridge model. Details are in Appendix I. A common interface does not imply that an estimator or radius transfers across adapters.

Given residuals $r_i = |\hat{\Delta}_i - \Delta_i^{\text{cell}}|$, ordered as $r_{(1)} \leq$

$\dots \leq r_{(n)}$, the order-statistic rule is

$$k = \lceil (n+1)(1-\alpha) \rceil, \quad \varepsilon = \begin{cases} r_{(k)}, & k \leq n, \\ +\infty, & k > n. \end{cases} \quad (11)$$

Split-conformal coverage requires joint exchangeability of the calibration and next-unit scores under the fitting and candidate-construction protocol [36, 37]. An independently fitted predictor and exchangeable calibration/deployment units are one sufficient route. The rank counts evaluation units, not the images inside them.

The paper distinguishes current outcome-disjoint calibration, Protocol B, from the historical leave-one-cell-out replay, Protocol A.

Protocol B: Three-way cell-outcome-disjoint cross-fitting (used in the current primary CIFAR table, interval diagnostics, the 108-cell ImageNet-C population replay, and the Appendix B.3 27-cell audit). For each score fold, its entire complement is partitioned into strictly disjoint estimator-fit and residual-calibration subsets, ensuring that scored cell outcomes enter neither predictor training nor conformal calibration. Historical saved LOO predictions and radii do not determine these current CIFAR results.

Protocol A: Leave-one-cell-out empirical order-statistic calibration (used in historical controlled grids). Benefit predictions $\hat{\Delta}_i$ were cross-fitted on the remaining $n-1$ cells, and each scored residual $r_i = |\hat{\Delta}_i - \Delta_i^{\text{cell}}|$ was excluded from the pool used to compute its own radius ε_i . However, this historical leave-one-out construction is peer-dependent: for $j \neq i$, the model h_{-j} was fitted on training data containing Δ_i^{cell} , so the scored cell’s outcome can indirectly influence other residuals r_j in its calibration pool through model fitting. Where historical stress grids retain pooled LOO residuals (Protocol A), they are reported with this peer-dependence caveat rather than claiming full outcome disjointness. Natural protocols use separated development, calibration, and target partitions where the archived design records that separation.

For example, prediction 0.04 and radius 0.01 give [0.03, 0.05] and ADAPT; prediction -0.04 gives $[-0.05, -0.03]$ and FREEZE; prediction 0.005 gives $[-0.005, 0.015]$ and ABSTAIN. These illustrate the rule, not measured results.

Candidate adaptation, frozen and candidate inference, evidence extraction, and shadow-state storage all contribute to runtime. The tree prediction and interval comparison are inexpensive, but the whole procedure is not one extra forward pass. Historical synchronized timing and peak-memory evidence

Algorithm 1 KGA for one proposed update

Require: Frozen f_0 , adapter $\mathcal{A}$, unlabeled X , frozen h_θ and calibration artifacts

- 1: Create $f_a \leftarrow \mathcal{A}(f_0, X)$ in isolated shadow state
 - 2: Extract $Z = \phi(X, f_0, f_a)$ and validate the assessment inputs
 - 3: **if** the assessment is unavailable **then**
 - 4: retain f_0 , log the reason, and return ABSTAIN
 - 5: **end if**
 - 6: Obtain ε from the calibrated residual rule in Eq. (11)
 - 7: $\hat{\Delta} \leftarrow h_\theta(Z)$; $L \leftarrow \hat{\Delta} - \varepsilon$; $U \leftarrow \hat{\Delta} + \varepsilon$
 - 8: **if** $L > 0$ **then**
 - 9: commit f_a and return ADAPT
 - 10: **else if** $U < 0$ **then**
 - 11: discard f_a and return FREEZE
 - 12: **else**
 - 13: retain f_0 , log uncertainty, and return ABSTAIN
 - 14: **end if**
-

was not recovered; the retained literals are therefore audit records rather than performance claims (Table 45). Workload-specific measurements reported in the supplement have a distinct scope, and comparable end-to-end deployment latencies across the historical benchmark tracks remain unestablished.

7 Primary Experimental Protocols

7.1 Two Primary Questions and the Evidence Hierarchy

The controlled study asks whether the gate distinguishes useful from harmful updates. The CCT-20 study asks whether a frozen gate avoids measured candidate harm under a natural camera-location shift. These are different claims. We predefine the interpretation of each study before presenting its results in Table 6.

CIFAR-10-C [23] supplies 2,160 recorded cells per candidate: five archived ResNet-18 run-seed groups, each evaluated across 432 corruption/composition/update conditions. The current gate is refitted from the saved label-free features Z and measured cell outcomes under Protocol B, rather than reusing historical stored predictions or radii. For this canonical evaluation, the analysis uses three-way cell-outcome-disjoint cross-fitting: each scored fold has separate estimator-fit and residual-calibration subsets, so its outcome enters neither its fitted prediction nor its radius. This is still a retrospective opened

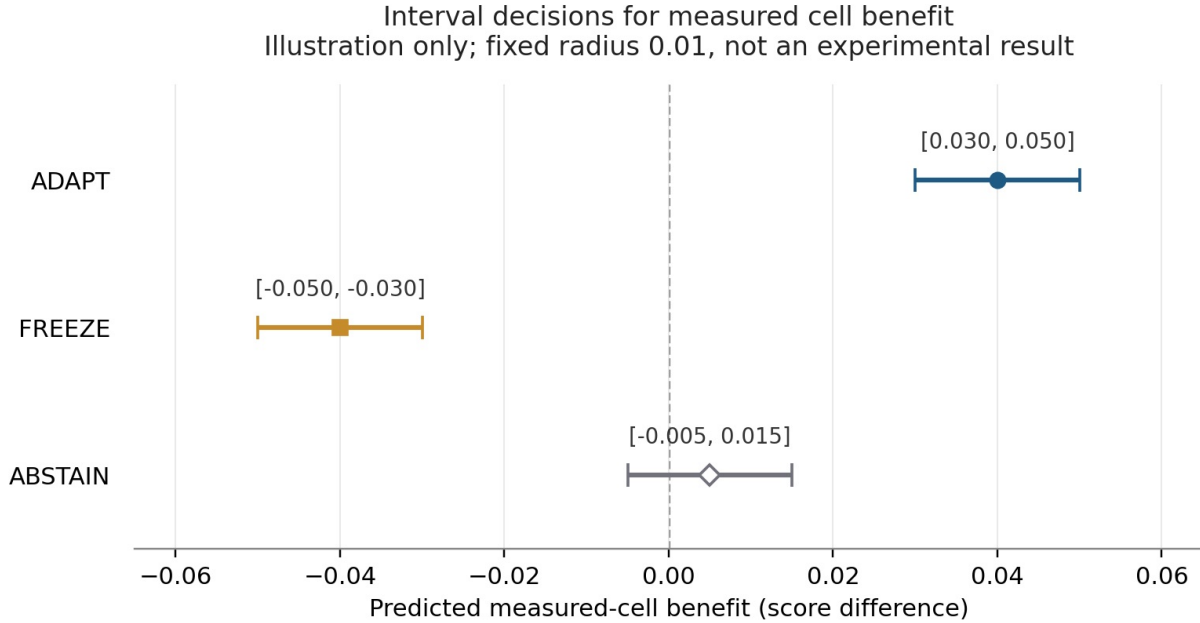


Figure 3: Illustration of the three interval decisions for measured cell benefit. Only an interval strictly above or below zero supports a commitment. The figure illustrates the rule; coverage must be established for the intended evaluation unit.

Table 6: Experiment status and role. Primary presentation does not imply prospective confirmation. All recorded unfavorable outcomes remain disclosed.

Evidence	Design and status	Question it can answer
CIFAR-10-C: Tent, EATA, SAR	opened controlled grids; cross-fitted empirical calibration	candidate-specific routing on the evaluated cells
CCT-20: Tent	internally sealed protocol; five checkpoints and nine target camera locations	conservative retention under the stated score and predeclared utility-retention rule
So2Sat-LCZ42 v1/v2	v1 development stop; v2 calibration-screen failure; targets unopened	whether the respective pre-target eligibility checks were met
Other replayable tracks	opened natural/semantic and candidate-configuration diagnostics	transfer limits and one-sided retention; full results in the supplement
PACS; iWildCam; historical ports; focused Office-Home route	partial replay, withheld metric, supervised policy, or invalid route	audit history only where the corresponding contract is incomplete or invalid

panel, not an untouched environment test; historical stored fields remain audit-only and do not determine the current decisions.

7.2 CCT-20 Design and Utility Retention

The CCT-20 camera-location study [48] crosses 5 independent source checkpoints with 9 target locations, giving 45 checkpoint–location cells. It uses 6950 probe images and 16325 evaluation images (23275 total). The primary metric is set-membership top-1 accuracy: a prediction is correct if it matches any distinct archived category for that image. It is not an official single-label leaderboard reproduction. The candidate uses the official Tent implementation, one

non-episodic Adam step at learning rate 10^{-3} , and state reset per cell.

The protocol was internally sealed before source training and outcome access, not publicly registered. Aggregate target label metadata had been inspected during candidate ranking; cell-level outcomes remained unopened until one-shot scoring. This is outcome-blind execution, not a label-unopened study. The source ridge fit used 10 rows from two camera locations; calibration used 45 rows aggregated to nine cis-test locations. Transfer to trans-test locations remains an assumption: the protocol treats camera-location clusters as the population unit and assumes the nine trans-test clusters are exchangeable with the

declared target population under the locked sampling rule. This is a design assumption, not an empirically proven fact about future deployments.

For either fixed policy b , define G_b as mean baseline regret minus KGA regret. Positive G_b means lower regret for KGA. The *utility-retention* rule is a predeclared, limited harm-avoidance check. Using nominal pointwise 95% bootstrap lower endpoints, it passes when

$$L_{\text{adapt}} > 0, \quad L_{\text{freeze}} > -0.005. \quad (12)$$

The second threshold is the frozen-policy noninferiority margin, on the accuracy-proportion scale. This rule can pass when KGA always retains f_0 . The stronger routing criterion additionally requires improvement over both fixed policies at the declared inference level and nontrivial ADAPT/FREEZE exposure. This endpoint evaluates utility retention under the declared protocol; it is not a population-safety guarantee.

7.3 Scoring, Dependence, and Statistical Inference

All policies use the same evaluation records, and evaluation labels are joined only after actions are fixed. The candidate TTA step can be transductive: episodic candidates update on the unlabeled evaluation inputs, and online candidates may use evaluation-batch BatchNorm statistics. Development/calibration separation therefore does not imply independent fresh-sample inference.

We report oracle regret, action counts, commitment rate, and false-commit frequencies. Empirical interval inclusion and interval width are reported separately from commitment. A paired regret contrast always means baseline regret minus KGA regret, so positive values favor KGA. Accuracy and macro-F1 regrets are separated rather than pooled.

Bootstrap levels are nominal throughout. CIFAR sensitivity resamples six corruption families, conditional on an archived checkpoint. CCT-20 uses a paired two-way product bootstrap over checkpoint rows and location columns; sign-flip calculations instead use nine location means. For a vector G of cluster gaps, the sign-flip reference model is

$$G \stackrel{d}{=} s \odot G \quad \text{for every } s \in \{-1, +1\}^m. \quad (13)$$

Independent zero-symmetric cluster gaps suffice. Exchangeability, marginal symmetry, or a single global sign symmetry alone is not sufficient [38]. Enumeration removes Monte Carlo error but does not verify

this null model. Retrospective Holm adjustment over the six prospectively named contrasts is reported, but Holm adjustment requires valid input p -values; nominal family intervals also require valid marginal coverage of their constituent intervals. The full inference specification and retrospective search status are collected in Appendix F.

8 Primary Results

8.1 Controlled Routing Depends on the Candidate

Table 7 reports all three CIFAR-10-C candidates under the same current rule. Tent and EATA have lower measured point regret than always-adapt and always-freeze. SAR is the counterexample: always-adapt has lower regret than KGA. The mixed panel therefore supports candidate-specific routing, not a claim that gating improves every adapter.

Tent regret is 0.0018, compared with 0.0080 for always-adapt and 0.1239 for always-freeze. EATA’s values are 0.0016, 0.0033, and 0.1313. In the canonical primary panel (Table 7), their actions are 1,094/334/732 and 1,207/118/835, respectively; both have zero observed false adaptations ($\text{FA}_u = 0.0000$). SAR commits 1,430/0/730 actions with regret 0.0017, versus 0.0003 for always-adapt ($\text{FA}_u = 0.0005$, 1 false adaptation).

On the recorded CIFAR SAR panel, always-adapt has very low paired-oracle regret. KGA’s higher regret reflects the net benefit forgone on cells where it retains the frozen predictor. The current results do not isolate whether those decisions arise mainly from prediction error, conservative calibration, or candidate-update dynamics. Normalization and sample-filtering effects remain hypotheses rather than established causes.

In contrast, Tent and EATA suffer degradation under severe corruptions, allowing KGA’s selective freezing and abstention to achieve net regret reductions over always-adapt (+0.0062 for Tent, +0.0017 for EATA on the canonical primary panel; matching the synchronized baseline study). Figure 4 illustrates the effect of changing helpful and harmful mixture weights in a preserved historical Tent seed-0 run. Its constructed mixture curve is separate from the current Protocol B aggregate above; it is not an additional deployment or a new cross-fitting evaluation.

A retrospective cluster-bootstrap sensitivity analysis across the six observed CIFAR-10-C corruption families (Table 34) indicates an estimated paired regret reduction for Tent of +0.0062

Table 7: Primary controlled CIFAR-10-C results. Regret is the mean accuracy shortfall from the per-cell oracle, on the proportion scale; lower is better. A/F/U denotes ADAPT/FREEZE/ABSTAIN. Commitment rate is $(A + F)/n$, not interval coverage. Cells share checkpoints and residual pools.

Candidate	n	KGA	Adapt	Freeze	A/F/U	FA_u	Commitment rate
Tent	2160	0.0018	0.0080	0.1239	1094/334/732	0.0000	0.6611
EATA	2160	0.0016	0.0033	0.1313	1207/118/835	0.0000	0.6134
SAR	2160	0.0017	0.0003	0.1405	1430/0/730	0.0005	0.6620

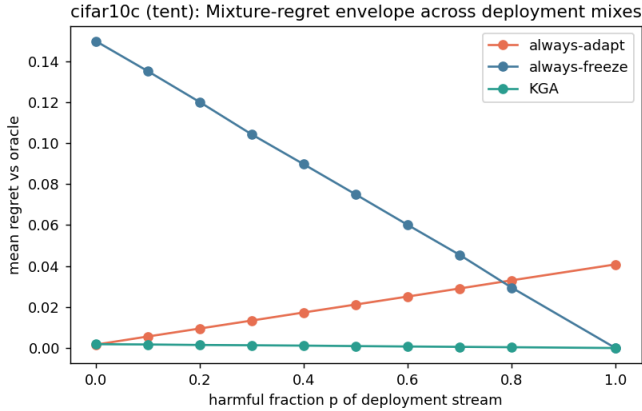


Figure 4: Historical CIFAR-10-C Tent seed-0 mixture illustration, using the preserved `stress_grid_multiseed_v1/seed0` curve. Mean regret is plotted against the constructed harmful fraction p for always-adapt, always-freeze, and KGA. The curve illustrates changing mixture weights within that saved run; it does not report the current Protocol B aggregate or independently sampled deployments.

over always-adapt (95% CI $[+0.0023, +0.0095]$) and $+0.1221$ over always-freeze (95% CI $[+0.0630, +0.1777]$). Across five stream and evaluation runs (seeds 0–4) on the archived source checkpoint, per-seed regret reductions for Tent are $+0.0073, +0.0063, +0.0058, +0.0061, +0.0055$, strictly positive across each evaluated run. These repetitions evaluate stream variation rather than variation across independently trained source models. Historical run-seed bootstrap vectors from earlier development cycles are cataloged separately in the supplement. Corruption-family sensitivity breakdowns appear in Table 34.

8.2 Observed Interval Inclusion Is a Separate Diagnostic

Commitment rate measures how often the interval excludes zero. Interval inclusion measures how often it contains the observed cell outcome. Table 8 reports these diagnostics separately under Protocol B (disjoint fit/calibrate/score cross-fitting) at nominal target $1 - \alpha = 0.90$.

Observed inclusion rates closely track the 90% nominal level across all three candidates: Tent includes 1,954 of 2,160 outcomes (90.46%), EATA includes 1,957 (90.60%), and SAR includes 1,933 (89.49%). These are observed out-of-fold inclusion rates on an opened, dependent benchmark; they do not establish interval coverage on unseen distributions. Evaluating directional errors relative to committed decisions reveals asymmetric behavior: EATA commits one false FREEZE among 118 FREEZE decisions ($FF_c = 0.0085$, occurring in Gaussian noise). Tent commits zero false FREEZE decisions among 334 FREEZE decisions ($FF_c = 0.0000$), while SAR commits zero FREEZE decisions and one false ADAPT among 1,430 ADAPT decisions ($FA_c = 0.0007$, in pixelate). These conditional denominators describe the observed directional errors among committed decisions on this panel, reflecting low empirical error rates on the evaluated records.

8.3 Synchronized Development Baseline Comparison

Table 9 compares KGA against fixed policies, scalar heuristics, and an uncalibrated point gate on 2,160 CIFAR-10-C Tent rows, using identical saved features and candidate-specific oracles under Protocol B.

KGA achieves lower point regret (0.0018, mean accuracy 0.7931) than either fixed policy (always-adapt 0.0080, always-freeze 0.1239), while committing zero false adaptations ($FA_u = 0.000$, zero false ADAPT decisions out of 1,094 ADAPT commitments). The same-prediction point gate achieves even lower point

Table 8: Retrospective interval diagnostics for the controlled cells under Protocol B (disjoint fit/calibrate/score cross-fitting), at nominal target $1 - \alpha = 0.90$. Observed inclusion concerns Δ^{cell} . Candidate/family breakdowns and directional-error denominators are provided in the supplement.

Candidate	n	Nominal	Inclusion	Mean width	Median width	Commitment	FA_u/FA_c	FF_u/FF_c
TENT	2160	0.9000	0.9046	0.0469	0.0483	0.6611	0.0000/0.0000	0.0000/0.0000
EATA	2160	0.9000	0.9060	0.0387	0.0378	0.6134	0.0000/0.0000	0.0005/0.0085
SAR	2160	0.9000	0.8949	0.0279	0.0277	0.6620	0.0005/0.0007	0.0000/-

Table 9: Current outcome-disjoint saved-feature comparison on 2,160 CIFAR-10-C Tent rows, computed from the same paired scores. All rows have the same oracle and weights; both fitted gates use identical predictions. Lower regret is better. FA_u is the fraction of all rows with ADAPT and $B \leq 0$; commitment is $(A + F)/n$.

Policy	Regret	Mean acc.	FA_u	Commitment	ADAPT	FREEZE	ABSTAIN
Always Adapt	0.0080	0.7869	0.334	1.000	2160	0	0
Always Freeze	0.1239	0.6709	0.000	1.000	0	2160	0
Confidence increase	0.0080	0.7869	0.250	1.000	1834	326	0
Entropy decrease	0.0081	0.7868	0.249	1.000	1812	348	0
Point benefit (same prediction)	0.0004	0.7945	0.041	1.000	1471	689	0
Current KGA	0.0018	0.7931	0.000	0.661	1094	334	732

regret (0.0004, mean accuracy 0.7945) by committing to adaptation whenever $\hat{\Delta} > 0$ without an uncertainty margin; on the evaluated panel, this rule records 88 nonpositive adaptations ($\text{FA}_u = 0.041$, 88/2,160). KGA trades point regret for greater directional conservatism: by abstaining on 732 cells whose intervals contain zero, it records zero false ADAPT decisions on these evaluated cells. The scalar heuristics (confidence increase and entropy decrease) achieve regret of 0.0080 and 0.0081, respectively, with substantial false adaptations ($\text{FA}_u = 0.250$ and 0.249) and lack formal calibration. An expanded nine-policy decision rule comparison—including drift monitors, ATC-style estimators, and fixed margins—is reported in Table 25.

8.4 CCT-20: Conservative Retention, Not Selective Routing

On the recorded CCT-20 cells, KGA served the frozen model throughout and avoided the measured loss of always adapting. Across 45 evaluation cells, KGA issued 0 ADAPT, 44 FREEZE, and 1 ABSTAIN decisions. Because every non-adapt decision serves the base model, KGA served frozen predictions across the entire benchmark.

KGA matched always-freeze and avoided the measured degradation of always-adapt. The predeclared utility-retention endpoint was therefore met. The stronger selective-routing criterion was not: KGA issued no ADAPT decisions and failed to select the single helpful candidate. All recorded development

cells for Tent had negative measured benefit.

A separate retrospective labeled finite-frame audit is reported in Appendix S. That audit uses a different interval construction and produces 0 ADAPT / 35 FREEZE / 10 ABSTAIN actions; these are audit-assisted decisions and do not revise the primary KGA actions reported here (0 ADAPT / 44 FREEZE / 1 ABSTAIN).

8.5 Development Stops and Adverse Diagnostics

The earlier v1 study on So2Sat-LCZ42 [28] tested 2 candidates on 9 development cities and 5 independent checkpoints. Tent had both helpful and harmful cities, but neither candidate passed all predeclared eligibility checks. The study stopped before gate calibration: no target inputs or labels were accessed, and there is no target natural-shift score. The development outcomes are now open, so retuning them cannot produce fresh confirmation. The separately locked v2 calibration study is reported in Section 8.9; it also stopped before target access, after failing its fixed calibration eligibility screen.

The auxiliary studies also show limitations. ImageNet-R and PACS favor always-adapt; ImageNet-C is sensitive to the candidate and its operating point. Office-Home and RxRx1 mainly retain the frozen model. The opened one-sided Camelyon17 panel adapts throughout because all 18 cells are helpful.

Table 10: CCT-20 utility-retention endpoint defined in Eq. (12). Contrasts are baseline regret minus KGA regret on the set-membership accuracy-proportion scale. Contrasts report nominal pointwise 95% two-way bootstrap intervals. Location-level records are detailed in the supplement.

Comparator	Mean contrast	Nominal 95% CI	Required lower bound
Always adapt	0.1819	[0.0937, 0.2914]	$L > 0.0000$
Always freeze	0.0000	[0.0000, 0.0000]	$L > -0.0050$

A retrospective 72-condition Camelyon17 diagnostic contains both helpful and harmful candidate effects but issues 0 ADAPT, 2 FREEZE, and 70 ABSTAIN decisions, matching always-freeze regret (0.0438; Appendix B.5). Together with CCT-20 and RxRx1, it provides retention evidence without ADAPT exposure. These studies do not establish prospective selective routing on unseen natural shifts.

8.6 Additional Tests of Decision Value

The following analyses ask three related questions at different levels. First, does residual calibration add decision value beyond a shared point predictor? Second, can paired multiclass transport make a decision under an explicitly declared structural assumption? Third, does a natural-shift calibration screen provide enough evidence to authorize target scoring? Sections 8.7–8.9 examine these questions separately; Section 8.10 describes the local deployment integration. None turns a conditional or retrospective result into a general deployment guarantee.

8.7 What Does Calibration Add to a Shared Predictor?

We conducted a new retrospective comparison on the already opened CIFAR, ImageNet, and mixed-checkpoint records. Every arm shares the same fitted GBRT within a fold. CIFAR and ImageNet folds exclude complete scored corruption types from fitting, tuning, and calibration. Checkpoint-only folds exclude the scored source model and rotate held-out semantic groups; joint folds exclude both the scored source model and the complete corruption type. This tests transfer beyond the earlier cell-wise splits, while retaining the historical data’s provenance and dependence limits. It is not a new independent deployment study. The loss grid, fold construction, and policy family were fixed before the new fitting; held-out outcomes enter scoring only after decisions have been saved.

The primary loss uses a prespecified harm weight

of five:

$$L_5 = \frac{1}{N} \sum_{i=1}^N [(1 - a_i)(B_i)_+ + 5a_i(-B_i)_+], \quad (14)$$

where B_i is measured paired cell benefit and a_i is one for ADAPT and zero otherwise. A fixed margin minimizes this loss on tune-only groups. Constant and locally scaled residual intervals use the exact conformal rank, without clipping a rank that exceeds the calibration sample size. We report $100L_5$ in percentage points, so lower values indicate lower weighted decision loss. The full study includes harm weights 1, 5, 20, nominal levels 80, 90, 95%, group-maximum residuals, exposure-matched controls, and binomial risk-selection arms.

Table 11 does not establish a general advantage for calibration. Scaled calibration reduces CIFAR Tent loss from 0.2867 to 0.1429 percentage points versus point prediction, and from 0.2264 versus the tuned margin. Its loss at matched exposure is only modestly lower than point ranking (0.1429 versus 0.1610). Simple margins do better on CIFAR EATA and SAR; point or margin policies do better in the other cohorts. Within a fixed scoring fold, subtracting the same finite radius from the same point predictions preserves their ordering. At matched ADAPT exposure, the constant-radius rule selects the same cells as the point-ranking control, using the same tie rule. This does not compare rankings across separately fitted folds or studies; locally varying radii can change the ordering. The favorable Tent comparison is a point estimate on six observed corruption types. Omitting JPEG only as an influence diagnostic nearly removes the advantage over point prediction and reverses the comparison with the tuned margin; all six corruption types remain in the primary estimate. Paired bootstrap ranges across these types describe sensitivity, not a population confidence interval or multiplicity-adjusted discovery (Appendix R).

Nominal 90% constant-cell intervals include only 50.51–60.69% of held-out CIFAR benefits; scaling and group maxima do not restore nominal inclusion. Decision loss and interval inclusion answer different

Table 11: Retrospective shared-predictor comparison. Losses are $100L_5$ in percentage points (lower is better). Margin is selected on tuning data; Constant and Scaled use nominal 90% cell residual intervals. Inclusion C/S is observed held-out interval inclusion for those two arms, not a coverage guarantee. The same 108 mixed-checkpoint records appear in two evaluation designs; they are not 216 independent observations.

Cohort / outer hold-out	Cells	Point	Margin	Constant	Scaled	Inclusion C	Inclusion S
CIFAR EATA / corruption	2160	0.1077	0.0663	0.1556	0.0914	54.35%	51.67%
CIFAR SAR / corruption	2160	0.1677	0.0465	0.1291	0.0844	50.51%	48.19%
CIFAR Tent / corruption	2160	0.2867	0.2264	0.2150	0.1429	60.69%	58.24%
ImageNet EATA / corruption	135	0.0481	0.4443	0.6939	1.8787	91.11%	92.59%
ImageNet SAR / corruption	135	0.0000	0.2115	2.5989	1.8754	97.04%	93.33%
ImageNet Tent / corruption	135	2.1913	1.3756	1.4465	1.4465	93.33%	96.30%
Mixed Tent / checkpoint	108	0.3236	0.4032	0.7954	5.2708	85.19%	90.74%
Mixed Tent / checkpoint + corruption	108	0.5829	0.4611	0.9403	2.6329	79.63%	87.96%

questions: a lower observed loss does not establish nominal coverage, and observed inclusion does not establish useful routing. All independently held-out environment-calibration arms abstain: this split provides no calibration environments independently excluded from fitting and tuning. The 10% conditional group-risk arm also abstains throughout. Even with zero observed errors, its 13-threshold family needs 53 exposed calibration groups at within-fold confidence error .05, exceeding the largest partition of 45. These are information and transfer limits, rather than evidence that every possible policy is unsafe. The 20% sensitivity selects some CIFAR updates; its held-out group error rates differ materially from cell rates (Appendix O). B6 remains descriptive: this study does not recover its missing original execution identities or repair its historical rank clipping.

8.8 A Conditional Paired-Transport Certificate

The population frontier leaves open how extra structural information can make a decision possible. We instantiate one such assumption regime for fixed multiclass predictors. Let $r = (f_0(X), f_a(X))$ index their joint prediction bins, $A_{ry} = P_S(r \mid Y = y)$, and $q_r = P_T(r)$. With unknown target prior π , declare

$$\sum_y \pi_y \text{TV}(P_T(r \mid y), A_{.y}) \leq \rho. \quad (15)$$

The budget ρ requires external justification; unlabeled evidence cannot generally validate it. Both predictors must be fixed independently of the source and target evidence samples. In particular, adapting on the same target examples later used as independent evidence does not satisfy this construction.

For R bins and K classes, simultaneous binomial

intervals for the $RK + R$ source-conditional and target-marginal probabilities define a linear feasible set. The variables are target joint masses t_{ry} , transported source masses $s_{ry} = A_{ry}\pi_y$, priors, and absolute-value slacks for Equation (15). Source interval endpoints multiply π_y , so the resulting constraints remain linear. Minimize and maximize

$$\Delta(t) = \sum_{r,y} (\mathbf{1}\{f_a(r) = y\} - \mathbf{1}\{f_0(r) = y\})t_{ry}. \quad (16)$$

On the simultaneous interval event, the true table is feasible and these extrema enclose its benefit. Under the stated sampling and transport assumptions, strict interval thresholding therefore has unconditional wrong-direction probability at most α . It does not give acceptance-conditional or repeated-use error control. Detected infeasibility, solver failure, or an interval touching zero produces ABSTAIN. The full constraints, proof, and numerical scope appear in Appendix P.

This construction applies existing interval-constrained LP inference [57] to a paired contrast. Joint optimization can retain information lost by separately bounding the two accuracies: in one exact three-class construction, the paired interval is $[.1, .4]$, whereas subtracting separate extrema on the same feasible set gives $[-.1, .6]$. An unidentifiable prior can therefore coexist with an identified positive action. This is a concrete use of dependence, not a first claim for action identification [56] or new LP machinery.

The paired procedure adapts in all 60 cancellation trials while the separate accuracy relaxation always abstains. Near a small positive benefit, it avoids the plug-in rule’s 22 wrong FREEZE decisions out of 60 at the smaller sample size; its price is abstention. Larger sensitivity budgets likewise reduce commitment. Most importantly, the same observed evidence

Table 12: Selected diagnostic regimes from 4,860 synthetic trial configurations (nine scenarios, three sample sizes, three sensitivity budgets, 60 trials each). A/F/U count ADAPT/FREEZE/ABSTAIN over 60 trials. Separate subtracts accuracy extrema on the same polytope; it is not an implementation of MaC-LP. The same-evidence violation row deliberately violates the declared assumption and reuses the paired-cancellation row’s observable count streams. These archived outputs retain the earlier numerical qualification described in Appendix P.

Synthetic regime	n/class	ρ	True Δ	Paired A/F/U	Separate A/F/U	Plug-in A/F/U
Paired cancellation	100	0.0	0.16	60/0/0	0/0/60	60/0/0
Same evidence, violated assumption	100	0.0	-0.50	60/0/0	0/0/60	60/0/0
Near null	100	0.0	0.02	0/0/60	0/0/60	38/22/0
Near null	10000	0.0	0.02	20/0/40	0/0/60	60/0/0
Helpful	1000	0.0	0.20	60/0/0	60/0/0	60/0/0
Helpful, wider sensitivity	1000	0.1	0.20	0/0/60	0/0/60	60/0/0
Harmful	1000	0.0	-0.20	0/60/0	0/60/0	0/60/0

can have true benefit $-.5$ under a violated transport assumption: the paired procedure then makes 60 harmful ADAPT decisions. The diagnostic demonstrates both conditional value and the impossibility of obtaining the missing assumption from unchanged unlabeled observations. It does not establish useful natural-shift adaptation.

8.9 Natural-Shift Calibration and Execution Costs

The separate So2Sat v2 study fixes the controller derived from nine already opened development cities and uses five independently trained source checkpoints. Its calibration unit is a city: the score is the maximum absolute residual across all five checkpoints, with the 18th order statistic among 19 city scores at $\alpha = .10$. Under exchangeability, this construction targets simultaneous coverage of the five checkpoints for one new city. It does not give simultaneous coverage of ten target cities, coverage conditional on passing an eligibility screen, or a demonstrated guarantee under the held-out geographic shift. The chronology documents the study sequence but does not establish general prospective validity.

The locked calibration completed all 95 city-checkpoint cells. Helpful and harmful development effects were both present, but the uncertainty interval was too wide to meet the pre-target action-exposure criterion. Its radius was 0.14224 (14.224 percentage points), and its decisions were 2 ADAPT, 0 direct FREEZE, and 93 ABSTAIN. Both accepted updates were helpful in the calibration records, but they occurred in only one city; no city supplied a direct FREEZE. Thus both prespecified requirements of at least seven direct-action cities failed, and the study

stopped with zero target pixels or labels read. No target authorization or target score was produced. The observed mean calibration utility was 0.0989 percentage points for the gate, 0.1939 for always-adapt, 0.1264 for the checkpoint-only point baseline, and zero for always-freeze. These same calibration outcomes set the radius and screen, so these utilities are descriptive and cannot demonstrate independent target benefit.

A posthoc diagnosis of those opened records found 51 helpful, 43 harmful, and one tied candidate effect. The radius was set by São Paulo checkpoint 0: its predicted benefit was 14.4215 percentage points but its observed benefit was only 0.1976 points. The resulting radius exceeded every negative prediction magnitude (at most 2.6067 points), preventing direct FREEZE. Thus the failed screen reflects insufficient prediction accuracy and interval informativeness, rather than an absence of helpful and harmful candidates. This diagnosis did not refit the controller, alter the screen, or authorize target access.

Calibration took 36.55 minutes. The controller requires five shadow candidates before routing a city, so retaining the frozen model does not avoid candidate-construction cost. In a separate profile on one opened development city, that process took a median 207.44 seconds, excluding subsequent serving inference and cleanup. These measurements are workload-specific; calibration acquisition, energy, and long-duration maintenance costs remain unmeasured. Stage timings, memory measurements, and measurement conditions are reported in Appendix J.1.

8.10 A Local Deployment Integration

We also implemented a local fail-closed deployment lifecycle that isolates the candidate from the active checkpoint, rejects malformed or outcome-bearing evidence, and retains the frozen model when assessment fails. The implementation records calibration and model identities and exercises rollback and persistence failures. These tests evaluate software behavior, not the statistical validity of the certificate or field reliability. Detailed fault tests, integrity records, and workload-specific latency measurements are reported in Appendix J.1.

9 Limitations and Broader Impact

9.1 Assumptions, Not an Unconditional Safety Guarantee

The population frontier is exact within its declared rich binary class. A credible residual bound must come from domain knowledge, target labels, or a stated transfer assumption. A narrower class may permit stronger decisions, while an unsupported bound cannot justify a strict population claim. The useful object is the class-dependent feasible set, not the algebraic inequality alone.

Justifying the population residual bound. A development-domain proxy does not resolve the population residual bound required by Theorem 3. Within the declared class, β must come from external domain knowledge, target labels, or an explicit transfer assumption. Processing the same label-free evidence cannot supply that missing restriction. The supplementary discussion records the withdrawn proxy analysis (Appendix B.11).

For KGA, empirical interval coverage requires separate support at the intended decision unit. A larger radius reduces commitment but can increase regret when the frozen model is worse. A smaller radius increases exposure and makes calibration failures more consequential. Changing the adapter, checkpoint, feature schema, or update settings changes the prediction problem and requires recalibration or a justified transfer argument. The new held-out-corruption comparison directly exhibits this failure: nominal 90% calibration transfers poorly on CIFAR, and the best useful policy depends on the candidate and harm weight. The paired-transport certificate supplies a different sufficient assumption regime; it does not validate either that calibration transfer or its own

transport budget from unlabeled target data.

Repeated deployment needs a separate guarantee. A predeclared policy does not by itself supply exchangeability or time-uniform safety. Reusing calibration data, changing the frozen baseline after accepted updates, and selecting later candidates from earlier outcomes change the joint process. Adaptive conformal inference studies online coverage frequency with feedback [39]; that is not time-uniform control of KGA’s false commitments. Selection-conditional and group-conditional guarantees likewise require additional methods, as discussed in Section 2.

9.2 What the Experiments Establish—and What Remains Open

Four limitations determine how these experiments should be interpreted:

1. **Retention vs. Opportunistic Routing under Natural Shift:** On real-camera distribution shifts (CCT-20), KGA served the frozen model across all 45 cells and avoided the measured degradation of always-adapt in that study. However, it made zero ADAPT decisions. Prospectively demonstrating selective opportunistic routing on an unopened natural-shift target remains an open empirical challenge. The retrospective Camelyon17 diagnostic in Appendix B.5 similarly exhibits conservative retention (zero ADAPT decisions under an outcome-blind decision rule).
2. **Retrospective Benchmarks vs. Prospective Deployment:** The controlled CIFAR-10-C and ImageNet-C experiments evaluate retrospective, opened panels under their track-specific declared protocols (including Protocol A and Protocol B). While these report empirical commitment and inclusion diagnostics on opened records, prospective evaluation on unopened environments is required to assess distribution-transfer properties.
3. **Checkpoint and architecture transfer.** The primary CIFAR-10-C panel uses one source checkpoint. Additional checkpoint comparisons are descriptive: B6’s clipped calibration rank does not establish nominal coverage, and incomplete execution provenance limits independent replication claims. Prospective transfer to unseen model families and environments remains untested (Appendix B.3.3).
4. **External method comparisons.** The shared-feature baselines and scoped native components isolate specific decision rules, but do not con-

stitute complete reproductions of all published learners or a full official-method comparison (Appendix B.3.4).

The controlled studies quantify candidate-specific utility and commitment tradeoffs. CCT-20 provides retention evidence without ADAPT exposure. These experiments do not establish population coverage or protection on unseen environments. A unified statistical synthesis matrix consolidating sample sizes, evaluation units, nominal confidence intervals, multiplicity adjustments, and verdicts across the principal empirical tracks summarized here is reported in Table 44.

9.3 Operational Scope and Potential Harm

A rejected update can prevent adaptation harm, but retaining a frozen model is not proof that the model is accurate or suitable for a high-stakes setting. Misinterpreting an abstention as a certificate would be particularly risky. In high-stakes applications, an unresolved assessment calls for an application-specific fallback, with domain-expert review where appropriate; the choice depends on the consequences of continuing with the frozen model. The reported experiments do not validate clinical, safety-critical, or autonomous operation. No held-out physical camera deployment or common end-to-end latency result is claimed.

An unavailable assessment must remain distinct from evidence of negative benefit: both retain the frozen model, but only the latter supports a strict FREEZE decision. The local integration checks this behavior; its tests do not establish calibration validity or field reliability. Detailed implementation and reproducibility information is provided in the supplement.

10 Conclusion

The update decision depends on what the available evidence can distinguish. For the declared binary target class, K-Bound identifies the feasible population-benefit interval and its exact strict-decision frontier. The evidence-fibre audit establishes the accompanying limit: unchanged label-free observations cannot justify an arbitrarily tighter residual bound. The paired-transport construction illustrates a separate multiclass route under an explicit conditional-shift budget and sampling assumptions; its usefulness depends on the credibility of those assumptions.

KGA makes the empirical question concrete by predicting paired evaluation-cell benefit and calibrating an interval around that prediction. Its value is candidate- and operating-point-dependent. Controlled routing gains coexist with simpler-rule wins. In the shared-predictor comparison, scaled calibration has lower observed CIFAR Tent harm-weighted loss, but the advantage narrows at matched exposure and depends on the corruption mix. Nominal 90% constant-cell inclusion falls to 50.51–60.69% on held-out CIFAR corruptions. Useful decisions and reliable intervals must therefore be evaluated separately.

Natural-shift evidence is more limited: CCT-20 demonstrates retention without ADAPT exposure, and So2Sat stops before target scoring. The present studies do not establish population-risk protection or useful selective routing on unseen natural environments. The next empirical test is whether independently supported calibration can accept beneficial updates and reject harmful ones under a fixed protocol. Where the available information cannot support either strict sign, abstention records the unresolved decision; it does not certify that the retained model is safe.

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A A Rigorous Bridge from Cell Coverage to Population Coverage

The formal theory of K-Bound (Section 4 through Section 5) characterizes the exact population partial-identification frontier under a declared bound β on the hidden disagreement-conditional residual γ . In contrast, the empirical gate KGA (Section 6) predicts measured evaluation-cell benefit Δ^{cell} and constructs a prediction interval $[\hat{\Delta} - \varepsilon, \hat{\Delta} + \varepsilon]$. This appendix provides the rigorous mathematical bridge between these two formulations, corrects a common dimensional misconception, provides a formal counterexample to naive transfer, and establishes sufficient conditions under which an enlarged conformal interval provably covers population benefit Δ .

A.1 Scale and Structural Differences Between Deductive and Inductive Bounds

It is tempting to conjecture that an empirical gate could directly enforce the population frontier by setting its empirical conformal radius equal to the population residual bound, $\varepsilon = \beta$. This conjecture is fundamentally false: ε and β represent different mathematical objects living on different scales.

Recall the exact disagreement-region reduction under binary 0/1 loss (Lemma 2):

$$\Delta = 2d(M + \gamma), \quad d = P_T(D) > 0, \quad (17)$$

where $D = \{x : f_0(x) \neq f_a(x)\}$ is the disagreement region, $M = \mathbb{E}[s(X) \mid D] - \frac{1}{2} \in [-\frac{1}{2}, \frac{1}{2}]$ is the observable disagreement margin, and $\gamma = \mathbb{E}[\eta_a(X) - s(X) \mid D]$ is the hidden calibration residual. Under the externally declared residual class $\mathcal{C}_\beta = \{P_T : |\gamma(P_T)| \leq \beta\}$, the feasible population benefit set satisfies:

$$|\Delta - 2dM| = 2d|\gamma| \leq 2d\beta. \quad (18)$$

Consequently, the unclipped population-benefit envelope centered at $2dM$ has radius

$$r_{\text{pop}} = 2d\beta, \quad (19)$$

not β ; the exact population benefit set $\Delta(\mathcal{C}_\beta)$ in Theorem 2 includes clipping to $[-d, d]$: $\Delta(\mathcal{C}_\beta) = 2d[\max(-\frac{1}{2}, M - \beta), \min(\frac{1}{2}, M + \beta)]$. Here $d = P_T(D) \in (0, 1]$ is the disagreement probability, so the scaling factor $2d \in (0, 2]$ is twice that probability.

For example, with typical values $d = 0.20$, $M = 0.15$, and $\beta = 0.05$, the population-benefit interval is:

$$\begin{aligned} 2(0.20)[0.15 - 0.05, 0.15 + 0.05] &= 0.40[0.10, 0.20] \\ &= [0.04, 0.08]. \end{aligned}$$

The interval center is 0.06 and its half-width on the risk-reduction scale is 0.02. The declared residual budget is $\beta = 0.05$, whereas the benefit-scale radius is 0.02. Equating ε and β conflates two different targets: β bounds the absolute target disagreement-conditional calibration residual $|\gamma|$, whereas ε is a calibrated prediction-error radius for the declared empirical cell-benefit target. The audit floor in Theorem 3 rules out a uniformly tighter bound from the same label-free evidence over the declared class. It does not rule out a conservative bound or a tighter conclusion after adding justified information or restricting the class. KGA calibrates prediction error for measured cell benefit; this empirical radius does not estimate the population residual bound.

A.2 Counterexample: Perfect Cell Prediction Does Not Imply Population Coverage

We now prove that even an ideal empirical predictor with zero cell-prediction error ($\varepsilon = 0$) can fail to cover population benefit and can commit to false adaptation with substantial probability.

Proposition 4 (Failure of automatic cell-to-population transfer). *There exists a data-generating process, a fixed model pair (f_0, f_a) , and a benefit predictor $\hat{\Delta}$ such that $\mathbb{P}(|\hat{\Delta} - \Delta^{\text{cell}}| = 0) = 1$ (perfect cell coverage with $\varepsilon = 0$), yet the cell gate selects ADAPT with probability 0.45 despite the true population benefit being strictly negative ($\Delta = -0.10$).*

Proof. Fix binary predictors $f_0(x) \equiv 0$ and $f_a(x) \equiv 1$. Let the target input space consist of two points, $\mathcal{X} = \{a, b\}$, with input distribution

$$\mathbb{P}(X = a) = 0.45, \quad \mathbb{P}(X = b) = 0.55,$$

and deterministic target labels $Y(a) = 1$ and $Y(b) = 0$. The zero-one population risks are:

$$R_T(f_0) = \mathbb{P}(f_0(X) \neq Y) = \mathbb{P}(X = a) = 0.45,$$

$$R_T(f_a) = \mathbb{P}(f_a(X) \neq Y) = \mathbb{P}(X = b) = 0.55.$$

The true population benefit of candidate adaptation is strictly negative:

$$\Delta = R_T(f_0) - R_T(f_a) = 0.45 - 0.55 = -0.10 < 0.$$

Now consider an evaluation cell consisting of a single independent target sample ($m = 1$). The measured cell benefit is

$$\begin{aligned}\Delta^{\text{cell}} &= \mathbf{1}\{f_a(X) = Y\} - \mathbf{1}\{f_0(X) = Y\} \\ &= \begin{cases} +1, & X = a, \\ -1, & X = b. \end{cases}\end{aligned}$$

Define the fixed label-free cell predictor $h(a) = +1$ and $h(b) = -1$. This predictor observes only the unlabeled input X and achieves perfect prediction of the cell outcome:

$$|\hat{\Delta} - \Delta^{\text{cell}}| = |h(X) - \Delta^{\text{cell}}| = 0 \quad \text{almost surely.}$$

A conformal interval with radius $\varepsilon = 0$ achieves 100% marginal coverage of Δ^{cell} . However, whenever $X = a$ (which occurs with probability 0.45), $h(a) - \varepsilon = +1 > 0$, triggering an ADAPT decision. The probability of false adaptation is therefore:

$$\mathbb{P}(g = \text{ADAPT}, \Delta \leq 0) = \mathbb{P}(X = a) = 0.45.$$

Specializing this to residual notation with constant score $s(x) \equiv 1/2$, the disagreement region is $D = \{a, b\}$, $d = 1$, $M = 0$, and $\gamma = \mathbb{E}[\eta_a(X) - s(X) \mid D] = 0.45 - 0.50 = -0.05$. The tightest valid residual bound is $\beta = 0.05$, whereas the empirical cell-prediction radius is $\varepsilon = 0$. Thus $\varepsilon = 0 \neq 0.05 = \beta$. $\square$

Proposition 4 demonstrates why empirical cell-level validity cannot be equated with population-level safety: a gate that accurately predicts small-batch or single-cell advantages can repeatedly commit to globally harmful adaptations.

A.3 Formal Protocol for Cell-to-Population Transfer

To rigorously bridge cell prediction to population benefit, we specify a formal statistical protocol:

- (P1) Independent Estimator Fitting:** The benefit predictor h and feature extractor ϕ are fitted on historical development episodes $\mathcal{D}_{\text{dev}}$ and frozen prior to calibration and deployment.
- (P2) Representative Episode Exchangeability:** Conditional on the frozen development data $\mathcal{D}_{\text{dev}}$ (which determines the fitted predictor h and feature representation ϕ), the N calibration episodes $(Z_1, B_1), \dots, (Z_N, B_N)$ and the deployment episode (Z_*, B_*) are exchangeable draws from a deployment meta-distribution $\mathcal{Q}$. Each episode e defines an environment P_e .

(P3) Pre-Evaluation Candidate Isolation: For each episode e , candidate $f_{a,e} = \mathcal{A}(f_0, X_{e,\text{adapt}})$ is constructed from its unlabeled adaptation window, and its parameter and normalization states are strictly frozen prior to scoring.

(P4) Fresh Evaluation Sample: Evaluation cell e comprises m fresh, conditionally i.i.d. draws $(X_{e,j}, Y_{e,j})_{j=1}^m \sim P_e$ that are independent of the adaptation data.

(P5) Matched Risk Metric: The cell outcome is the sample paired accuracy difference $B_e = \Delta_e^{\text{cell}} = \frac{1}{m} \sum_{j=1}^m W_{e,j}$, where $W_{e,j} = \mathbf{1}\{f_{a,e}(X_{e,j}) = Y_{e,j}\} - \mathbf{1}\{f_0(X_{e,j}) = Y_{e,j}\} \in [-1, 1]$, matching the population risk reduction $\Delta_e = R_{P_e}(f_0) - R_{P_e}(f_{a,e})$.

A.4 Theorem and 4-Step Proof of Population Risk Control

Under protocol **(P1)–(P5)**, we prove that an enlarged decision interval achieves finite-sample population-benefit coverage and strict directional error control.

Theorem 4 (Finite-sample population benefit certificate). *Under protocol **(P1)–(P5)**, let $\alpha_{\text{cell}} \in (0, 1)$ and $\delta \in (0, 1)$ be predeclared error budgets such that $\alpha_{\text{cell}} + \delta < 1$. Let $N \geq 0$ and $m \geq 1$ be integers giving the calibration-episode count and evaluation sample size. Let ε be the order-statistic split-conformal radius over N calibration residuals:*

$$k = \lceil (N+1)(1 - \alpha_{\text{cell}}) \rceil, \quad \varepsilon = \begin{cases} r(k), & k \leq N, \\ +\infty, & k > N. \end{cases} \quad (20)$$

Define the finite-sample concentration radius

$$b(m, \delta) = \min \left\{ 2, \sqrt{\frac{2 \log(2/\delta)}{m}} \right\}. \quad (21)$$

Then for a new deployment episode $* \sim \mathcal{Q}$, the compound interval $I_{\text{population}} = [\hat{\Delta}_* - (\varepsilon + b), \hat{\Delta}_* + (\varepsilon + b)]$ satisfies:

$$\mathbb{P}(\Delta_* \in I_{\text{population}}) \geq 1 - \alpha_{\text{cell}} - \delta. \quad (22)$$

Furthermore, the three-way decision rule

$$g_* = \begin{cases} \text{ADAPT}, & \hat{\Delta}_* - (\varepsilon + b) > 0, \\ \text{FREEZE}, & \hat{\Delta}_* + (\varepsilon + b) < 0, \\ \text{ABSTAIN}, & \text{otherwise,} \end{cases} \quad (23)$$

controls the marginal probability of committing to an incorrect strict direction:

$$\begin{aligned} \mathbb{P}(\{g_* = \text{ADAPT}, \Delta_* \leq 0\} \cup \{g_* = \text{FREEZE}, \Delta_* \geq 0\}) \\ \leq \alpha_{\text{cell}} + \delta. \end{aligned} \quad (24)$$

Proof. The proof proceeds in four discrete steps:

Step 1: Conformal coverage of measured cell benefit. By (P1) and (P2), the residuals $r_e = |h(Z_e) - B_e|$ for $e \in \{1, \dots, N, *\}$ are exchangeable random variables conditional on the frozen development data $\mathcal{D}_{\text{dev}}$ (and hence conditional on h and ϕ). By standard split-conformal quantile properties [36], the k -th order statistic guarantees marginal coverage:

$$\mathbb{P}(\mathcal{E}_{\text{conf}}) \geq 1 - \alpha_{\text{cell}},$$

where $\mathcal{E}_{\text{conf}} = \{|h(Z_*) - B_*| \leq \varepsilon\}$.

Step 2: Hoeffding concentration of sample benefit to population benefit. Conditioned on environment P_* and the frozen model pair $(f_0, f_{a,*})$, the variables $W_{*,j} = \mathbf{1}\{f_{a,*}(X_{*,j}) = Y_{*,j}\} - \mathbf{1}\{f_0(X_{*,j}) = Y_{*,j}\}$ are independent and bounded in $[-1, 1]$ with range $c = 1 - (-1) = 2$. Their conditional expectation is the true population benefit:

$$\mathbb{E}[W_{*,j} \mid P_*, f_0, f_{a,*}] = R_{P_*}(f_0) - R_{P_*}(f_{a,*}) = \Delta_*.$$

By Hoeffding's inequality for the sample average $B_* = \frac{1}{m} \sum_{j=1}^m W_{*,j}$:

$$\begin{aligned} \mathbb{P}(|B_* - \Delta_*| > t \mid P_*, f_0, f_{a,*}) &\leq 2 \exp\left(-\frac{2m^2 t^2}{m(2)^2}\right) \\ &= 2 \exp\left(-\frac{mt^2}{2}\right). \end{aligned}$$

Setting the right-hand side equal to δ yields $t = \sqrt{2 \log(2/\delta)/m}$. Since B_* and Δ_* both lie in $[-1, 1]$, their difference cannot exceed 2. Hence, defining $b(m, \delta) = \min\{2, \sqrt{2 \log(2/\delta)/m}\}$, we have unconditionally:

$$\mathbb{P}(\mathcal{E}_{\text{conc}}) \geq 1 - \delta,$$

where $\mathcal{E}_{\text{conc}} = \{|B_* - \Delta_*| \leq b(m, \delta)\}$.

Step 3: Compound interval coverage. On the intersection event $\mathcal{E}_{\text{conf}} \cap \mathcal{E}_{\text{conc}}$, the triangle inequality gives:

$$|h(Z_*) - \Delta_*| \leq |h(Z_*) - B_*| + |B_* - \Delta_*| \leq \varepsilon + b(m, \delta).$$

Applying the union bound to the complements:

$$\begin{aligned} \mathbb{P}(\Delta_* \in I_{\text{population}}) &\geq \mathbb{P}(\mathcal{E}_{\text{conf}} \cap \mathcal{E}_{\text{conc}}) \\ &\geq 1 - \mathbb{P}(\mathcal{E}_{\text{conf}}^c) - \mathbb{P}(\mathcal{E}_{\text{conc}}^c) \\ &\geq 1 - \alpha_{\text{cell}} - \delta. \end{aligned}$$

Crucially, this step requires no independence assumption between $\mathcal{E}_{\text{conf}}$ and $\mathcal{E}_{\text{conc}}$.

Step 4: Strict directional error bound. Suppose $g_* = \text{ADAPT}$, which requires $h(Z_*) - (\varepsilon + b) > 0$. On the coverage event $\{\Delta_* \in I_{\text{population}}\}$, this implies $\Delta_* \geq h(Z_*) - (\varepsilon + b) > 0$. Therefore, a false adaptation ($g_* = \text{ADAPT}$ and $\Delta_* \leq 0$) can occur only on the complement event $\Delta_* \notin I_{\text{population}}$. Symmetrically, a false freeze ($g_* = \text{FREEZE}$ and $\Delta_* \geq 0$) also requires $\Delta_* \notin I_{\text{population}}$. Since the actions ADAPT and FREEZE are mutually exclusive, the union of false commitments satisfies:

$$\begin{aligned} &\{g_* = \text{ADAPT}, \Delta_* \leq 0\} \cup \{g_* = \text{FREEZE}, \Delta_* \geq 0\} \\ &\subseteq \{\Delta_* \notin I_{\text{population}}\}. \end{aligned}$$

Taking probabilities yields the claimed bound $\leq \alpha_{\text{cell}} + \delta$. $\square$

A.5 Sample Size and Abstention Under the Compound Bound

Theorem 4 quantifies the additional sampling allowance required to transfer a valid cell interval to population benefit under its stated assumptions. It does not attribute the observed empirical abstentions to population uncertainty. For this transfer construction, the total interval half-width is $\varepsilon + b(m, \delta)$. Table 13 computes the Hoeffding sampling radius $b(m, \delta)$ at confidence level $1 - \delta = 0.95$ across representative evaluation sample sizes m .

Table 13: Sampling error radius $b(m, \delta)$ for paired zero-one accuracy differences at $\delta = 0.05$ (95% concentration confidence) across evaluation sample sizes m .

Sample Size m	Hoeffding Radius $b(m, \delta = 0.05)$	Total Radius $(\varepsilon = 0.015)$
64	0.3396	0.3546
200	0.1921	0.2071
512	0.1200	0.1350
1,000	0.0859	0.1009
5,000	0.0384	0.0534
10,000	0.0272	0.0422
20,000	0.0192	0.0342

As an illustrative scenario, consider an evaluation cell with $m = 512$ samples (a common benchmark batch size), a forecast benefit of $\hat{\Delta} = +0.04$ (a 4% point gain), conformal prediction error $\varepsilon = 0.015$, and concentration risk budget $\delta = 0.05$. Here the Hoeffding radius is $b = 0.1200$, yielding a total half-width $\varepsilon + b = 0.1350$. Even with a positive point prediction, the population lower bound is:

$$\hat{\Delta} - (\varepsilon + b) = 0.04 - 0.1350 = -0.0950 < 0.$$

The interval spans zero, forcing the population decision rule to retain the frozen predictor and record **ABSTAIN**. Holding the prediction $\hat{\Delta} = 0.04$ and cell radius $\varepsilon = 0.015$ fixed, certifying a positive population benefit requires $\hat{\Delta} - \varepsilon - b(m, \delta) > 0$, or:

$$0.04 - 0.015 - \sqrt{\frac{2 \log(2/\delta)}{m}} > 0 \iff m > \frac{2 \log(40)}{(0.025)^2} \approx 11,804.41.$$

Thus $m = 11,805$ is the smallest integer satisfying this specific condition. We emphasize that this is an illustrative calculation from this particular Hoeffding bound holding the prediction and cell radius fixed, not an experimental finding or a universal sample requirement for certifying a 4% gain under other procedures or assumptions.

This calculation illustrates the potential commitment cost of the proposed population interval. It does not identify the cause of abstention or freezing in the reported benchmarks, whose gates did not use this sampling correction. Establishing a necessary sample-size requirement would require a separate lower-bound argument. Furthermore, the empirical outcome on CCT-20 was not universal abstention, but conservative retention: KGA issued 44 FREEZE decisions, 1 ABSTAIN, and 0 ADAPT. When evidence of positive benefit is insufficient, a prudent gate must retain the frozen baseline and record ABSTAIN; FREEZE specifically requires a supported negative interval, and uncertainty alone does not justify a directional freeze claim.

A.6 Alternative Route: Direct Empirical Sampling of the Population Frontier

Finally, for completeness, we formalize an alternative operationalization that directly samples the theoretical population frontier $|M| > \beta$ rather than training a learned benefit regressor $\hat{\Delta} = h(Z)$.

Proposition 5 (Sampled population frontier). *Let (f_0, f_a) be fixed binary predictors and let $s : \mathcal{X} \rightarrow [0, 1]$ be a fixed source-fitted score. For an integer $m \geq 0$, suppose target inputs $X_1, \dots, X_m \sim \mu_T$ are drawn independently of one another and of the fitting data and randomness used to select this pair and score. Let $D_m = \{i \in \{1, \dots, m\} : f_0(X_i) \neq f_a(X_i)\}$ be the observed disagreement indices with count $n_D = |D_m|$. If $n_D > 0$, define the empirical disagreement margin:*

$$\widehat{M} = \frac{1}{n_D} \sum_{i \in D_m} s(X_i) - \frac{1}{2}. \quad (25)$$

For confidence level $1 - \alpha \in (0, 1)$, define the margin concentration radius:

$$r_M(n_D, \alpha) = \sqrt{\frac{\log(2/\alpha)}{2n_D}}. \quad (26)$$

Under the declared residual class $\mathcal{C}_\beta = \{P_T : |\gamma(P_T)| \leq \beta\}$, the decision rule

$$g_{\text{margin}} = \begin{cases} \text{ADAPT}, & \widehat{M} - r_M > \beta, \\ \text{FREEZE}, & \widehat{M} + r_M < -\beta, \\ \text{ABSTAIN}, & \text{otherwise (or if } n_D = 0), \end{cases} \quad (27)$$

satisfies $\mathbb{P}(g_{\text{margin}} = \text{ADAPT}, \Delta \leq 0) \leq \alpha$ and $\mathbb{P}(g_{\text{margin}} = \text{FREEZE}, \Delta \geq 0) \leq \alpha$.

Proof. For each $k \geq 1$ with $\mathbb{P}(n_D = k) > 0$, conditional on $n_D = k$, the disagreement samples in their original order are i.i.d. draws from $\mu_T \mid D$. Since $s(X) \in [0, 1]$, Hoeffding's inequality for variables of range 1 gives:

$$\begin{aligned} \mathbb{P}(|\widehat{M} - M| > r_M \mid n_D) &\leq 2 \exp\left(-\frac{2n_D^2 r_M^2}{n_D(1)^2}\right) \\ &= 2 \exp(-2n_D r_M^2) = \alpha. \end{aligned}$$

On the coverage event $|\widehat{M} - M| \leq r_M$, the condition $\widehat{M} - r_M > \beta$ implies:

$$M \geq \widehat{M} - r_M > \beta \geq |\gamma| \geq -\gamma \implies M + \gamma > 0.$$

By the disagreement reduction (Lemma 2), $\Delta = 2d(M + \gamma) > 0$. Thus, conditional on each positive count, a false adaptation requires $|\widehat{M} - M| > r_M$, which has probability at most α . Averaging over the counts preserves the bound, since $n_D = 0$ always gives abstention. The freeze case is identical by symmetry. $\square$

This direct route demonstrates the additive relationship between uncertainties:

$$\begin{aligned} \text{decision threshold} &= \underbrace{r_M(n_D, \alpha)}_{\text{finite-sample margin uncertainty}} \\ &+ \underbrace{\beta}_{\text{structural calibration uncertainty}}. \end{aligned}$$

This rule controls sampling uncertainty in the observed margin while retaining the externally declared residual budget β in Theorem 2. It does not tighten the unresolved residual range on the same evidence fibre.

A.7 Explicit Implementation Boundaries and Operational Scope

We distinguish three procedures and their evaluation scope.

1. **Empirical KGA** ($\hat{\Delta} \pm \varepsilon$): The cell-benefit prediction interval uses the residual construction specified for each study (Section 6). Current primary CIFAR results, the 108-cell ImageNet-C replay, and the corrected 27-cell SAR audit use Protocol B’s separate fitting, calibration, and scoring roles. Historical pooled leave-one-out analyses retain Protocol A’s outcome-dependence limitation; natural studies retain their separately documented data partitions. These are cell-benefit decisions, not population certificates.
2. **Population Extension** ($\hat{\Delta} \pm (\varepsilon + b)$): The compound decision interval incorporating finite-sample Hoeffding concentration $b(m, \delta)$ (Theorem 4). This procedure is exercised under controlled synthetic diagnostic simulations with exact ground truth (an observed 100% empirical population-inclusion rate and zero wrong-direction commitments in the finite trials); it is not a population guarantee for the historical empirical benchmark tables, which evaluate transductive evaluation-cell benefit rather than independent population draws. The separate 108-cell benchmark replay below exercises the enlarged interval as a sensitivity analysis, not as fresh-sample validation.
3. **Sampled Frontier Gate** ($\hat{M} \pm r_M$ under external β): The direct finite-sample operationalization of the knowability frontier (Proposition 5). This provides a formal alternative requiring an externally supplied residual bound β ; it has no real-data benchmark instantiation in this paper.

Neither the population transfer theorem nor the sampled frontier retroactively converts the empirical benchmark results into population-safety claims.

Synthetic population simulation protocol and evidence. To empirically evaluate the compound interval $I_{\text{population}} = [\hat{\Delta} - (\varepsilon + b), \hat{\Delta} + (\varepsilon + b)]$ under exact ground truth, we conducted a diagnostic simulation. In each episode, an environment covariate state $Z \in \{-1, 0, 1\}$ is drawn with equal probability (1/3 each) and held fixed while m test examples are sampled with conditional label distributions $\mathbb{P}(Y = 1 \mid Z = -1) = 0.20$ ($\Delta = -0.60$, harmful), $\mathbb{P}(Y = 1 \mid Z = 0) = 0.50$ ($\Delta = 0.00$, tie/null), and $\mathbb{P}(Y = 1 \mid Z = 1) = 0.80$ ($\Delta = +0.60$, helpful). The benefit estimator $\hat{\Delta}(Z) = 0.6Z$ is the exact Bayes-

optimal predictor. Calibration draws N_{cal} independent episodes to form residual pool $\{r_c = |\hat{\Delta}(Z_c) - b_c|\}$. Risk parameters are $\alpha_{\text{cell}} = 0.05$ and $\delta_{\text{sampling}} = 0.05$ (total population error budget $\alpha_{\text{pop}} = 0.10$). Across calibration sizes $N_{\text{cal}} \in \{18, 19, 39, 99\}$ and evaluation batch sizes $m \in \{32, 128, 512, 2048\}$ over 300 independent trials per configuration (Table 15), empirical coverage and action distributions confirm: (1) at $N_{\text{cal}} = 18$, exact rank $k = \lceil 19 \times 0.95 \rceil = 19 > 18$ is infeasible, setting $\varepsilon = \infty$ and 100% abstention (failed rank protection); (2) at $m = 32$, the sampling radius $b(32, 0.05) = 0.4802$ combined with calibration radius $\varepsilon_{\text{cell}} \approx 0.28$ yields compound width $\varepsilon + b \approx 0.76 > 0.60$, bracketing zero across all three states and producing 100% abstention under this compound-bound construction; (3) for finite-radius configurations with $m \geq 128$, the gate commits on nonzero-benefit states without observed wrong-direction decisions in these trials; Table 15 reports the corresponding abstention rates, while the $N_{\text{cal}} = 18$ configuration continues to abstain throughout; (4) wrong-direction error is 0.0000 across all 16 configurations ($\leq \alpha_{\text{pop}} = 0.10$). An observed 100% population coverage rate on finite simulation trials reflects consistency under the simulated model, not a mathematical proof of universal coverage. Full summary records are archived in the supplementary repository.

Non-oracle population diagnostic. To evaluate the setting where the benefit predictor $\hat{\Delta}$ is not an oracle Bayes rule but an imperfect empirical model fitted on finite development episodes, we conducted an extended non-oracle diagnostic (1,000 independent trials per configuration; archived in the supplementary repository). In each trial, an imperfect linear predictor is fitted on 15 independent development episodes ($m = 32$) subject to estimation noise. We then evaluate three decision mechanisms across the (N_{cal}, m) grid (Table 14): (1) **Point Rule** ($\text{sign}(\hat{\Delta})$): Blindly committing to adaptation whenever $\hat{\Delta} > 0$ achieves 0% abstention, but incurs a 31%–36% wrong-direction commitment error rate; crucially, 100% of these errors occur on null/tie states ($Z = 0, \Delta = 0$) due to random estimation noise around zero, with strictly zero errors on harmful states ($\Delta = -0.6$). (2) **Cell Interval** ($\hat{\Delta} \pm \varepsilon_{\text{cell}}$): In the finite-radius configurations, conformal residual calibration over the imperfect predictions records empirical cell inclusion near the nominal level (93.9%–96.5% against $1 - \alpha_{\text{cell}} = 0.95$) and zero observed wrong-direction decisions in these trials (0.0000). Observed population inclusion is approximately 100% in this synthetic

model. At $N_{\text{cal}} = 18$, the unbounded interval has inclusion 1.000 and abstains throughout. (3) **Population Interval** ($\hat{\Delta} \pm (\varepsilon_{\text{cell}} + b)$): Incorporating the Hoeffding sampling allowance $b(m, \delta)$ produces additional abstention under this compound-bound construction. The interval records 100% empirical population inclusion and zero observed wrong-direction decisions in these finite trials. At $m = 32$, abstention is 99.9%–100% with $b(32) = 0.4802$. For finite-radius configurations with $m \geq 128$, it commits on nonzero-benefit states without observed wrong-direction decisions in these trials; abstention is approximately 32%–34%, near the 33.3% null-state frequency. The $N_{\text{cal}} = 18$ configuration abstains throughout. These observations do not prove universal coverage.

Table 14: Non-oracle population diagnostic (1,000 trials/cell; $\alpha_{\text{cell}} = \delta = 0.05$, $\alpha_{\text{pop}} = 0.10$). An imperfect benefit predictor is fitted on 15 independent development episodes ($m = 32$). Entries report **Cell Coverage / Wrong-Direction Error / Abstention Rate**. The point rule has 31%–36% wrong-direction error, entirely on null states $Z = 0$, and 0% abstention. In finite-radius configurations, the cell interval records 93.9%–96.5% empirical cell inclusion against nominal 95%, approximately 100% population inclusion, and zero observed wrong-direction decisions. The compound interval records 100% empirical population inclusion and zero observed wrong-direction decisions. Its abstention is 99.9%–100% at $m = 32$ and approximately one third for finite-radius configurations with $m \geq 128$. At $N_{\text{cal}} = 18$, the unavailable calibration rank gives an unbounded interval, inclusion 1.000, and 100% abstention at every batch size. These finite-simulation observations do not prove universal coverage. Artifacts: see supplementary repository.

N_{cal}	$m = 32$ ($b = 0.480$)	$m = 128$ ($b = 0.240$)	$m = 512$ ($b = 0.120$)	$m = 2048$ ($b = 0.061$)
<i>Point Rule</i> ($\text{sign}(\hat{\Delta})$): <i>Err / Abstain</i>				
All N	0.309–0.356 / 0.000	0.313–0.332 / 0.000	0.314–0.353 / 0.000	0.326–0.340 / 0.000
<i>Cell Interval</i> ($\hat{\Delta} \pm \varepsilon_{\text{cell}}$): <i>CellCov / Err / Abstain</i>				
18	1.000 / 0.000 / 1.000	1.000 / 0.000 / 1.000	1.000 / 0.000 / 1.000	1.000 / 0.000 / 1.000
19	0.961 / 0.000 / 0.327	0.948 / 0.000 / 0.313	0.948 / 0.000 / 0.314	0.940 / 0.000 / 0.336
39	0.960 / 0.000 / 0.348	0.946 / 0.000 / 0.319	0.956 / 0.000 / 0.345	0.952 / 0.000 / 0.326
99	0.961 / 0.000 / 0.357	0.947 / 0.000 / 0.325	0.939 / 0.000 / 0.321	0.965 / 0.000 / 0.326
<i>Population Interval</i> ($\hat{\Delta} \pm (\varepsilon_{\text{cell}} + b)$): <i>PopCov / Err / Abstain</i>				
18	1.000 / 0.000 / 1.000	1.000 / 0.000 / 1.000	1.000 / 0.000 / 1.000	1.000 / 0.000 / 1.000
19	1.000 / 0.000 / 0.999	1.000 / 0.000 / 0.342	1.000 / 0.000 / 0.314	1.000 / 0.000 / 0.336
39	1.000 / 0.000 / 1.000	1.000 / 0.000 / 0.338	1.000 / 0.000 / 0.345	1.000 / 0.000 / 0.326
99	1.000 / 0.000 / 0.999	1.000 / 0.000 / 0.340	1.000 / 0.000 / 0.321	1.000 / 0.000 / 0.327

108-Cell ImageNet-C development replay. As a separate release replay, we instantiated the same outcome-blind action construction on 108 existing ImageNet-C development cells (five-fold cross-fitting, 36 cells each for Tent, EATA, and SAR; sample-size parameter $m = 2,000$; Table 16). The parameter

Table 15: Synthetic population diagnostic under exact Bayes-optimal prediction (300 trials/cell; $\alpha_{\text{cell}} = \delta = 0.05$, $\alpha_{\text{pop}} = 0.10$). Entries report empirical Cell Coverage / Abstention rate. Across all 16 cells, the finite simulation observed empirical population inclusion of 1.000 and wrong-direction error of 0.000; this is not a universal coverage guarantee. Sampling radii: $b(32) = 0.4802$, $b(128) = 0.2401$, $b(512) = 0.1200$, $b(2048) = 0.0600$. Artifacts: see supplementary repository.

N_{cal}	$m = 32$	$m = 128$	$m = 512$	$m = 2048$
18	1.000 / 1.000	1.000 / 1.000	1.000 / 1.000	1.000 / 1.000
19	0.953 / 1.000	0.963 / 0.330	0.950 / 0.333	0.973 / 0.333
39	0.963 / 1.000	0.977 / 0.370	0.953 / 0.340	0.973 / 0.353
99	0.967 / 1.000	0.967 / 0.397	0.950 / 0.397	0.960 / 0.347

$m = 2,000$ is a sensitivity input to the enlarged-radius formula, not a count of independently drawn evaluation samples. All 108 cells are existing development records; this replay supplies no new independent sampling. The authenticated input and output receipts are preserved in the supplementary reproducibility archive. This is a benchmark replay, not independent sampling or prospective population evidence.

B Supplementary Results and Diagnostic Studies

This section preserves the evidence outside the two primary protocols. The presentation separates ordinary accuracy from balanced accuracy; no value is pooled across metrics, datasets, candidates, or protocols. The record identities and checksums are retained in `reconciled_panels_v1/source_manifest.json`. Historical, incomplete, and invalid studies are identified where their results appear.

B.1 Accuracy-Based Diagnostic Replays

Table 17 collects the remaining ordinary-accuracy replays; the dataset-specific interpretations follow below.

B.2 Balanced-Accuracy Diagnostic Replays

Table 18 reports the balanced-accuracy replays on their own metric scale, with each recorded candidate and evaluation unit kept distinct.

Table 16: Outcome-blind ImageNet-C 108-cell population-transfer replay across three candidates (36 cells each, five-fold cell-outcome-disjoint cross-fitting; $\alpha_{\text{cell}} = 0.05$, $\delta_{\text{samp}} = 0.05$, $\alpha_{\text{pop}} = 0.10$, sensitivity parameter $m = 2,000$, giving uniform sampling radius $b = 0.0607$). For each fold, 7–8 cells are scored while the remaining 28–29 complement cells are split into 19 residual-calibration cells and 9–10 estimator-fit cells. Finite ε indicates non-degenerate calibration (108/108 cells). Decisions report counts of ADAPT/FREEZE/ABSTAIN before (cell gate, $\hat{\Delta} \pm \varepsilon$) versus after (compound population interval, $\hat{\Delta} \pm (\varepsilon + b)$). Across all 108 cells, $\text{FA}_u = 0.000$ (zero false adaptations). Regret reports mean oracle shortfall on cell outcomes for Cell Gate vs. Compound Population Interval ($0.0167 \rightarrow 0.0374$, illustrating the utility cost of the added sampling allowance). Population truth is not established on historical development replays. Receipts: preserved in the supplementary repository.

Candidate	N	Cell radius ε : mean [min, max]	Finite ε
Tent	36	0.0519 [0.0331, 0.0659]	36 / 36
EATA	36	0.0292 [0.0196, 0.0393]	36 / 36
SAR	36	0.1184 [0.0365, 0.2106]	36 / 36
Total / Panel	108	0.0665 [0.0196, 0.2106]	108 / 108

Candidate	Regret cell / compound	Cell gate A / F / U	Compound A / F / U	FA_u	Population truth
Tent	0.0162 / 0.0468	18 / 0 / 18	1 / 0 / 35	0.000	Not established
EATA	0.0095 / 0.0386	20 / 0 / 16	0 / 0 / 36	0.000	Not established
SAR	0.0245 / 0.0267	2 / 11 / 23	0 / 10 / 26	0.000	Not established
Total / Panel	0.0167 / 0.0374	40 / 11 / 57	1 / 10 / 97	0.000	Not established

Table 17: Supplementary accuracy-regret rows. Lower regret is better. Commitment rate counts ADAPT or FREEZE and is not interval coverage. These are protocol-specific point summaries: Office-Home’s primary and stream-seed rows are different from its invalidated five-checkpoint audit; ImageNet-C is configuration-sensitive; PACS has aggregate-only replay. For PACS, $n = 12$ counts domain-seed inference units, whereas action frequencies use 216 repeated decisions.

Protocol	n	KGA	Adapt	Freeze	FA_u	Commitment rate
Office-Home primary	35	0.0158	0.0468	0.0158	0.0000	0.3143
Office-Home stream seeds	54	0.0215	0.0458	0.0217	0.0000	0.2778
ImageNet-C Tent	135	0.0139	0.0191	0.0145	0.0000	0.0074
ImageNet-C EATA	135	0.0031	0.0001	0.0342	0.0000	0.8000
ImageNet-C SAR	135	0.0348	0.0529	0.0319	0.0148	0.0741
PACS (aggregate)	12	0.0431	0.0176	0.0446	0.0093	0.8148
CIFAR-10.1	48	0.0017	0.0190	0.0017	0.0000	0.4583

Table 18: Supplementary balanced-accuracy-regret rows, kept separate from ordinary accuracy. The archived runner contracts name balanced accuracy for Camelyon17, ImageNet-R, and RxRx1; ordinary accuracy can coincide only under the recorded class-balancing conditions. ImageNet-R aggregates candidate backbones rather than one deployed gate. Camelyon17 OOD was already opened; its B-v2 rows are separate within-seed diagnostics. RxRx1’s displayed row is model seed 0.

Protocol	n	KGA	Adapt	Freeze	FA_u	Commitment rate
ImageNet-R backbones	480	0.0136	0.0064	0.0325	0.0063	0.4417
Camelyon17 OOD	18	0.0000	0.0000	0.1381	0.0000	1.0000
Camelyon17 B-v2 Tent	108	0.0399	0.0097	0.0820	0.0185	0.3241
Camelyon17 B-v2 EATA	108	0.0235	0.0040	0.0911	0.0741	0.5648
Camelyon17 B-v2 SAR	108	0.0240	0.0016	0.1001	0.0093	0.4630
RxRx1 model seed 0	60	0.0000	0.2531	0.0000	0.0000	1.0000

B.3 Large-Scale Corruption: ImageNet-C

ImageNet-C is candidate-dependent. Across the 135 aligned conditions (5 seeds $\times$ 27 cells), the canoni-

cal reconciled panel (Table 17 and Table 33) yields KGA/adapt/freeze regrets of 0.0348/0.0529/0.0319 (0.0348 vs. 0.0529 vs. 0.0319) for SAR (8 ADAPT, 2 FREEZE, 125 ABSTAIN, commitment 7.4%, 2 false adap-

tations, $FA_u = 0.0148$); 0.0139 vs. 0.0191 vs. 0.0145 for Tent (1 ADAPT, 0 FREEZE, 134 ABSTAIN, commitment 0.7%, $FA_u = 0.0000$, beating always-adapt by +0.0052 and beating always-freeze by +0.0006); and 0.0031 vs. 0.0001 vs. 0.0342 for EATA (107 ADAPT, 1 FREEZE, 27 ABSTAIN, commitment 80.0%, $FA_u = 0.0000$). An earlier historical LOO diagnostic logged alternative tuning snapshots (e.g., higher abstention under conservative calibrations), but the reconciled canonical panel preserves the unified evaluation contract. These results show why ImageNet-C must be interpreted at the candidate-policy level rather than through one dataset-wide success label.

Completed Corrected-BN Five-Stream-Seed Comparison (135 Cells). The corrected series completed all 135 cells: the same 27 declared conditions for each of stream seeds 0–4, around one shared source checkpoint (ResNet-50). Saved-record digests, adjacent execution manifests, condition order, and raw-score projections were authenticated before aggregation. These checks do not authenticate the model and code bytes originally loaded by the historical producer processes. The maintained head-to-head CLI was run on the final seed, repeated byte-identically, and rerun after changing one scored candidate accuracy within its valid range. That cell’s prediction, radius, interval endpoints, threshold, and both gate actions remained unchanged. Seeds 0–3 retain their separately verified CLI replays. This is saved-feature, cell-outcome-disjoint verification, not environment-held-out or prospective validation. All point-policy decisions use KGA’s identical fitted predictions and the same zero threshold.

Table 19 shows a lower KGA point regret than either fixed policy (0.0262 vs. 0.0275 for always-adapt and 0.0320 for always-freeze), but substantially better point accuracy and regret for uncalibrated benefit regression (0.4334 acc, 0.0047 regret). KGA’s one false ADAPT occurs among 16 ADAPT decisions; the point gate’s six occur among 108. Thus fewer false events do not establish a lower conditional error rate or overall superiority. The declared bootstrap first averages five seeds within each of 27 aligned conditions. Its two-comparison Holm check supports lower regret than always-freeze ($p = 0.0496$), but not always-adapt ($p = 0.9397$); the latter regret-difference interval crosses zero. These are conditional, exploratory uncertainty summaries: they do not resample source models or establish independence among related corruption, severity, and composition conditions. The two-comparator success criterion is not met. Full

input hashes, replay/intervention outputs, lossless normalization, and aggregation are preserved in the supplementary reproducibility archive. The aggregator’s legacy Protocol-A schema label does not describe the bound current three-way gate replay; the replay receipts specify that construction.

Historical 60-Cell Matched-Control Matrix (Seed 0). To resolve whether KGA’s decision value on SAR was confounded by an aggressive setting, an earlier development-only matched control matrix was reported on the same ImageNet-C source checkpoint (ResNet-50; shot and impulse noise, all five severities, three batch regimes with batch sizes $\{16, 32, 64\}$, 64 images per cell, two sample windows per condition, 60 cells per arm, seed 0). The raw 60-cell records and reconciliation manifest for that report are not recovered in the maintained evidence package, so it is retained as historical context and is not release-promoted. Table 20 defines the reported five-arm configuration matrix isolating learning rate and parameter mask factors. Table 21 reproduces that historical summary; it is not a current authenticated benchmark result.

The historical matrix was intended to isolate the rate-versus-mask question. Its reported reference SAR follows the official upstream implementation [5] on ResNet-50: SGD with learning rate $\eta = 2.5 \times 10^{-4}$, momentum 0.9, entropy threshold $E_0 = 0.4 \times \ln(1000)$, and gradient updates restricted to affine parameters in layers 1–3 (freezing layer 4), with per-batch model resets. The two aggressiveness settings isolate: (1) learning rate ($\eta = 2.5 \times 10^{-4}$ reference vs. 4.0×10^{-3} aggressive), and (2) parameter mask (layers 1–3 adapted vs. all layers 1–4 adapted). The historical report stated that across 60 cells per arm, action exposure was: Reference SAR issued 46 ADAPT, 10 FREEZE, 4 ABSTAIN (zero false ADAPT; all 10 non-positive cells were frozen); Rate only issued 42 ADAPT, 14 FREEZE, 4 ABSTAIN; Mask only issued 45 ADAPT, 11 FREEZE, 4 ABSTAIN; Historical combined issued 42 ADAPT, 14 FREEZE, 4 ABSTAIN; and BN-statistics only issued 50 ADAPT, 6 FREEZE, 4 ABSTAIN. In that historical report, relative to reference SAR, the rate-only arm drops always-adapt accuracy by 0.0359 and increases its regret by 0.0344, whereas the mask-only arm changes adaptation accuracy by only -0.0008 . The historical combination lowers always-adapt accuracy by 0.0419. The historical report further stated that under the published reference SAR parameters, KGA achieves mean accuracy 0.4086 (vs. 0.3953 for Always Adapt and 0.3734 for Always Freeze) with regret 0.0023, achieving lower

Table 19: Completed corrected-BN SAR development comparison: five stream seeds over 27 aligned ImageNet-C conditions, not five independently trained models. Accuracy and paired-oracle regret average the 135 seed-condition records. A/F/U denote ADAPT/FREEZE/ABSTAIN; false A means a non-positive measured benefit on an ADAPT decision. FREEZE and ABSTAIN both serve the frozen model. This new comparison is separate from the historical 135-cell panel and 60-cell arm study.

Policy	Accuracy	Oracle regret	A / F / U	False A
Always freeze	0.4062	0.0320	0 / 135 / 0	0
Always adapt	0.4107	0.0275	135 / 0 / 0	21
Point-benefit gate	0.4334	0.0047	108 / 27 / 0	6
KGA	0.4120	0.0262	16 / 6 / 113	1

Table 20: Historical configuration matrix for the five ImageNet-C SAR matched-control arms (ResNet-50, seed 0). The raw 60-cell records are not recovered in the maintained evidence package, so this matrix is not release-promotable. All reported arms perform test-time Batch Normalization statistics updates with per-batch model resets. Learning rate η , adapted affine parameters, and layer 4 status define each arm.

Arm	Description	Learning Rate η	Adapted Layers	Layer 4 State
1	BN-statistics only	0.0 (no grad)	None	Frozen
2	Reference SAR (published settings)	2.5×10^{-4}	Layers 1-3 affine	Frozen
3	Rate only	4.0×10^{-3}	Layers 1-3 affine	Frozen
4	Mask only	2.5×10^{-4}	Layers 1-4 affine	Adapted
5	Historical combined	4.0×10^{-3}	Layers 1-4 affine	Adapted

empirical regret than either fixed policy on that development seed-0 ResNet-50 checkpoint. Importantly, Table 21 also reported that the BN-statistics-only arm + KGA achieves mean accuracy 0.4125 (zero oracle regret), yielding higher observed accuracy than reference SAR + KGA (0.4086). If its underlying records are recovered and re-authenticated, this would confirm the provenance of the historical BN-only comparison. Even with authenticated provenance, observing that normalization adaptation performs comparably to reference SAR does not by itself establish a causal attribution that normalization accounts for the bulk of SAR’s gains; rather, it documents that in this historical noise setting, test-time normalization updates alone yielded comparable observed accuracy. Cross-candidate regret must not be pooled as though candidates share a single oracle; task accuracy on common records provides the valid basis for comparison. Historical source pointer (raw 60-cell manifest not resident): preserved in the supplementary research archive.

Historical Pre-Repair 27-Cell Calibration Audit (Seed 0). As a calibration-audit check, we replayed the completed seed-0 reference-SAR records through the canonical three-way cell-outcome-disjoint

cross-fitting implementation (Protocol B), without altering historical records or the matched-control table above. This saved-feature audit evaluates the 27 completed conditions from the historical, pre-repair seed-0 grid (3 corruptions [gaussian, shot, impulse noise] $\times$ 3 severities [1, 3, 5] $\times$ 3 compositions [iid, imbalanced, single_class]; small batch regime; baseline $a_0 = 0.3976$). Table 22 reports the full performance metrics across all decision policies.

On these 27 audit cells, Clean Cross-Fit KGA (Protocol B) achieves accuracy 0.4301 (shortfall of only 0.0041 regret vs. Always Adapt), zero false adaptations ($FA_u = 0.000$), and captures 19 of 25 beneficial updates (76.0%), issuing 19 ADAPT, 0 FREEZE, and 8 ABSTAIN decisions. Historical LOO KGA (Protocol A) achieves accuracy 0.4318 (0.0024 regret, 20/25 helpful captured, actions 20/0/7). A saved-feature perturbation check changed one scored outcome by a large amount while leaving that cell’s fitted prediction, radius, and action unchanged. This supports outcome exclusion for the tested cell and replay path; it does not establish exclusion for every cell or for upstream feature construction. The resident 27-cell audit in Table 22 and the historical 60-cell matched-control summary in Table 21 represent distinct experimental configurations: the historical summary described 2 noise types $\times$ 5 severities $\times$ 3 batch sizes $\times$ 2 windows (and is not currently raw-record authenticated), whereas the resident audit evaluates 3 noise types $\times$ 3 severities $\times$ 3 compositions at small batch regime. The 27-cell panel includes a saved-feature outcome-perturbation check; it evaluates a distinct configuration and does not establish that seeds 1–4 are complete. An additional September 12 replay retains the original audit’s literal condition-string sample identifiers. It exactly reproduces every seed-0 prediction, radius, and action. For seeds 0/1/2, Protocol-B accuracy is 43.01/43.20/41.07%, with ADAPT/FREEZE/ABSTAIN counts 19/0/8,

Table 21: Historical, non-promotable ImageNet-C SAR control summary on 60 development cells per arm (ResNet-50, shot/impulse noise [2 types], all five severities [5 levels], three batch sizes {16, 32, 64} [3 sizes], evaluated across 2 distinct sample windows per condition: $2 \times 5 \times 3 \times 2 = 60$ cells; seed 0). The raw cell records and reconciliation manifest are not recovered in the maintained evidence package. The 5 arms and Protocol A definitions are retained for historical context only; the separate 27-cell saved-feature audit is reported in Table 22.

Arm	Always Adapt	Always Freeze	KGA Acc	Regret A	Regret F	Regret KGA
BN-statistics only	0.3984	0.3734	0.4125	0.0141	0.0391	0.0000
Reference SAR (2.5×10^{-4})	0.3953	0.3734	0.4086	0.0156	0.0375	0.0023
Rate only (4×10^{-3})	0.3594	0.3734	0.4031	0.0500	0.0359	0.0063
Mask only (incl. layer4)	0.3945	0.3734	0.4055	0.0161	0.0372	0.0052
Historical combined	0.3534	0.3734	0.4031	0.0560	0.0359	0.0063

Table 22: Saved-feature outcome-disjoint calibration audit on 27 historical, pre-repair seed-0 local SAR records (ResNet-50, reference-matched learning rate $\eta = 2.5 \times 10^{-4}$ and layers 1–3 affine; 3 corruptions $\times$ 3 severities $\times$ 3 compositions = 27 cells; small batch regime, 25 helpful $B > 0$, 0 harmful $B < 0$, 2 neutral $B = 0$). This replay does not repair upstream BatchNorm behavior. Clean Cross-Fit KGA uses Protocol B (5-fold outcome-disjoint cross-fitting, $\alpha = 0.10$); Historical LOO KGA uses Protocol A (leave-one-cell-out exact rank, $\alpha = 0.10$). $FA_u = \mathbb{P}[\text{ADAPT}, B \leq 0]$. Helpful Captured is the fraction of the 25 helpful cells ($B > 0$) receiving ADAPT. Baseline accuracy $a_0 = 0.3976$.

Policy	Accuracy	Regret	FA_u	Helpful Captured	Actions (A / F / U)
Always Adapt	0.4342	0.0000	0.074 (2/27)	25 / 25 (100.0%)	27 / 0 / 0
Always Freeze	0.3976	0.0366	0.000 (0/27)	0 / 25 (0.0%)	0 / 27 / 0
Historical LOO KGA (Protocol A)	0.4318	0.0024	0.000 (0/27)	20 / 25 (80.0%)	20 / 0 / 7
Clean Cross-Fit KGA (Protocol B)	0.4301	0.0041	0.000 (0/27)	19 / 25 (76.0%)	19 / 0 / 8

21/0/6, and 13/0/14. For each of the 81 cells individually, replacing its saved benefit by a value within $[-1, 1]$ and rerunning fitting and calibration leaves its prediction, radius, interval endpoints, and action unchanged; the decision boundary remains fixed at zero. This verifies exclusion from the saved-feature replay, not upstream feature construction, environment-held-out validity, or prospective coverage. None of these gates achieves lower empirical regret than always-adapt. The full replay is stored in the September 12 condition-ID replay receipt in the archived September 12 evidence record.

Development Replay of Corrected-BN Seed 0. An execution-path audit also found that these historical local runs excluded the final block from both affine optimization and BatchNorm reconfiguration. The pinned SAR source instead configures all BatchNorm layers to use batch statistics and excludes final-block parameters separately from optimization. Consequently, historical train-mode scoring could update final-block running buffers before eval-mode evidence extraction. The local implementation now separates these operations, with a regression-

tested checkpoint-version guard preventing continuation from old records. The corrected series’ seed 0 completes the same declared 27 condition identifiers, with 23 positive and four negative candidate benefits. Its saved features and outcomes were replayed through the maintained eight-fold three-way construction, with $\alpha = 0.10$, fixed zero decision threshold, and the existing 11-feature gradient-boosted regressor (250 trees, depth 2, learning rate 0.05, subsample 0.8, fit seed 0). Each fold has nine residual-calibration cells and rank nine; the other roles are 14 fit/four score cells or 15 fit/three score cells. The point-benefit policy uses exactly the same fitted predictions as KGA. KGA issues 5 ADAPT, 1 FREEZE, and 21 ABSTAIN decisions, with accuracy 0.4146 and paired-oracle regret 0.0271. Always-adapt accuracy is 0.4186 (regret 0.0232), always-freeze accuracy is 0.4095 (regret 0.0322), and the same-prediction point policy reaches 0.4290 (regret 0.0128). KGA therefore does not achieve lower regret than both fixed policies on this split. One of its five ADAPT decisions is false ($FA_u = 1/27$, $FA_c = 1/5$); the point policy makes three false ADAPT decisions among 23 ADAPT actions. For every scored cell, replac-

ing its candidate accuracy by a different endpoint in $\{0,1\}$ with the frozen accuracy and features fixed, then rerunning all fits and calibration, leaves that cell’s prediction, radius, interval, threshold, and action unchanged. These 27 synthetic validity probes do not create new observations. The original records remain unchanged. This is a dependent, opened-grid development replay, not a native-method comparison, environment-held-out result, or independent source-training repetition. Full-precision records, software identities, and the executable replay are preserved in the supplementary reproducibility archive. The maintained head-to-head CLI now also accepts an explicitly hash-bound saved-row checkpoint and a fresh output directory. Its independent CLI replay reproduces the same predictions, radii, actions, and paired-outcome metrics. Its fallback POEM/AETTA rows remain local ports and are not external native evidence.

Five-Arm Matched SAR Control Study and Provenance Scope.

To evaluate whether earlier SAR routing collapse was driven by learning rate ($\eta = 4.0 \times 10^{-3}$), layer-4 affine parameter mask unfreezing, or BatchNorm test statistics, we report a five-arm matched ablation on the shared declared 27-condition ImageNet-C noise panel ($n = 27$ cells, ResNet-50 source checkpoint, small batch regime $b = 16$, seed 0; 135 total evaluations). Table 23 presents the comparative evaluation. On this matched 27-condition panel, BN-statistics-only adaptation matched reference SAR candidate accuracy to displayed precision, while $16\times$ learning-rate scaling and layer-4 affine unfreezing changed mean candidate accuracy by at most 0.01 percentage points. Under corrected BatchNorm handling, the tested learning-rate and layer-mask interventions produced negligible differences in candidate accuracy. The historical BatchNorm execution mismatch therefore remains a plausible explanation for the earlier discrepancy, but this experiment does not isolate that mismatch causally. This conclusion is specific to the evaluated panel and does not imply that SAR gradients, learning rate, or parameter masks are generally irrelevant. Protocol B KGA achieves competitive routed accuracy (44.05%–44.26%), near-zero regret (0.00–0.20 pp), and zero observed false adaptations ($FA_u = 0.0000$) across the five evaluated arms. Saved manifests and raw outputs are preserved; the execution-identity limitations below bound the authentication claim.

Each arm record in the raw audit ledger contains a recorded path/label sample hash

(`sample_hash`), pre- and post-adaptation model state hash fields (`candidate_state_hash_before` and `candidate_state_hash_after`), and execution flags (`bn_statistics_active`, `gradient_update_active`, and `learning_rate`) describing the declared intervention. The sample hash does not bind decoded pixels or the complete adaptation stream, and the fields alone do not reconstruct executed-source identity. Across all five arms (27 condition cells each, 135 total records), Arm 1 (BN-only) records `bn_statistics_active=True`, `gradient_update_active=False`, and $\eta = 0.0$ ($\|\Delta\theta\|_2 = 0.0000$). Arm 2 (reference SAR) records active gradients with $\eta = 2.5 \times 10^{-4}$ and layers 1–3 unmasked ($\|\Delta\theta\|_2 = 0.0011$). Arms 3–5 record the rate-scaled and layer-unfrozen configurations with update norms reaching 0.0241. Per-record hashes, raw accuracies, and 11-dimensional Z feature vectors are archived in the supplementary repository.

Distinct SAR execution identities. The matched reference arm uses SAR reference parameters in a local execution; the saved records do not establish fidelity to the complete native execution. The corrected five-seed run and this seed-0 matched-arm study are not one interchangeable execution. Joining their 27 seed-0 condition identifiers gives equal frozen scores in every pair (mean 40.9528%), but different candidate means: 41.8620% for corrected seed 0 versus 44.2509% for the matched reference arm, with four versus zero harmful conditions. Their candidate-specific oracles therefore also differ. The saved configurations specify 50 versus one pass over a 128-image adaptation stream at batch size 16, allowing 400 versus eight minibatch updates; these are configured budgets, not independently reconstructed update traces. Saved configurations and current producers also differ in evaluation batching and other details; current code is not proof of executed source. The saved arrays establish different candidates, not the causal contribution of any one setting. Raw-arm hashes agree with the matched manifest, but complete executed-source, checkpoint-state, and decoded-image/stream bindings are not recovered for both runs. Equal frozen scores do not establish those identities. Both remain post-adaptation diagnostics, not authenticated native prequential reproductions, and the matched arm does not replace the completed five-seed comparison.

Table 23: **Matched SAR Control Study on ImageNet-C**. Five-arm matched ablation varying learning rate η and parameter mask (with layer 4 frozen vs adapted) on a shared 27-condition ImageNet-C noise panel ($n = 27$ cells, ResNet-50 source checkpoint, small batch regime $b = 16$, seed 0). Candidate accuracy a_{cand} , benefit $\Delta = a_{\text{cand}} - a_0$, parameter update norm $\|\Delta\theta\|_2$, Protocol B KGA routed accuracy a_{KGA} , oracle regret R_{KGA} , unconditional false adapt rate FA_u , and decision counts (Adapt / Freeze / Abstain) under exact rank conformal routing ($\alpha = 0.10, \tau = 0$). Always Freeze regret is 3.30–3.31 pp across arms (3.30 pp for Arms 1–4, 3.31 pp for Arm 5); Always Adapt regret is 0.00 pp. Accuracy is reported as a percentage; benefit Δ and regret are reported in percentage points. Saved records support these comparisons; complete execution identities remain unavailable. Per-condition records and declared execution fields are indexed in the accompanying reproducibility manifest.

Arm	Configuration	η	Adapted Layers	a_{cand}	Δ	$\ \Delta\theta\ _2$	a_{KGA}	R_{KGA}	FA_u	Decisions (A/F/Abs)
Arm 1	BN-only (zero grad)	–	None	44.25%	+3.30%	0.0000	44.05%	0.20%	0.0%	23/0/4
Arm 2	SAR reference parameters; local execution	2.5×10^{-4}	Layers 1–3	44.25%	+3.30%	0.0011	44.11%	0.14%	0.0%	24/0/3
Arm 3	Rate only	4.0×10^{-3}	Layers 1–3	44.26%	+3.30%	0.0182	44.06%	0.19%	0.0%	23/0/4
Arm 4	Mask only	2.5×10^{-4}	Layers 1–4	44.25%	+3.30%	0.0015	44.25%	0.00%	0.0%	27/0/0
Arm 5	Historical combined	4.0×10^{-3}	Layers 1–4	44.26%	+3.31%	0.0241	44.26%	0.00%	0.0%	27/0/0

B.3.1 ImageNet-C Decision Baselines Head-to-Head Comparison

To directly evaluate whether 1D heuristic gates (ATC confidence, predictive entropy progress, EATA reliable-sample filtering, and drift monitors) can match KGA on large-scale vision corruptions, we replayed all decision rules across the 108 ImageNet-C noise conditions (36 conditions each for Tent, EATA, and SAR) using identical sample streams and feature vectors. Table 24 reports the head-to-head results. The Drift Monitor row is a protocol-matched port monitoring prediction-distribution drift under the synchronized streaming harness, not native upstream POEM execution.

On the Tent and EATA panels, the harmful-update base rates are 6% and 3%, respectively. On the SAR panel, the harmful-update base rate is 44%; ATC confidence, entropy progress, EATA filtering, and drift monitoring each have observed $\text{FA}_c = 1.000$, meaning all their ADAPT commitments on this panel decrease accuracy. KGA has no observed false ADAPT decisions across these three candidate panels, with $\text{FA}_u = 0.000$ and $\text{FA}_c = 0.000$. On SAR its commitment rate is 0.39 (61% abstention). These are descriptive results on the evaluated cells, not proof of protection on future shifts.

We also examine the point-benefit regressor ($\varepsilon = 0$). On the SAR panel, uncalibrated benefit regression achieves mean accuracy 0.431 and regret 0.0010 with $\text{FA}_u = 0.028$ ($\text{FA}_c = 0.050$, deploying harmful updates on 1 of 16 harmful conditions), whereas KGA achieves accuracy 0.409 and regret 0.0229 with zero observed false adaptation ($\text{FA}_u = 0.000$). On this specific SAR diagnostic, the point-benefit predictor achieves lower regret by accepting a non-zero false

adaptation risk. The interval gate adds conservatism (issuing 2 ADAPT, 12 FREEZE, and 22 ABSTAIN decisions out of 36 conditions, corresponding to an adapt rate of 0.06 and commitment rate of 0.39), with no observed false ADAPT on this panel and a higher abstention rate. Zero observed false ADAPT under two ADAPT decisions is descriptive rather than large-sample confirmation. The interval rule controls the declared marginal directional error when its coverage premise holds. This diagnostic does not establish that premise under arbitrary drift.

B.3.2 CIFAR-10-C Decision-Rule Comparison

To comprehensively evaluate decision rules on the identical deployment testbed, we evaluated nine same-prediction decision rules across the 2,160 authoritative Protocol B Tent CIFAR-10-C cells (comprising the primary and protocol subsets defined in the condition manifest). Table 25 reports performance and action counts; Table 26 reports directional error rates and benefit capture. Heuristic 1D gates (confidence and entropy) suffer high false-adaptation rates ($\text{FA}_u \approx 0.25$, committing harmful adaptation updates on 75% of harmful cells). Uncalibrated point-benefit regression achieves low regret (0.0004) but accepts a 4.1% false adaptation rate ($\text{FA}_u = 0.0407$, deploying updates on 12.2% of cells with $B \leq 0$). In contrast, KGA records zero observed false ADAPT decisions on this evaluated panel ($\text{FA}_u = 0.0000$, 0% harmful adaptation) with competitive regret (0.0018) while executing 1,094 ADAPT commitments on 1,439 helpful cells (76.0% cell capture). On this Tent panel, an exploratory fixed ± 0.02 margin (chosen near the empirical mean half-width) achieves a similar operat-

Table 24: ImageNet-C head-to-head decision baseline comparison across 36 noise conditions per candidate ($N = 108$ total evaluations). Regret is shortfall relative to per-condition oracle; lower is better. Adapt rate is $\mathbb{P}[\text{ADAPT}]$. Directional commitment rate is $\mathbb{P}[\text{ADAPT or FREEZE}]$. $\text{FA}_u = \mathbb{P}[\text{ADAPT}, B \leq 0]$; $\text{FA}_c = \mathbb{P}[B \leq 0 \mid \text{ADAPT}]$. Harmful adapted is the proportion of all harmful conditions ($B < 0$) that received adaptation updates.

Candidate	Policy	Mean Acc	Regret	FA_u	FA_c	Harmful Adapted	Worst Case	Adapt Rate	Commitment Rate
Tent (Oracle: 0.455, Harmful: 6%)	Always Adapt	0.454	0.0007	0.056	0.056	1.00	0.164	1.00	1.00
	Always Freeze	0.405	0.0494	0.000	—	0.00	0.088	0.00	1.00
	ATC (confidence)	0.443	0.0114	0.056	0.067	1.00	0.088	0.83	1.00
	Entropy Progress	0.446	0.0089	0.056	0.062	1.00	0.088	0.89	1.00
	EATA Filter	0.454	0.0007	0.056	0.056	1.00	0.164	1.00	1.00
	Drift Monitor (POEM port)	0.452	0.0028	0.056	0.057	1.00	0.088	0.97	1.00
	Benefit Regression ($\varepsilon = 0$)	0.455	0.0000	0.000	0.000	0.00	0.164	0.94	1.00
	KGA (Ours)	0.449	0.0060	0.000	0.000	0.00	0.164	0.69	0.69
EATA (Oracle: 0.444, Harmful: 3%)	Always Adapt	0.444	0.0000	0.028	0.028	1.00	0.169	1.00	1.00
	Always Freeze	0.405	0.0386	0.000	—	0.00	0.088	0.00	1.00
	ATC (confidence)	0.411	0.0328	0.028	0.100	1.00	0.088	0.28	1.00
	Entropy Progress	0.412	0.0321	0.028	0.091	1.00	0.088	0.31	1.00
	EATA Filter	0.444	0.0000	0.028	0.028	1.00	0.169	1.00	1.00
	Drift Monitor (POEM port)	0.441	0.0023	0.028	0.029	1.00	0.088	0.97	1.00
	Benefit Regression ($\varepsilon = 0$)	0.444	0.0000	0.028	0.028	1.00	0.169	1.00	1.00
	KGA (Ours)	0.439	0.0047	0.000	0.000	0.00	0.169	0.75	0.75
SAR (Oracle: 0.432, Harmful: 44%)	Always Adapt	0.319	0.1124	0.444	0.444	1.00	0.061	1.00	1.00
	Always Freeze	0.405	0.0267	0.000	—	0.00	0.088	0.00	1.00
	ATC (confidence)	0.365	0.0673	0.111	1.000	0.25	0.088	0.11	1.00
	Entropy Progress	0.355	0.0765	0.139	1.000	0.31	0.088	0.14	1.00
	EATA Filter	0.395	0.0369	0.028	1.000	0.06	0.088	0.03	1.00
	Drift Monitor (POEM port)	0.394	0.0382	0.056	1.000	0.12	0.088	0.06	1.00
	Benefit Regression ($\varepsilon = 0$)	0.431	0.0010	0.028	0.050	0.06	0.103	0.56	1.00
	KGA (Ours)	0.409	0.0229	0.000	0.000	0.00	0.088	0.06	0.39

ing point, illustrating that conservative gating can be approximated by a calibrated scalar threshold on this specific benchmark; KGA automates this thresholding via data-driven conformal radii without manual margin tuning.

B.3.3 Independent-Checkpoint Diagnostics and Calibration Audit

Historical routing authority qualification. Table 27 retains historical reported aggregates. Three recovered JSON receipts preserve B6 fold identities, 108 per-cell predictions/radii/actions and aligned AETTA/LOO/B6 decisions. All four input hashes match the resident saved files; independent arithmetic checks reproduce the B6 action counts and point regrets without refitting. These receipts and their scope are preserved in `release/recovered_gemini_evidence_v1/`. The producer and completed execution log also report 108 outcome-perturbation checks; those historical fitting tests were not rerun in this release pass.

A separate limitation remains: each B6 fold uses eight calibration residuals. At $\alpha = 0.10$, the exact split-conformal rank is $\lceil 9(0.9) \rceil = 9$, exceeding the available eight. The historical producer clipped that rank to eight and used the maximum residual. Its finite radii therefore do not establish nominal 90% split-conformal coverage. The recovered point estimates

are descriptive retrospective results, not exact-rank or prospective confirmation.

The separate helpful-only presentation remains unresolved: its current evaluator uses measured frozen accuracy as a label-derived feature, and no recovered receipt establishes that evaluator as the producer of the historical helpful-panel values. The mixed-sign receipt recovery does not authenticate that separate panel or establish population transfer.

The repository checkpoint inventory verifies that CIFAR-10 evaluations in the original panel originate from a single source model checkpoint (`resnet18_cifar.pt`, clean test accuracy 0.8737). The evaluation seeds in historical controlled grids represent stream/order permutations over this shared initial checkpoint rather than independently initialized and trained neural networks. To address this, we executed an independent source-model replication suite. Two independent ResNet-18 models were trained from scratch with distinct random initializations (seeds 101 and 102) matching the canonical SGD + OneCycleLR protocol over 12 epochs, achieving clean test accuracies of 0.8953 and 0.8956.

We then evaluated Tent adaptation across six representative corruptions (*gaussian noise*, *gaussian blur*, *fog*, *contrast*, *pixelate*, *jpeg compression*) at moderate and severe shifts (severities 3 and 5) across three stream seeds (0, 1, 2) across all three source models

Table 25: Systematic comparison of nine decision rules on the authoritative Protocol B CIFAR-10-C Tent panel ($N = 2,160$ cells): **performance and action counts**. Regret is mean accuracy shortfall to oracle. A/F/U are Adapt/Freeze/Abstain counts. Commitment = $(A + F)/N$; Abstention = U/N . Denominators: $A+F+U = 2,160$ for all rows.

Policy	Mean Score	Regret	A / F / U	Commitment	Abstention
Always Adapt	0.7869	0.0080	2160 / 0 / 0	1.0000	0.0000
Always Freeze	0.6709	0.1239	0 / 2160 / 0	1.0000	0.0000
Confidence gate	0.7869	0.0080	1834 / 326 / 0	1.0000	0.0000
Entropy gate	0.7868	0.0081	1812 / 348 / 0	1.0000	0.0000
Drift/KL gate	0.6709	0.1239	0 / 2160 / 0	1.0000	0.0000
ATC-style estimator	0.7914	0.0035	1467 / 693 / 0	1.0000	0.0000
Point-benefit gate	0.7945	0.0004	1471 / 689 / 0	1.0000	0.0000
Fixed-margin three-way gate	0.7930	0.0018	1090 / 328 / 742	0.6565	0.3435
KGA calibrated interval gate	0.7931	0.0018	1094 / 334 / 732	0.6611	0.3389

Table 26: Systematic comparison of nine decision rules: **directional errors and benefit capture** (same $N = 2,160$ Tent panel as above). $FA_u = \mathbb{P}[\text{ADAPT}, B \leq 0]$ (divides by $N = 2,160$); FA_c divides by ADAPT count (undefined when ADAPT=0). $FF_u = \mathbb{P}[\text{FREEZE}, B \geq 0]$ (divides by N); FF_c divides by FREEZE count. Nonpos. Adapt = fraction of $B \leq 0$ cells ($N_{B \leq 0} = 721$; $N_{B < 0} = 714$, $N_{B=0} = 7$) that received ADAPT, satisfying $FA_u \times N = \text{Nonpos. Adapt} \times 721$. Positive Benefit Captured is the fraction of oracle positive adaptation benefit preserved: $\sum_{i:g_i=\text{ADAPT}} \max(0, B_i) / \sum_{i=1}^N \max(0, B_i)$; Forgone is $1 - \text{Captured}$. KGA executed ADAPT on 1,094 of 1,439 helpful cells (76.0% cell capture), abstaining on the remainder where empirical intervals overlap zero; zero nonpositive cells received ADAPT ($FA_u = 0.0000$).

Policy	FA_u	FA_c	FF_u	FF_c	Nonpos. Adapt	Benefit Captured	Benefit Forgone
Always Adapt	0.3338	0.3338	0.0000	—	1.0000	1.0000	0.0000
Always Freeze	0.0000	—	0.6694	0.6694	0.0000	0.0000	1.0000
Confidence gate	0.2505	0.2950	0.0699	0.4632	0.7503	0.9958	0.0042
Entropy gate	0.2486	0.2964	0.0782	0.4856	0.7448	0.9948	0.0052
Drift/KL gate	0.0000	—	0.6694	0.6694	0.0000	0.0000	1.0000
ATC-style estimator	0.1111	0.1636	0.1009	0.3146	0.3329	0.9926	0.0074
Point-benefit gate	0.0407	0.0598	0.0273	0.0856	0.1221	0.9989	0.0011
Fixed-margin three-way gate	0.0000	0.0000	0.0005	0.0030	0.0000	0.9851	0.0149
KGA calibrated interval gate	0.0000	0.0000	0.0000	0.0000	0.0000	0.9855	0.0145

(108 condition cell runs total). An empirical variance decomposition reveals a mean between-model standard deviation of $\sigma_{\text{between}} = 0.0211$ (2.11%) compared to a within-model stream ordering standard deviation of $\sigma_{\text{within}} = 0.00135$ (0.14%), yielding a standard deviation ratio of $\sigma_{\text{between}}/\sigma_{\text{within}} = 15.60\times$. Across the three evaluated source models, adaptation-benefit direction was consistent away from the near-zero region, while absolute benefit magnitude exhibited nontrivial model-to-model variation.

Furthermore, to evaluate whether KGA routing behavior replicates within independently initialized source weights, we evaluated leave-one-cell-out (LOO) exact-rank empirical residual routing across two complementary 36-condition panels ($N = 108$ evaluations each across the three models; Table 27). We note that while the main 2,160-cell CIFAR panel employs 5-fold three-way cell-outcome-disjoint cross-fitting

(Protocol B), this compact 36-condition independent-checkpoint replication uses leave-one-cell-out (LOO) exact-rank empirical residual calibration. Because peer-dependent LOO calibration does not automatically inherit ordinary split-conformal coverage theorems, we report empirical cell metrics descriptively.

In **Panel A** (the helpful-only reduced panel), adaptation benefit is uniformly positive ($\Delta > 0$), so Always Adapt achieves zero regret (0.00 pp) and zero false adaptations. LOO exact-rank empirical residual KGA achieves low regret (0.41 pp) and zero observed false adaptation ($FA_u = 0.0000$), capturing 30 of 36 beneficial adaptation opportunities (83.3% cell capture, 97.7% positive benefit mass capture) while abstaining on 6 mild-shift cells where empirical confidence intervals overlap zero.

To evaluate whether KGA reproduces selective adapt-versus-freeze routing across independently

trained checkpoints when harmful adaptation is genuinely present, **Panel B** reports a historical 36-condition mixed-sign paired-outcome panel deliberately containing 39–42% negative-benefit conditions ($\Delta < 0$, severe collapse induced by single-class or class-imbalanced streaming under small/tiny batches on mild shifts); separately, 11–14% of conditions had absolute measured benefit within 1 percentage point of zero ($|\Delta| \leq 0.01$). Away from this near-zero margin ($|\Delta| > 0.01$), the sign of adaptation benefit was strictly consistent across all three independently initialized checkpoints, whereas three low-margin mild-shift conditions exhibited minor sign flips across model initializations (e.g., $|\Delta| \leq 0.0075$ on mild defocus blur, contrast, and fog), illustrating that routing margins can be checkpoint-specific. The historical summaries report that exact-rank leave-one-out empirical residual KGA routing achieves lower point oracle regret than both fixed baselines on each evaluated checkpoint: $R_{\text{KGA}} = 0.08\%$ on Model 0 (vs. $R_{\text{AF}} = 12.67\%$, $R_{\text{AA}} = 1.39\%$), $R_{\text{KGA}} = 0.09\%$ on Model 101 (vs. $R_{\text{AF}} = 14.12\%$, $R_{\text{AA}} = 1.01\%$), and $R_{\text{KGA}} = 0.23\%$ on Model 102 (vs. $R_{\text{AF}} = 13.71\%$, $R_{\text{AA}} = 1.01\%$). The unconditional observed false adapt rate remained low across all checkpoints ($\text{FA}_u = 0.0000$ on Model 0, 0.0278 on Model 101, and 0.0000 on Model 102, below the nominal $\alpha = 0.10$ benchmark threshold), while actively rejecting harmful updates via 5–9 FREEZE and 9–13 ABSTAIN decisions. Recovered aligned per-cell decisions reproduce this point-regret ordering across the three checkpoints. They do not independently establish original fitted-model/environment identity or prospective transfer.

The historical supplementary held-out-model transfer summary reports that applying the canonical KGA calibrated interval (fitted on seed-0 model weights, without recalibration) directly to seeds 101 and 102 achieves regret = 0.0041, $\text{FA}_u = 0.0000$, and 30 ADAPT / 0 FREEZE / 6 ABSTAIN decisions identically across all three models under zero-shot gate transfer on Panel A. The reported transfer summary lacks recovered per-cell decisions and predictor/transfer receipts; it is not an authenticated zero-shot transfer execution.

B.3.4 Published Baseline Reproducibility Audit and Historical Scope

We surveyed third-party published baseline implementations. Authenticated native Kim et al. (TTA / ALine) source was unavailable in the recorded comparison. This does not establish why the advertised

repository was unavailable.

To provide a synchronized head-to-head comparison against a closely related published comparator on the exact same conditions, we evaluated a deployment recovery decision rule built from the authenticated native AETTA accuracy estimator [18] (official AETTA implementation [18]) across all 36 conditions of the mixed-sign panel across all three independent ResNet-18 source checkpoints (108 evaluations total; Table 27, Panel C). The comparison equalizes source model, adapted candidate, target conditions, evaluation records, and test-time label access, but not the development-information budget: KGA uses historical labeled shift outcomes for benefit-model fitting and calibration, whereas AETTA uses its published label-free accuracy-estimation mechanism. The AETTA estimator was evaluated using its official 10-draw MC-dropout accuracy estimator ($p = 0.4$), and the deployment recovery decision rule retains updates (ADAPT) only if post-adaptation accuracy is estimated not to degrade relative to the frozen model ($\hat{a}_{\text{adapt}} \geq \hat{a}_{\text{frozen}}$) and estimated accuracy remains above the absolute hard-reset threshold ($100 \times \hat{a}_{\text{adapt}} \geq 20.0$); otherwise the model recovers to the frozen state (FREEZE).

As documented in Table 27 (Panel C): (1) Under leave-one-out calibration (within-model leave-one-condition-out calibration on the evaluated checkpoint, which pools peer cells from the same model), KGA achieves lower point oracle regret than the AETTA-based rule across every evaluated model: 0.08% vs. 0.36% on Model 0, 0.09% vs. 0.18% on Model 101, and 0.23% vs. 0.36% on Model 102 (a regret advantage of +0.09 to +0.28 pp), translating to higher realized test accuracy. The point ordering replicated across all three independently trained checkpoints; with three model instances, we treat this as replication evidence rather than population-level inference over model training. (2) Because this recovery rule operates as an unhedged binary choice without conformal uncertainty calibration, its accuracy estimator misjudges several severe collapse conditions, committing 4–5 observed false adaptations per model. Consequently, under the same descriptive 0.10 benchmark threshold used to summarize KGA, the AETTA-based rule yielded observed $\text{FA}_u = 0.1111$ –0.1389 across the three checkpoints (0.1389 on Model 0, 0.1111 on Model 101, and 0.1389 on Model 102). (3) In contrast, under LOO calibration, KGA leverages empirical residual calibration to abstain on borderline conditions (issuing 9–13 ABSTAIN decisions), with zero observed false adaptations on Models 0 and 102

Table 27: Historical independent-checkpoint summaries (36 conditions per model). Panel A is helpful-only and lacks recovered execution authority; it is not authenticated transfer evidence. Panels B and C retain the recovered mixed-sign arithmetic under peer-dependent LOO and B6 replay. B6 clips the required conformal rank nine to eight and is descriptive only, with no nominal 90% coverage claim. Panel C uses the native AETTA estimator with a local recovery rule; it does not establish fidelity of the complete published learner. KGA has lower historical point regret under LOO, but higher point regret and fewer false ADAPT decisions under B6. Original fitting/environment identities remain incomplete (Appendix B.3.3).

Source Model	$\mathbf{a}_{\text{clean}}$	n	$\mathbf{a}_0$	$\mathbf{a}_{\text{cand}}$	$\mathbf{R}_{\text{AF}}$	$\mathbf{R}_{\text{AA}}$	$\mathbf{a}_{\text{policy}}$	$\mathbf{R}_{\text{policy}}$	FA_u	Decisions (A/F/Abs)
<i>Panel A: Helpful-Only Reduced Panel (Uniformly Helpful, $\Delta > 0$)</i>										
Model 0 (Canonical)	87.37%	36	61.20%	79.65%	18.45%	0.00%	79.24%	0.41%	0.0000	30/0/6
Model 101 (Independent)	89.53%	36	63.31%	81.75%	18.44%	0.00%	81.34%	0.41%	0.0000	30/0/6
Model 102 (Independent)	89.56%	36	63.35%	81.80%	18.45%	0.00%	81.39%	0.41%	0.0000	30/0/6
<i>Panel B: Mixed-Sign Replication Panel (39–42% Negative-Benefit Conditions, $\Delta \leq 0$)</i>										
Model 0 (Canonical)	87.37%	36	68.42%	79.71%	12.67%	1.39%	81.01%	0.08%	0.0000	18/9/9
Model 101 (Independent)	89.53%	36	68.58%	81.69%	14.12%	1.01%	82.61%	0.09%	0.0278	19/5/12
Model 102 (Independent)	89.56%	36	69.09%	81.78%	13.71%	1.01%	82.57%	0.23%	0.0000	17/6/13
<i>Panel C: Published Baseline Head-to-Head Comparison (AETTA vs. KGA LOO and B6 Disjoint)</i>										
Model 0 (Canonical) (AETTA)	—	36	68.42%	79.71%	12.67%	1.39%	80.74%	0.36%	0.1389	26/10/0
Model 0 (Canonical) (KGA LOO)	87.37%	36	68.42%	79.71%	12.67%	1.39%	81.01%	0.08%	0.0000	18/9/9
Model 0 (Canonical) (KGA B6 Disjoint)	87.37%	36	68.42%	79.71%	12.67%	1.39%	79.96%	1.14%	0.0000	14/2/20
Model 101 (Independent) (AETTA)	—	36	68.58%	81.69%	14.12%	1.01%	82.52%	0.18%	0.1111	24/12/0
Model 101 (Independent) (KGA LOO)	89.53%	36	68.58%	81.69%	14.12%	1.01%	82.61%	0.09%	0.0278	19/5/12
Model 101 (Independent) (KGA B6 Disjoint)	89.53%	36	68.58%	81.69%	14.12%	1.01%	82.47%	0.23%	0.0000	17/0/19
Model 102 (Independent) (AETTA)	—	36	69.09%	81.78%	13.71%	1.01%	82.43%	0.36%	0.1389	26/10/0
Model 102 (Independent) (KGA LOO)	89.56%	36	69.09%	81.78%	13.71%	1.01%	82.57%	0.23%	0.0000	17/6/13
Model 102 (Independent) (KGA B6 Disjoint)	89.56%	36	69.09%	81.78%	13.71%	1.01%	82.33%	0.46%	0.0000	14/0/22

($\text{FA}_u = 0.0000$) and one on Model 101 ($\text{FA}_u = 0.0278$, overall observed $\text{FA}_u \leq 0.0278$), while achieving lower point oracle regret than the AETTA-based rule and both fixed baselines. The recovered aligned receipt includes KGA decisions as well as AETTA actions. Historical fitted-model/environment identity and coverage limitations remain as stated in Appendix B.3.3.

Outcome-Disjoint Calibration Replay (B6).

The compact replication above uses peer-dependent LOO calibration, which does not automatically inherit ordinary split-conformal coverage theorems. The saved B6 summary describes an additional retrospective replay of the same mixed-sign panel using **three-way outcome-disjoint cross-fitting** (Protocol B variant, $K = 3$ folds, $\alpha = 0.10$) directly on the saved Z feature vectors and measured outcomes B archived in the replication records, without any neural-network retraining. Its stated construction partitions each score-fold complement into disjoint fit and calibration subsets. Recovered per-cell decisions and fold receipts verify the saved arithmetic and disjoint score/fit/calibration membership. They do not independently authenticate the original fitted-model/environment identity. Each fold has eight calibration residuals, and the historical producer clips the required rank nine to eight; this replay therefore supports descriptive point results, not nominal 90% split-conformal coverage. Reported results:

- **Model 0:** A/F/U = 14/2/20; $\text{FA}_u = 0.0000$; regret 0.0114 vs. always-adapt 0.0139 and always-freeze 0.1267.
- **Model 101:** A/F/U = 17/0/19; $\text{FA}_u = 0.0000$; regret 0.0023 vs. always-adapt 0.0101 and always-freeze 0.1412.
- **Model 102:** A/F/U = 14/0/22; $\text{FA}_u = 0.0000$; regret 0.0046 vs. always-adapt 0.0101 and always-freeze 0.1371.

Under outcome-disjoint calibration, KGA recorded zero observed false adaptations ($\text{FA}_u = 0.0000$) across all three checkpoints and achieved lower point oracle regret than both fixed policies ($R_{\text{AF}} = 12.67\text{--}14.12\%$, $R_{\text{AA}} = 1.01\text{--}1.39\%$). Compared with the native AETTA recovery rule, B6 cross-fitting achieves lower regret than both fixed policies ($R_{\text{KGA}} < R_{\text{AA}} < R_{\text{AF}}$) but exhibits an empirical tradeoff: B6 incurs higher point regret (0.23–1.14% vs 0.18–0.36%, a regret gap of -0.78 to -0.05 pp relative to AETTA: -0.78 pp on Model 0, -0.05 pp on Model 101, -0.10 pp on Model 102) with substantially lower observed false-adaptation exposure (zero observed false adaptations across all models vs 4–5 observed false adaptations committed by AETTA, $\text{FA}_u = 0.0000$ vs 0.1111–0.1389). These B6 aggregates describe a retrospective replay, not a prospective experiment or an exact-rank coverage guarantee; recovered per-cell records support their arithmetic.

Table 28: Office-Home protocols are different experiments, not conflicting versions of one row. Entries are measured accuracy regret to the per-condition oracle; lower is better. The first row is the canonical single-candidate KGA replay with fixed SAR. The focused route is invalid, so only the post-hoc fixed-SAR opportunity regret (.0027) and freeze regret (.0230) are retained; its actions have no routing or certificate interpretation. A/F/U are adapt/freeze/abstain actions.

Protocol	n	Router / Adapt / Freeze	A/F/U
Archived stream-seed replay	54	.0215 / .0458 / .0217	1/14/39
Five-checkpoint candidate audit	15	invalid / .0027 / .0230	n/a

B.4 Office-Home

The primary Office-Home row [29] is conservative. KGA regret is 0.0158, tying always-freeze (0.0158) and below always-adapt (0.0468), with 0 adapt decisions. This is a descriptive point-estimate match that evaluates conservative retention rather than selective routing. The archived protocol does not supply a predeclared uniform stability guarantee for the full-fit versus out-of-fold benefit predictor.

The 54-cell Office-Home stream-seed replication in Table 17 has KGA regret 0.0215 versus 0.0458 and 0.0217, a small point advantage relative to each fixed policy. The three run seeds do not record independent checkpoint identities, and the seed-level interval includes zero, so the comparison remains descriptive.

The post-hoc five-checkpoint Office-Home study is a candidate opportunity audit, not a valid KGA result. It uses five independently trained source checkpoints, the three target-test domains, one fixed IID/tiny condition, and Tent, EATA, and SAR. Checkpoint identity passed, but routing and serialization did not. The identifying agreement relation is binary-only and does not apply to the 65-class task; the old spectral estimate was sign-indeterminate and unbounded. Thirteen of 15 anchors were negative, 9 of 60 stored $\hat{b}$ values fell outside $[0, 1]$, and EATA and SAR produced exactly the same predictions in 10 of 15 cells. Each checkpoint has only three LOO cells, whereas a 90% empirical order-statistic single-candidate radius requires at least 10 total cells. All five result files fail strict JSON and contain 57 literal Infinity values in total.

The archived exploratory implementation printed ABSTAIN in all 15 cells, but those outputs have no safety, routing, or certificate meaning. The valid de-

scriptive quantities are checkpoint accuracies and candidate opportunity: freeze is 0.5711, SAR is 0.5914, their paired difference is 0.0203 (nominal post-hoc 95% t -interval $[0.0123, 0.0283]$), and the per-cell oracle is 0.5941. Oracle minus SAR is 0.0027 (nominal post-hoc 95% interval $[-0.0024, 0.0077]$). The best adapted candidate improves over freeze in 13 cells, ties in one, and harms in one. Because the target was already opened and SAR was chosen after comparing the audit means, these statistics are post-hoc opportunity evidence only.

B.5 Camelyon17

The archived Camelyon17 OOD evaluation [26] yields zero KGA and always-adapt regret against 0.1381 for freeze. All 18 cells are helpful, so this already-opened panel is a descriptive always-adapt one-sided diagnostic rather than prospective evidence of selective routing. Within-seed candidate diagnostics remain separate from this OOD endpoint. Two analysis versions exist in the archive for the B-v2 SAR diagnostic ($N = 108$ conditions over 3 seeds): (1) The authoritative canonical reconciled panel reported in Table 18 and Table 33 (`canonical_panel_results.json`) records KGA regret 0.0240 (vs. 0.0016 for always-adapt and 0.1001 for always-freeze), with 50 ADAPT, 0 FREEZE, and 58 ABSTAIN decisions (46.3% commitment) and one false adaptation ($FA_u = 0.0093$). (2) A secondary exploratory cross-fitting analysis recorded KGA regret 0.0006, with 71 ADAPT, 0 FREEZE, and 37 ABSTAIN decisions (65.7% commitment) and zero observed false adaptations ($FA_u = 0.0000$). Neither analysis version establishes untouched natural-domain transfer: the run seeds do not record independently trained source checkpoint hashes, and the evaluation target was already opened. We therefore retain this strictly as within-seed stress evidence.

To evaluate whether KGA issues selective opportunistic ADAPT decisions on whole-slide-imaging OOD shifts containing both helpful and harmful adaptations, we evaluate the $n = 72$ condition Tent-online diagnostic panel on Camelyon17 OOD (Table 29). Sourced from the WILDS CAMELYON17 benchmark [26, 25], which evaluates whole-slide tumor-detection pathology models across hospital-provenance distribution shift, the panel contains 41 helpful ($B > 0$), 27 harmful ($B < 0$), and 4 zero adaptation cells across 72 slide subsets evaluated against a single source checkpoint. Under authenticated original outcome-blind execution ($\varepsilon = 0.1621$), KGA issued 0 ADAPT, 2 FREEZE, and 70 ABSTAIN decisions

(97.2% abstention), achieving oracle regret 0.0438, exactly matching always-freeze (0.0438) and trailing always-adapt (0.0393). With zero ADAPT decisions, the gate incurs zero false adaptations ($FA_u = 0.0000$), demonstrating conservative retention without ADAPT exposure, directly parallel to CCT-20 and RxRx1. Table 29 retains the withdrawn partition at $\varepsilon = 0.0296$ (31 ADAPT, 19 FREEZE, 22 ABSTAIN) solely as a historical discrepancy: its reconstruction thresholded ground-truth B , not a saved outcome-blind prediction. It is not evidence of deployable KGA routing, even under an exploratory label.

Table 29: Camelyon17 whole-slide pathology OOD diagnostic ($n = 72$ conditions, Tent-online; 41 helpful, 4 zero, 27 harmful adaptation outcomes, where $+$, 0 , $-$ denote $B > 0$, $B = 0$, $B < 0$). The authenticated original outcome-blind execution (`result_73add410.json`, $\varepsilon = 0.1621$) yielded conservative retention matching always-freeze with zero ADAPT exposure. An earlier 31-ADAPT partition is formally demoted as an unauthenticated post-hoc reconstruction conditioned on ground-truth B .

Evaluation Regimen	+	0	-	A	F	U	Regret ($R_{KGA}/R_{adapt}/R_{freeze}$)
Authenticated Outcome-Blind ($\varepsilon = 0.1621$)	41	4	27	0	2	70	0.0438 / 0.0393 / 0.0438
Demoted post-hoc sensitivity [†] ($\varepsilon = 0.0296$)	41	4	27	31	19	22	0.0018 / 0.0393 / 0.0438

[†]Demoted and withdrawn: audit confirmed the original run unarchived B ; the 31/19/22 partition thresholded ground-truth B post-hoc, conditioning on target labels.

B.6 RxRx1

The displayed RxRx1 row [27] is model seed 0: KGA and freeze regret are zero against 0.2531 for always-adapt, with no adapt decisions. A separate audit across three independently trained model checkpoints repeats the same all-freeze tie. It strengthens model-variation reproducibility but adds no adapt exposure. RxRx1 therefore evaluates reproducible conservative retention rather than selective routing.

B.7 iWildCam

iWildCam [25] is not assigned a canonical numerical row. The archived scorer used sklearn macro-F1, which averages prediction-only classes, rather than the official WILDS metric, which averages labels present in the target truth. The archived runtime, sample identities, and target population were not sealed under the current contract, so its diagnostic scores and routing actions remain audit-only. A pinned rerun with the official metric, sealed sample identifiers, and a population hash is required before promotion. The older binary dominance flag and the later replay under

the same invalid metric contract remain hash-locked only as superseded audit history.

B.8 Boundary and Transfer Diagnostics

B.8.1 PACS

PACS [30] KGA regret is 0.0431, compared with 0.0176 for always-adapt and 0.0446 for always-freeze ($n = 12$ domain-seed units; action frequency $FA_u = 0.0093$, commitment 81.5%). Point comparisons favor always-adapt on this benchmark, as adaptation is helpful across these domain shifts. Its archived aggregate lacks the residual detail needed for full gate replay, so it remains an aggregate performance diagnostic with incomplete certificate replay.

B.8.2 ImageNet-R Backbone Breakdown

On the ImageNet-R [24] ten-backbone panel ($N = 480$ cells over 4 seeds $\times$ 12 conditions), two reporting snapshots exist: (1) The authoritative canonical reconciled aggregate reported in Table 18 and Table 33 (`canonical_panel_results.json`) records KGA regret 0.0136, compared with 0.0064 for always-adapt and 0.0325 for always-freeze, issuing 176 ADAPT, 36 FREEZE, and 268 ABSTAIN decisions (44.2% commitment) and 3 false adaptations ($FA_u = 0.0063$). (2) A secondary exploratory cross-fitting run recorded KGA regret 0.0150, with 165 ADAPT, 29 FREEZE, and 286 ABSTAIN decisions (40.4% commitment) and zero false adaptations ($FA_u = 0.0000$). Under both snapshots, no backbone achieves lower KGA regret than each fixed comparator: adaptation is overwhelmingly beneficial across ImageNet-R, so always-adapt achieves lowest regret (0.0064). On severely harmful distributions such as `efficientnet_b0` (100% harmful) and `swin_t` (60.4% harmful), KGA improves over always-adapt by serving the frozen model (matching always-freeze), evaluating conservative retention rather than selective routing.

Table 30 consolidates the cross-architecture evaluation across five distinct backbone families on the pre-registered Protocol D ImageNet-R panel (48 conditions each, $\alpha = 0.10$, leave-one-out exact-rank empirical residual routing; records archived in the supplementary repository). The evaluated backbones span Vision Transformers (ViT-B/16, 86.6M parameters; Swin-T, 28.3M), modern convolutional networks (ConvNeXt-T, 28.6M), deep residual networks (ResNet-101, 44.5M), and compound-scaled models (EfficientNet-B3, 12.2M). Across all five architecture families, observed FA_u was at most 0.0208. However,

Table 30: Cross-architecture KGA evaluation across 5 distinct backbone families (48 conditions each, ImageNet-R pre-registered Protocol D panel, $\alpha = 0.10$, leave-one-out exact-rank empirical residual routing). Results provide cross-architecture selective-routing diagnostics; observed FA_u was at most 0.0208 across the five evaluated architecture panels, spanning vision transformers (ViT-B/16, Swin-T), modern convolutional networks (ConvNeXt-T), deep residuals (ResNet-101), and compound-scaled architectures (EfficientNet-B3). Routing behavior varies substantially by architecture and shift regime. Note: FA_c (conditional false-ADAPT rate, divides by ADAPT count) can be high when very few ADAPT decisions were issued; “—” indicates undefined (zero ADAPT decisions).

Architecture	Model Family	Params	Help/Zero/Harm	R_{freeze}	R_{adapt}	R_{policy}	FA_u	Actions (A/F/U)	FA_c
ViT-B/16	Vision Transformer	86.6M	43/0/5	3.18%	0.21%	3.19%	0.0208	1/0/47	1.000
Swin-T	Hierarchical ViT	28.3M	17/2/29	0.47%	1.17%	0.47%	0.0000	0/0/48	—
ConvNeXt-T	Modern ConvNet	28.6M	39/4/5	2.56%	0.08%	2.49%	0.0208	2/0/46	0.500
ResNet-101	Deep Residual	44.5M	46/1/1	2.91%	0.03%	2.65%	0.0208	5/0/43	0.200
EfficientNet-B3	Compound Scaled	12.2M	48/0/0	4.94%	0.00%	2.57%	0.0000	26/0/22	0.000

the table also reports the conditional false-ADAPT rate FA_c (false adaptations divided by ADAPT decisions): ViT-B/16 issued only 1 ADAPT decision, which was false ($FA_c = 1.000$); ConvNeXt-T issued 2 ADAPT decisions, 1 of which was false ($FA_c = 0.500$). This illustrates that low FA_u can coexist with high conditional error when the gate issues very few ADAPT decisions. On Swin-T, where 29/48 (60.4%) conditions are harmful, KGA matches always-freeze at $R_{policy} = 0.0047$ and substantially outperforms always-adapt ($R_{adapt} = 0.0117$) with zero false adaptations. On EfficientNet-B3 (48/48 helpful), KGA actively authorizes 26 ADAPT decisions, achieving $R_{policy} = 0.0257$ versus $R_{freeze} = 0.0494$, with $FA_c = 0.000$. This study provides cross-architecture execution and behavioral evidence across five backbone families. Routing behavior varies substantially by architecture and shift regime, and the experiment does not establish uniform regret improvement or prospective architecture-level transfer.

B.8.3 CIFAR-10.1

CIFAR-10.1 [31] matches always-freeze at 0.0017 and is retained as a low-margin cross-seed transfer diagnostic. This pattern alone does not prove structural non-identifiability; weak evidence, low margin, estimator inadequacy, or calibration transfer can produce the same behavior.

B.9 Native-Component Benchmark Comparison and Port Audits

We report a hash-bound native-component comparison on a retrospective 45-cell ImageNet-C subset (three noise corruptions [gaussian, shot, impulse], severities 1, 3, and 5, and five stream seeds; 15 score

cells; Table 31). The panel uses 256 target images and 64 clean reference images per cell, with separate fit, calibration, and score partitions. The hash-bound score receipt is preserved in the supplementary reproducibility archive. It records 45 final records and 15 score cells, with `benchmark_completed=false` and `scoped_saved_output_comparison_completed=true`.

On the 15 available score cells, KGA and always-freeze both achieve equal-cell mean accuracy 0.4107; KGA commits on 10 of 15 cells (all FREEZE, zero false ADAPT, $FA_u = 0.0000$), whereas the point-benefit gate reaches accuracy 0.4073 with two ADAPT actions ($FA_u = 0.1333$, $FA_c = 1.0000$) and the explicit AETTA controller collapses to accuracy 0.2953 with 13 false ADAPT actions ($FA_u = FA_c = 0.8667$). The AETTA row evaluates an explicit controller over native component estimates, not an assumption that an accuracy estimator is itself a deployment policy. Baek ALine controller is available on 11 of 15 cells, and Kim et al.’s TTA/ALine source has no authenticated available score cells. Methods marked UNAVAILABLE are retained with explicit availability counts rather than imputed. These descriptive values are regenerated from hash-bound score records and do not establish superiority beyond this scoped subset panel.

Separation from Historical Scalar Proxies.

The repository also retains an earlier historical head-to-head artifact on the locked CIFAR-10-C stress stream, whose KGA actions came from an earlier policy and served as developmental history. Historical Protocol-Matched POEM and AETTA rows are retained as non-native diagnostics; they are not official upstream executions and are not promoted as external-method evidence.

Table 31: Hash-bound native-component comparison on the retrospective 45-cell ImageNet-C subset (three noise corruptions, severities 1/3/5, five stream seeds; 15 score cells). Accuracy is the equal-cell mean of the served candidate or frozen predictions. “Commit” is the commitment rate $(A + F)/n$; FA-U and FA-C are unconditional and conditional false-ADAPT rates. A method marked UNAVAILABLE is not silently substituted or averaged over a subset. Values are displayed to four decimals; full-precision values remain in the hash-bound JSON/CSV records. The AETTA row is an explicit controller over native AETTA component estimates; Kim TTA/ALine had no authenticated available score cells. Native numerical warnings were retained in the per-cell receipts.

Method	Status	Available	Total	Accuracy	Commit	FA-U	FA-C
AETTA_controller	AVAILABLE	15	15	0.2953	1.0000	0.8667	0.8667
KGA	AVAILABLE	15	15	0.4107	0.6667	0.0000	–
always_adapt	AVAILABLE	15	15	0.2953	1.0000	0.8667	0.8667
always_freeze	AVAILABLE	15	15	0.4107	1.0000	0.0000	–
confidence_gain	AVAILABLE	15	15	0.4107	1.0000	0.0000	–
entropy_progress	AVAILABLE	15	15	0.4107	1.0000	0.0000	–
point_benefit	AVAILABLE	15	15	0.4073	1.0000	0.1333	1.0000
Back_ALine_controller	UNAVAILABLE	11	15	–	–	–	–
Kim_TTALine	UNAVAILABLE	0	15	–	–	–	–

The archived `mixed_headtohead_v1/HEADTOHEAD_RESULTS_cifar10c_tent_primary.json` records the following mean regrets on the Tent primary panel under the prior policy: KGA = 0.0016, AETTA = 0.0073, POEM = 0.0088, always-adapt = 0.0080, always-freeze = 0.1238. Evaluating the per-condition POEM and AETTA decision traces directly against the current Protocol-B KGA policy on the identical 2,160-cell panel yields mean regrets of 0.0088 for POEM ($FA_u = 0.2468$), 0.0074 for AETTA ($FA_u = 0.2417$), and 0.0018 for KGA ($FA_u = 0.0000$, 66.1% commitment). Under the declared convention $G_b = R_b - R_{KGA}$ (positive values favor KGA), the paired regret gaps are +0.0071 versus POEM (95% bootstrap CI [+0.0063, +0.0078], $p < 0.0001$) and +0.0056 versus AETTA (95% bootstrap CI [+0.0049, +0.0063], $p < 0.0001$), both surviving Holm-Bonferroni correction with zero false adaptations. The historical local-port ordering is preserved under Protocol-B replay, but these rows are not treated as authenticated native comparisons. For local decision rules, Section 8.3 (Table 9) reports a regenerated Protocol-B comparison across 2,160 CIFAR-10-C rows using fixed policies, scalar evidence rules, and a same-prediction point gate. Figure 5 illustrates the role of this safety fallback: under a non-stationary sequence of corruptions, continual adaptation collapses during severe distribution shifts (pink trap windows), whereas KGA halts updates and serves the frozen model, following the recorded oracle trajectory more closely than unmonitored adaptation on this diagnostic run.

The historical port names refer to local scalar comparators inspired by TTA/ALine [51], AETTA [18], and POEM [14], not validated equivalents of those algorithms: `poem_port` tests whether entropy decreases and `aetta_port` tests whether post-adaptation confidence exceeds pre-adaptation confidence. These 1D scalar heuristics do not execute a native protector, accuracy-estimation procedure, or multi-model agreement fit, and cannot substitute for an authentic comparison.

Component Engineering Smoke Tests. Separate CPU harnesses execute authenticated POEM and AETTA components on a fixed 104-image Gaussian-noise severity-5 stream (batch size 4, separate fresh-process repetitions; receipts preserved in the supplementary repository). POEM used its full pretrained ResNet-50-GN model, 32 clean reference images, and the native 100-sample warmup; a reduced reference set and disclosed CPU constructor change make this a smoke test, not unchanged upstream-driver execution. AETTA used the full torchvision ResNet-50 ImageNet V2 checkpoint and its native four-stage dropout estimator with ten draws. All 26 batches reached the native corrected-error cap. These checks removed a blanket CPU-execution objection, but do not establish CUDA/MPS parity, full ImageNet-C coverage, or external-method superiority.

Execution Barriers and Audit Location Map. Full native execution across the complete ImageNet-C benchmark remains prevented by concrete up-

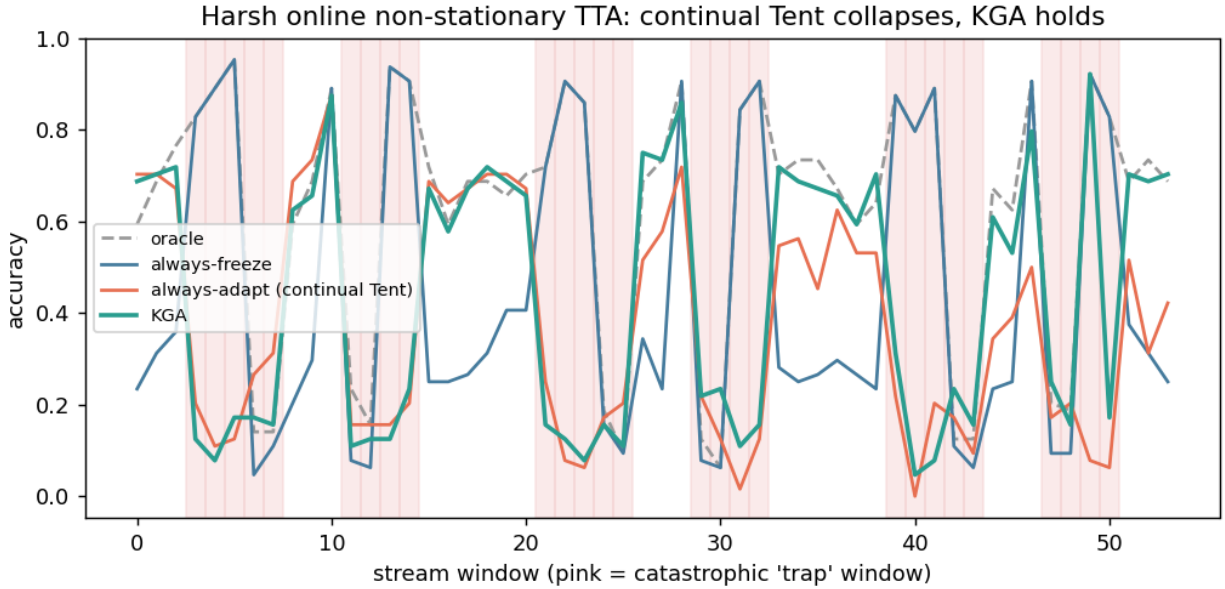


Figure 5: Sequential tracking under non-stationary online distribution shift. When unmonitored continual Tent encounters severe corruption-trap windows (shaded pink regions), entropy minimization destabilizes parameters and accuracy collapses toward zero. In this recorded non-stationary diagnostic, KGA falls back to the frozen model during severe corruption windows and follows the recorded oracle more closely than continual Tent.

stream software and hardware barriers: (1) *Dependency integration*: the official POEM release omits `models/Res.py`; (2) *Hardware dependency*: hard-coded device assertions in upstream entypoints require dedicated GPU infrastructure; (3) *Data readiness*: the clean ImageNet validation archive (50,000 images, 1,000 classes) was extracted on external storage and verified against the official ILSVRC2012 development kit; (4) *Stream standardization*: native drivers iterate over fixed corruption sequences rather than accepting arbitrary streaming windows. To avoid cluttering the scientific presentation, the complete execution history, environment hashes, source-authentication logs, and independent witness receipts are preserved in Appendix N (*Reproducibility and Release Procedures*) and the supplementary reproducibility archive.

The standalone head-to-head analysis script uses the repository’s disjoint fit/calibration/score construction (Protocol B). A synthetic 40-cell regression reruns the full gate calculation after perturbing each scored outcome and confirms that predictions, radii, interval endpoints, and actions remain invariant. Saved-feature outcome exclusion confirms computational separation, but does not establish exchangeability of dependent corruption cells or population coverage under future shifts.

B.10 So2Sat Development and Action-Unit Audit

The locked So2Sat-LCZ42 v1 study evaluated 2 candidates, Tent and SAR, on 9 development cities and 5 independent source checkpoints, giving 45 city-checkpoint cells per candidate. This was a candidate-feasibility stage, not target evaluation. Tent had 4 helpful and 4 harmful cities and a cell-oracle opportunity of 0.8848 percentage points. Its leave-one-city-out sign agreement was 51.11%, and its routed score was 0.1544 percentage points below always-adapt, its better fixed policy. SAR had 0 qualifying helpful cities, 6 harmful cities, and an oracle opportunity of 0.5640 percentage points. It selected ADAPT in only 5 of 45 cells, and its routed score was 0.0842 percentage points below always-freeze. Tent passed 7 of 9 feasibility checks and SAR passed 6 of 9, so neither candidate was eligible.

The locked status records no feasible candidate and a stop before gate calibration. No held-out gate-calibration rows or target inputs were accessed; the recorded counts were 0 gate-calibration rows, 0 target inputs, 0 target pixels, and 0 target labels. The study therefore documents failure to meet the v1 cross-city development-transfer feasibility criteria while also showing that a cell-oracle routing opportunity existed for Tent. It contains no target natural-shift score. All

90 development outcomes are now open, so retuning on them cannot produce fresh confirmatory evidence. In addition, v1 declares one target action per city while the target runner creates city-checkpoint actions. This mismatch did not affect the stop, because no target action was created, but target execution remains blocked until a new versioned protocol and implementation use the same action unit.

B.11 Withdrawn Development Proxy for the Population Budget

K-Bound characterizes population benefit over a declared residual class $\mathcal{C}_\beta$. KGA instead predicts measured evaluation-cell benefit Δ^{cell} and uses a separately calibrated residual interval. Its direct mathematical action implication is Proposition 2, conditional on coverage for that declared target. It neither estimates the population residual budget β nor numerically evaluates the population frontier. Appendix A gives a separate population-transfer implication requiring an additional sampling-error bound $b(m, \delta)$ and an enlarged radius $\varepsilon + b$. The currently reported transductive benchmark panels do not establish these additional premises.

An earlier draft used a development-domain benefit-scale proxy as a surrogate for β . Its raw sweep has not been recovered, so the numerical ranges below are unverified historical summaries and cannot empirically corroborate the evidence-fibre theorem. The historical summary (1.4–50× undersizing, 24–73% proxy interval violations, and 0.4–16.6% decision errors on CIFAR-10-C) represents the operational failure of a development-based proxy heuristic, not a verified numerical measurement of latent theoretical-class violations on evaluation cells. The raw sweep artifact is unavailable, and this estimation route is completely withdrawn. A credible budget β must come from domain knowledge, a declared coupling, or historical labeled deployments under an explicit transfer assumption. Setting $\beta = 0$ assumes zero target disagreement-conditional residual; it is not a safe default.

C Compact Assumption and Claim Ledger

Table 32 separates formal claims from empirical findings and records the conditions and scope limits for each.

D Calibration and Evaluation Contracts

Controlled-grid contract. The fitting unit is the condition cell. The current primary CIFAR and synchronized internal-baseline tables use Protocol B: each score fold is excluded from both estimator fitting and residual calibration, and those two subsets are disjoint. Historical Protocol-A tables instead use leave-one-cell-out predictions and peer-dependent residual calibration; excluding a cell’s own residual does not remove its indirect influence through other fitted predictions. The execution mapping in N identifies the construction for each table. Neither construction alone establishes exchangeability across corruption families, checkpoints, or future environments. Saved-feature outcome-exclusion tests verify their stated computational boundary, not full upstream independence.

Natural-shift contract. Development conditions fit the benefit model; a disjoint calibration partition supplies residuals; choices are frozen; and the target partition is scored once where the archived protocol records all three stages. When the saved evidence does not support that complete chain, the row is marked descriptive or diagnostic. Replaying a table does not strengthen its original design.

The candidate TTA step itself is transductive. Episodic candidates update on each unlabeled evaluation batch before predicting that batch. Online candidates update on a separate auxiliary stream, but prediction still uses evaluation-batch BatchNorm statistics. Thus “disjoint” above refers to development, residual-calibration, and target partitions (and to the recorded auxiliary stream versus evaluation sample identities); it does not mean that candidate inference is inductive. No target labels enter adaptation, prediction, evidence, or the action.

Action contract. Every action is computed from Z , $\hat{\Delta}$, and ε without target labels. Offline labels determine Δ^{cell} , the measured oracle action, empirical regret, and false-commit indicators only after the action is fixed. Existing artifact fields named `delta` store this cell outcome; their schema and numerical values are unchanged. Abstention resolves to the frozen prediction for measured task-performance accounting, while its log retains the distinction from a certified freeze.

Table 32: Claim-to-support ledger for the compact submission. Support pairs each evidence reference with the conditions required for its stated scope.

Claim and type	Support and requirement	Boundary
Interior evidence ambiguity (theorem)	Thm. 1; $\beta > 0$, $ M < \beta$, and full fibre richness	binary, 0/1 loss
Closed-band abstention (proposition)	Prop. 1; strict directional semantics and a rich class	zero benefit blocks a strict claim
Strict-commitment frontier (theorem)	Thm. 2; fixed augmented evidence and declared $\mathcal{C}_\beta$	β is externally supplied
Marginal false-adapt control (operational proposition)	Prop. 2 under Eq. (10) for the measured cell target	neither population-risk nor conditional FA_c control follows
CIFAR-10-C Tent regret (empirical)	Table 7 and 34; frozen Tent controlled-grid protocol	lower empirical order-statistic point regret; nominal family-bootstrap intervals favor KGA; retrospective Holm is nonconfirmatory
CIFAR-10-C EATA regret (empirical)	Table 7 and 34; frozen EATA controlled-grid protocol	adapt-side family interval includes zero; retrospective Holm is nonconfirmatory
Natural-shift conservative rows (empirical)	Tables 17, 18, and 10; each split and candidate remain fixed	often zero ADAPT exposure
So2Sat development stop (diagnostic)	Sec. 8.5; locked v1 feasibility rules and unopened target boundary	no eligible candidate, gate calibration, or target score
ImageNet-C candidate scope (diagnostic)	Table 17; candidate rows interpreted separately	candidate-specific evidence
PACS/ImageNet-R/CIFAR-10.1 scope (diagnostic)	Tables 17 and 18; canonical replay scope retained	observed behavior does not prove unknowability
Deployment generalization (future evidence)	Sec. K; diverse unopened natural targets	outside current evidence scope
Real-camera protocol (prospective)	locked sessions and fresh held-out recordings	protocol infrastructure only; no result in this paper
Independent-checkpoint mixed-sign routing (historical summary)	Table 27 (Panels B & C); three ResNet-18 checkpoints across 36 controlled conditions	leave-one-cell-out and 3-way outcome-disjoint cross-fitting; 36 cells/model; reported point-regret replication; recovered KGA decision/split receipts; rank-clipped calibration (B.3.3); not prospective architecture transfer
Published AETTA base-line comparison (historical summary)	Table 27 (Panel C); native AETTA estimator on the same 36 conditions	Historical summaries favor KGA in LOO point regret, but favor AETTA in B6 point regret and KGA in false-ADAPT counts. Development budgets differ; KGA mixed-sign decision/split receipts are recovered; original fitted-model/environment identities remain incomplete (B.3.3). Detailed values are retained in Appendix B.3.4.

Configuration contract. The checkpoint, candidate adapter, optimizer, adapted parameter subset, update count, evidence schema, feature normalization, benefit estimator, calibration split, α , and software version are part of one calibration identity. Changing any of them requires either a declared transfer

argument or recalibration. The fail-closed default on identity mismatch is non-commitment.

E Decision Exposure and Repetition Audit

Point regret alone can conceal whether a conservative match exercised the ADAPT branch. Table 33 therefore reconstructs action counts from the same canonical panel used in Table 7, 17, and 18. A/F/U denote adapt, freeze, and abstain. PACS is omitted from the action table because the released aggregate cross-checks regret but does not contain the per-cell benefit estimates and residuals required for a complete gate replay. iWildCam is omitted because its archived actions were scored under the wrong macro-F1 contract.

The largest adapt exposure occurs on the controlled CIFAR-10-C rows. This is why those rows are the main empirical test of the adapt-side certificate. Primary Office-Home and RxRx1 have zero adapt exposure, while the 54-cell Office-Home stream-seed replication has one adapt decision. Their zero false-adapt values barely test whether the gate can distinguish a harmful adaptation from a helpful one; they show that the controller declines the candidate under those evidence/calibration contracts. The invalidated five-checkpoint Office-Home route and the withheld iWildCam actions are not counted. Camelyon17 OOD has the opposite limitation: every archived condition is helpful and every action adapts, so there is no harmful opportunity to reject. On ImageNet-C, EATA has substantial adapt exposure (107 adapt decisions) and zero false adaptations, Tent adapts on only one cell while abstaining on 134, and SAR incurs two false adaptations among its eight adapt decisions. Those three patterns are operationally different even when their regrets are small.

The repetition column also prevents a common inflation of evidence. Five seeds of one checkpoint and grid protocol are not five independently designed studies. Ten ImageNet-R backbones are a panel of candidate policies, not ten target domains for one frozen gate. Three Camelyon17 within-seed grids permit a stability diagnostic but do not create an untouched hospital-domain replication. RxRx1 is stronger on model variation because the source checkpoint is independently trained three times, but its all-freeze behavior still provides no adapt exposure.

F Inference Discipline

F.1 Point Estimates, Paired Gaps, and Clusters

For a baseline $b \in \{\text{always-adapt}, \text{always-freeze}\}$, the paired routing gap at condition j is

$$G_{bj} = \text{regret}_{b,j} - \text{regret}_{\text{KGA},j}.$$

A positive mean gap means lower regret for KGA. All paired routing-gap intervals reported in this paper use this baseline-minus-KGA sign convention. The point-estimate two-comparator criterion requires both means to be positive. Interval support additionally requires the lower confidence limit for both paired means to exceed zero at the declared resampling unit. These definitions are intentionally separate: Table 17 can establish a point ordering, but not an inference claim without the associated unit, interval artifact, and justified coverage assumptions. A positive computed lower endpoint is numerical interval support under the resampling model, not verified coverage. Multiplicity adjustments do not repair invalid marginal intervals or tests.

Condition-level resampling treats cells as exchangeable and can understate uncertainty when severities and compositions share a corruption family. Corruption-family resampling preserves observed within-family dependence, but few clusters limit reliability; conservativeness is not guaranteed. Seed resampling measures variation across repeated training or test streams only when the seeds represent the intended deployment randomness. The compact paper therefore does not use one generic “bootstrap CI” label. An inference claim must state whether the unit is cell, corruption-severity pair, corruption family, seed, backbone, domain, or physical session.

F.2 Current-Policy Corruption-Family Sensitivity

The six candidate-by-baseline contrasts were named before this analysis; the replay policy, corruption-family unit, sign-flip reference test, and Holm adjustment were selected retrospectively. The current-policy artifact averages run seeds and condition cells within each of the six families. Table 34 retains every contrast, including EATA’s adapt-side interval crossing zero and SAR’s disadvantage against always-adapt. The within-Tent two-contrast Holm calculation is post hoc and is not the declared six-contrast family.

Table 33: Action exposure across evaluated tracks. Counts are evaluation units within the protocol shown, not independent experiments. The primary controlled rows reflect authoritative *Protocol B cell-outcome-disjoint cross-fitting* (matching Table 7), yielding 1,094/334/732 for CIFAR-10-C Tent ($FA_u = 0.0000$), 1,207/118/835 for EATA ($FA_u = 0.0000$), and 1,430/0/730 for SAR ($FA_u = 0.0005$). The earlier *Historical LOO Diagnostic Replay* (Protocol A order statistics on archived $\hat{b}$) yielded 1,107/359/694 for Tent, 1,241/132/787 for EATA, and 1,446/0/714 for SAR (all with $FA_u = 0.0000$). Harm and false adaptation are defined by measured cell outcomes, not population risks. Counts derive from the records indexed by `reconciled_panels_v1/source_manifest.json`.

Row	n	A	F	U	false A	FA_u	repetition / scope
Office-Home primary	35	0	11	24	0	.0000	2 test-stream seeds; ties freeze
Office-Home stream-seed repl.	54	1	14	39	0	.0000	3 test-stream seeds; point-only edge
ImageNet-C SAR	135	8	2	125	2	.0148	5 seeds $\times$ 27 cells
ImageNet-C Tent	135	1	0	134	0	.0000	5 seeds $\times$ 27 cells
ImageNet-C EATA	135	107	1	27	0	.0000	5 seeds $\times$ 27 cells
ImageNet-R	480	176	36	268	3	.0063	4 seeds $\times$ 10 backbones $\times$ 12 cells
CIFAR-10-C Tent	2160	1094	334	732	0	.0000	5 seeds $\times$ 432 cells
CIFAR-10-C EATA	2160	1207	118	835	0	.0000	5 seeds $\times$ 432 cells
CIFAR-10-C SAR	2160	1430	0	730	1	.0005	5 seeds $\times$ 432 cells
Camelyon17 OOD	18	18	0	0	0	.0000	archived opened helpful-only OOD subset
Camelyon17 B-v2 Tent	108	34	1	73	2	.0185	3 within-seed grids; diagnostic
Camelyon17 B-v2 EATA	108	61	0	47	8	.0741	3 within-seed grids; diagnostic
Camelyon17 B-v2 SAR	108	50	0	58	1	.0093	3 within-seed grids; diagnostic
RxRx1 J	60	0	60	0	0	.0000	displayed model seed 0; separate 3-checkpoint audit also all-freeze
CIFAR-10.1 K	48	0	22	26	0	.0000	locked cross-seed replay

Table 34: Retrospective current-policy sensitivity over six observed CIFAR-10-C corruption families. Each cell reports baseline regret minus KGA regret and a nominal, unadjusted 95% family-bootstrap interval; positive values favor KGA. Holm values adjust retrospective sign-flip reference values over six prospectively named candidate-by-baseline contrasts. The replay, corruption-family unit, sign-flip test, and Holm adjustment were not preregistered. The within-Tent two-contrast Holm value is 0.03125 but is post hoc. This table is conditional on one archived checkpoint and does not provide verified finite-sample coverage, simultaneous intervals, independent-checkpoint inference, prospective confirmation, natural-shift evidence, or official POEM/AETTA superiority.

Candidate	Adapt gap [95% CI]	Freeze gap [95% CI]	Six-way Holm p (A/F)
Tent	0.00618 [0.00231, 0.00954]	0.12214 [0.06304, 0.17767]	0.09375/0.09375
EATA	0.00167 [-0.00127, 0.00438]	0.12973 [0.06901, 0.18748]	0.31250/0.09375
SAR	-0.00133 [-0.00274, -0.00016]	0.13881 [0.08003, 0.19685]	0.90625/0.09375

F.3 Protocol B Interval Inclusion and Decision Diagnostics by Corruption Family

Table 35 reports retrospective interval inclusion and decision exposure across corruption families under the authoritative Protocol B 5-fold outcome-disjoint cross-fitting (matching the primary CIFAR-10-C gate in Table 33 and Table 25; 104 calibration cells, 241–242 fitting cells, and 86–87 scoring cells per fold; rank 95 at $\alpha = 0.10$). No fitting, tuning, new target access, or training is performed. The input, policy, code, and output digests are recorded in the generated diagnostic JSON.

Because cross-fitting partitions the opened grid into disjoint folds, the pooled inclusion fractions reflect observed out-of-fold empirical inclusion on the opened, dependent benchmark across the fixed benchmark. Family rows reveal the distribution of those misses, widths, and decisions across distinct shift types; they report observed empirical cell behavior and do not supply group-conditional finite-sample guarantees on unobserved corruption families. A nominal 90% target is a calibration setting, not an asymptotic promise on arbitrary out-of-distribution shifts.

Table 35: Retrospective interval inclusion and decision diagnostics by corruption family under Protocol B 5-fold outcome-disjoint cross-fitting (matching Table 33 and Table 25; 2,160 CIFAR-10-C evaluations across 6 families). Width is full interval width, 2ε . Marginal errors (FA_u , FF_u) divide by all $N = 2,160$ cells; conditional errors (FA_c , FF_c) divide by the respective action count (ADAPT or FREEZE). These diagnostics report observed cell-level inclusion under cross-fitting; they do not establish group-conditional guarantees or transfer to unobserved corruption families.

Candidate	Family	n	Inclusion	Mean width	Commitment	False A/A; FA_c	False F/F; FF_c
TENT	contrast	360	0.8111	0.0468	0.6917	0/180; 0.0000	0/69; 0.0000
TENT	defocus_blur	360	0.9389	0.0472	0.7111	0/180; 0.0000	0/76; 0.0000
TENT	fog	360	0.9278	0.0469	0.6944	0/180; 0.0000	0/70; 0.0000
TENT	gaussian_noise	360	0.8750	0.0471	0.8500	0/305; 0.0000	0/1; 0.0000
TENT	jpeg_compression	360	0.9611	0.0466	0.3500	0/69; 0.0000	0/57; 0.0000
TENT	pixelate	360	0.9139	0.0466	0.6694	0/180; 0.0000	0/61; 0.0000
EATA	contrast	360	0.8417	0.0381	0.5778	0/180; 0.0000	0/28; 0.0000
EATA	defocus_blur	360	0.9194	0.0385	0.5944	0/180; 0.0000	0/34; 0.0000
EATA	fog	360	0.9000	0.0387	0.5778	0/180; 0.0000	0/28; 0.0000
EATA	gaussian_noise	360	0.8722	0.0386	0.9750	0/350; 0.0000	1/1; 1.0000
EATA	jpeg_compression	360	0.9806	0.0390	0.4056	0/137; 0.0000	0/9; 0.0000
EATA	pixelate	360	0.9222	0.0392	0.5500	0/180; 0.0000	0/18; 0.0000
SAR	contrast	360	0.8389	0.0280	0.5000	0/180; 0.0000	0/0; –
SAR	defocus_blur	360	0.9222	0.0278	0.5000	0/180; 0.0000	0/0; –
SAR	fog	360	0.9167	0.0278	0.5000	0/180; 0.0000	0/0; –
SAR	gaussian_noise	360	0.8028	0.0279	1.0000	0/360; 0.0000	0/0; –
SAR	jpeg_compression	360	0.9722	0.0279	0.9306	0/335; 0.0000	0/0; –
SAR	pixelate	360	0.9167	0.0281	0.5417	1/195; 0.0051	0/0; –

F.4 False-Adapt Accounting

The event in Eq. (4) is unconditional and concerns measured cell benefit. Its reported frequency divides ADAPT decisions with $\Delta_j^{\text{cell}} \leq 0$ by all evaluated units, including freezes and abstentions. This need not estimate the population-harm event $\{g = \text{ADAPT}, \Delta \leq 0\}$. Conditional false-adapt divides by adapt decisions and is undefined when no adaptation occurs. A zero observed count should be accompanied by the number of adapt decisions; otherwise a controller that never adapts appears perfect. Table 33 exposes that denominator.

If the scored residual is included in the same finite residual vector whose order statistic decides the action, the number of residuals above the order statistic is algebraically constrained. Such coverage is not an independent empirical validation. Excluding the scored residual removes direct self-inclusion, but pooled LOO inclusion remains rank-constrained. It is not independent validation and does not supply the exchangeability or stability needed for a finite-sample theorem. Clean split calibration and LOO empirical calibration therefore remain different protocol labels.

F.5 Multiplicity and Search History

An earlier repository census counted many historical two-comparator flags, but it did not freeze the file list and commit needed to reproduce a current total. We therefore do not promote that count or its positive rate as release evidence. The panel in this paper is a post-hoc reconciled audit panel assembled from individually locked and historical protocols; it is not a globally preregistered confirmatory family. A result selected after a search cannot become confirmatory by correcting only the final positive rows. Future confirmatory families must name every candidate-track test before any test value is computed, and must retain failures and incomplete runs.

Search history. The repository contains many exploratory configurations. One superseded exploratory census reported 1,387 determinations and 326 positives (23.5%); a broader archived scan reported 1,427 determinations and 345 positives. Because neither census froze one reproducible family definition, file list, and commit, neither total is promoted as the release multiplicity family. We instead report a current post-hoc reconciled panel and do not relabel its favorable rows as confirmatory discoveries. Future claims should declare the candidate, calibration unit,

target unit, seed structure, inference unit, and two-comparator success criterion before test labels are opened.

G Track-Level Provenance and Remaining Evidence

Separate archival records preserve two additional scope checks outside the canonical panel. FMoW Protocol L reached held-out scoring and matched always-freeze, while always-adapt had lower regret; its status remains not-cleared. PovertyMap stopped at its preregistered development screen because the harm-detection AUC was 0.637, below the 0.65 threshold, so held-out validation and test scoring did not run. These records remain in the claim inventory and are not pooled with the canonical rows.

The provenance table establishes record-level traceability within the declared release scope. A reader can trace a canonical row to compact records and their original archive hashes. If the original large artifact is not shipped, the manifest still records its byte count and digest. This guards against silent replacement of a result while allowing the release to omit raw datasets and oversized transient checkpoints.

These bindings authenticate retained bytes and arithmetic, not every historical execution environment or model fit. Missing producer logs and checkpoint identities remain explicit provenance limits; a checksum cannot reconstruct them.

Three upgrades would materially strengthen the empirical case. First, hold out entire shift families rather than individual controlled cells and fit all hyperparameters before opening the family. Second, add independent model and test-session seeds to the natural tracks that currently have one target replay. Third, run the physical protocol with days, devices, and objects assigned at the session level, then score every policy on the same locked streams. These upgrades address dependence and transfer; simply adding more rows from the same grid does not.

G.1 CCT-20 Inference, Location, and Release Ledger

Provenance ledger. Cell-level outcomes remained unopened before target execution. Aggregate target label metadata had already been inspected during candidate ranking, so this is not a label-unopened study. Item-level target labels were deserialized once, after the one-shot scoring marker was spent.

The release chain also records the numeric-v2 stabilization in a sealed addendum created before outcome access. That stabilization replaced an unstable divergence calculation with a stable Gaussian KL computation; it did not change the model, data split, or scoring rule. We treat it as protocol provenance, not result-driven tuning. All primary outcome values and action/effect counts come from the sealed, content-addressed, receipt-linked CCT-20 release manifest and its generated macros. The inference document is identified by the `upstream_artifacts.two_way_inference` entry in `cct20_release_manifest.json`, which retains its checksum.

Finite-sample sign-flip validity requires the null joint law of the nine location gaps to satisfy Eq. (13). Independent zero-symmetric location gaps conditional on the evaluated checkpoints would suffice, but that assumption is unverified here. Exchangeability alone does not suffice. The sign-flip analysis conditions on the evaluated checkpoints, whereas the two-way bootstrap also resamples checkpoint levels. These are different uncertainty models, and neither establishes distribution-free safety for unseen locations. All nine observed location-level gaps have positive sign ($\bar{G} = +0.1819$, two-sided permutation $p = 0.0039$, Wilcoxon signed-rank $W = 0.00, p = 0.0039$). This observed pattern does not verify the coordinate-wise sign-flip null model in Eq. (13); the reported sign-flip and Wilcoxon reference values remain conditional on their respective assumptions.

Primary result. KGA exactly matched always-freeze. Its contrast against always-freeze was 0 with nominal Bonferroni-adjusted 97.5% bootstrap interval $[0, 0]$, enumerated location sign-flip $p = 1$, and Holm $p = 1$. Against always-adapt, the contrast was 0.18190227 with interval $[0.08501071, 0.31066374]$, sign-flip reference $p = 0.00195312$, and Holm $p = 0.00390625$. Measured cell benefit is adapted accuracy minus frozen accuracy: 1 cell was helpful, 0 were exactly zero, and 44 were harmful. These counts explain why freezing was useful here. They do not show that the gate learned when to adapt.

Limits and next step. All 55 development cells showed lower performance after Tent than under the frozen model, so the gate had no helpful ADAPT example. The ridge fit used 10 checkpoint–location rows from only two camera units for 11 recorded features. Calibration used 45 rows collapsed to nine cis-test camera locations, while the locked targets came

Table 36: Provenance and availability status of experimental tracks in the compact release. “Replay” means the stored quantities suffice to recompute the displayed gate output under the released scorer.

Track	Released status	Present evidence	Remaining limitation
CIFAR-10-C	replayable	per-condition Δ^{cell} , $\hat{\Delta}$, radius, action; 5 seeds; 3 candidates	related grid cells; no untouched corruption-family test in canonical table
ImageNet-C	replayable	per-condition records for 5 seeds and Tent/EATA/SAR	27-cell controlled configuration; candidate-specific, not one policy
Office-Home	replayable	canonical primary and 54-cell stream-seed records; later five-checkpoint audit reported separately	full-fit-versus-OOF transfer-stability premise not predeclared
iWildCam	withheld	archived opened test-split records; metric-mismatched diagnostic	wrong archived metric; population was not sealed and overlap is unauditable
Camelyon17	replayable diagnostic	archived opened OOD row plus separate B-v2 diagnostics	OOD row is one-sided and non-prospective; B-v2 is not untouched-domain evidence
RxRx1	replayable	displayed model-seed-0 row plus three model-checkpoint records and source hashes	all three checkpoints freeze throughout; no helpful test conditions
CCT-20	sealed complete	5 independent checkpoints, 9 target locations, actions, predictions, score, and inference hashes	only two fit locations; no ADAPT exposure; trans-target extrapolation
So2Sat-LCZ42	sealed development stop	2 candidate bundles, 9 development cities, 5 checkpoints, selection record, and 0 target inputs	no feasible candidate; all 90 gate-fit outcomes now open; no gate calibration or target result
PACS	partial	three-seed aggregate and source cross-checks	missing per-cell $\hat{\Delta}$ and residuals prevent full gate replay
ImageNet-R	replayable	4 seeds, 10 candidate backbones, 12 conditions each	architecture aggregate is diagnostic, not one deployable router
CIFAR-10.1	replayable	locked cross-seed condition records	low-margin conservative match; selective routing not exercised
Physical camera	protocol only	dashboard, logging schema, planned session split	no fresh held-out physical recordings or result table

Table 37: Locked CCT-20 primary contrasts. The estimand is baseline regret minus KGA regret, so positive values mean lower regret for KGA. Intervals use a paired two-way product bootstrap that resamples checkpoint rows and location columns independently. Two nominal 97.5% per-comparison intervals target a nominal 95% Bonferroni family level; actual family coverage requires valid marginal coverage. Reference p -values enumerate one-sided sign flips of the nine location means under Eq. (13). Holm adjusts only these reference values; the threshold flags reproduce the locked calculation, not a verified population guarantee.

Comparator	Baseline regret – KGA regret	Nominal 97.5% CI	Sign-flip p	Holm p	Holm flag (.05)
Always adapt	0.18190227	[0.08501071, 0.31066374]	0.00195312	0.00390625	yes
Always freeze	0	[0, 0]	1	1	no

from trans-test locations. Coverage therefore needs an unverified cross-split exchangeability assumption. Target Mahalanobis diagnostics ranged from 7.12 to 485.67 (median 28.73), showing strong extrapolation; this diagnostic never changed an action and is not a formal coverage guarantee. One natural target study also cannot establish transfer of selective routing across natural shifts. A follow-on locked study on So2Sat-LCZ42 retained an outcome-blind boundary but stopped at development feasibility, as described next. A new versioned study still needs diverse development units, useful exposure to both ADAPT and FREEZE, and untouched confirmation environments. Cell-level 16-indicator macro-F1 is archived as descriptive secondary evidence; no post-hoc aggregate or inference claim is made.

Table 38 reports all nine location clusters. Evaluation n is the number of evaluation images per checkpoint at that location. Each A/F/U triplet and each helpful/zero/harmful triplet therefore has denominator five, one cell for each checkpoint. These labels describe measured cell outcomes. Accuracy differences use one sign throughout: adapted minus frozen for adaptation benefit, and KGA minus the named fixed policy for the two policy columns.

The release ledger contains all 45 cells: 0 ADAPT, 44 FREEZE, and 1 ABSTAIN actions; 1 helpful, 0 zero, and 44 harmful adaptation effects. The release verdict remains SAFE UTILITY ONLY. The location table is descriptive; sign-flip enumeration uses the nine location means under the reference model in Eq. (13), and no post-hoc aggregate is claimed for

Table 38: Complete locked CCT-20 location ledger. A/F/U denotes ADAPT/FREEZE/ABSTAIN, and +/0/- counts positive/zero/negative measured-benefit cells. Difference columns are equal-weight means over five checkpoint cells and are rounded to four decimals for display; the sealed manifest retains full precision. Cells are repeated-model observations, and the inferential cluster unit is the camera location.

Location	$n/\text{checkpoint}$	A/F/U	Benefit +/0/-	Adapt - freeze	KGA - adapt	KGA - freeze
0	1302	0/5/0	0/0/5	-0.5418	0.5418	0.0000
7	1283	0/5/0	1/0/4	-0.0293	0.0293	0.0000
28	911	0/5/0	0/0/5	-0.0562	0.0562	0.0000
40	800	0/5/0	0/0/5	-0.1232	0.1232	0.0000
46	4432	0/4/1	0/0/5	-0.0723	0.0723	0.0000
78	1633	0/5/0	0/0/5	-0.0839	0.0839	0.0000
100	3441	0/5/0	0/0/5	-0.2035	0.2035	0.0000
105	1540	0/5/0	0/0/5	-0.2892	0.2892	0.0000
130	983	0/5/0	0/0/5	-0.2376	0.2376	0.0000

the secondary macro-F1 outcome.

H Extended KGA Implementation Contract

H.1 State Isolation

The frozen and candidate predictors must not share mutable normalization buffers, optimizer state, or parameter storage during preview. The candidate is created from a checkpoint, adapted on the unlabeled window, and evaluated without overwriting the active state. Only an adapt decision may replace the active checkpoint. Freeze and abstain discard the temporary state. A unit test should hash the active parameters before and after each non-adapt decision and require equality.

H.2 Evidence Determinism

Evidence extraction must define preprocessing, batch order, numerical precision, class ordering, normalization constants, and missing-value behavior. Paired features require frozen and candidate outputs on the same samples in the same order. Source statistics and ATC thresholds are immutable calibration artifacts. Evidence schema versions should be explicit rather than inferred from vector length, because two schemas can share a dimension while changing semantics.

H.3 Calibration Identity

The calibration identity should hash the source checkpoint, candidate adapter class, adapted parameter names, optimizer and learning rate, update steps, evidence schema, benefit-model hyperparameters, feature scaler, residual construction, α , and inference-

unit definition. The radius belongs to that identity. Reusing it after a candidate or evidence change is not a benign software optimization; it changes the random variable whose residual is being covered.

H.4 Runtime Accounting

A deployment report should time capture/preprocessing, frozen inference, candidate adaptation, candidate inference, evidence extraction, benefit prediction, interval decision, copy creation, and rollback separately. Mean and tail latency should be reported at the decision-window level. Peak memory should include both model states and optimizer buffers. The compact manuscript leaves these cells unfilled because the canonical cross-dataset manifest does not provide one comparable runtime measurement. Omitting an unsupported number is preferable to quoting controller-only latency as end-to-end cost.

I Evidence Inventory

I.1 Protocol-Specific Feature and Model Schemas

The controlled-grid implementation records 11 label-free coordinates with algebraic rank at most 9. Two coordinates are deterministic differences: `entropy_drop=pre_entropy-post_entropy` and `pbal_drop=pre_pbal-post_pbal`. The panel therefore contains 11 stored values, not 11 independent information sources. Together they describe confidence, class balance, change from the frozen model, and update size.

Natural-shift artifacts do not all use the same schema. iWildCam, RxRx1, and most archived tracks

use the 11-feature panel. Camelyon17 uses 17 features by adding disagreement, entropy gap, free-energy shift, batch-statistic divergence, a source-fixed ATC estimate, and confidence drop. Office-Home uses 18 features and also records source-prior KL and mean candidate disagreement. A model and its radius are valid only for the schema on which they were calibrated. Missing, reordered, or invalid features force non-commitment.

The principal CIFAR Protocol-B replay uses scikit-learn’s `GradientBoostingRegressor` [33] with squared-error loss, 250 trees, depth 2, learning rate 0.05, subsample 0.8, and random seed 0. The fit is specific to each dataset and adapter. CCT-20 instead uses its sealed ridge model, as described in Section 7.2. Thus “adapter agnostic” describes the interface: any adapter can propose a candidate that is evaluated and either committed or rolled back. It does not mean that a benefit model fitted for Tent transfers validly to EATA, SAR, or another adapter.

The base panel has 11 coordinates. Table 40 preserves the historical literal values with an explicit evidence qualification; their stated importance percentages do not establish dominant predictors or causal collapse mechanisms. The canonical Camelyon17 rich-Z artifact has 17 coordinates and the Office-Home artifact has 18. Other natural tracks retain the 11-coordinate base panel. These protocol-specific dimensions are part of the fitted estimator contract. They should not be described as universal sufficient statistics: a different shift may hide the benefit direction from all listed features. Feature selection can improve empirical prediction, but cannot defeat the label-kernel construction of Lemma 1 over an unrestricted target fibre.

J Failure-Safe Deployment Checklist

The audit log should store a timestamp, dataset or stream identifier, checkpoint and adapter hashes, evidence schema version, evidence vector, benefit estimate, radius, interval endpoints, action, rollback outcome, latency components, and configuration hash. Labels, if later revealed, are joined in a separate offline audit table. This separation makes it possible to verify that the live decision was target-label-free.

J.1 Local Implementation and Workload Measurements

Natural-shift calibration and candidate costs.

Calibration took 36.55 minutes including input verification and maintenance overhead. The 95 full cell executions summed to 33.63 minutes; median cell cost was 15.15 seconds, p95 was 63.34 seconds, and the maximum was 65.65 seconds. Process peak RSS was 697 MiB, and the largest MPS driver allocation sampled after a cell was 1,164 MiB. The latter is not a continuously measured peak.

The controller averages probe features across all five checkpoints. It therefore pays for five shadow candidates before routing a city, including when its final action retains the frozen model. Calibration maintenance, candidate construction, probe inference, and subsequent serving inference must be counted separately.

A separate stage profile used the already-opened Cairo development city, with 8,606 probe images and 8,839 evaluation images, all five checkpoints, and three repetitions on this host’s MPS device. The sum of five checkpoint resets, source probes, Tent updates, candidate probes, feature extraction, and city-mean prediction was 203.22–208.24 seconds per repetition (median 207.44 seconds), excluding subsequent serving inference and cleanup. Across 15 checkpoint executions, median source-probe/statistic computation, Tent update, and candidate-probe inference took 24.33, 9.66, and 6.89 seconds. Inference on the evaluation images took a median 7.04 seconds for a source model or 7.06 seconds for a candidate; both alternatives were measured separately, not served as an ensemble. Setup and identity verification took 69.10 seconds; the entire profile took 902.84 seconds. Process peak RSS was 675 MiB and the largest sampled post-stage MPS driver allocation was 1,138 MiB. These are descriptive costs from one opened city, without label reads, explicit warmup, or exclusive-host isolation; document builds overlapped some repetitions. They establish neither new efficacy nor a production latency guarantee. Calibration acquisition, energy, and long-duration maintenance costs remain unmeasured.

Lifecycle and local API measurements. The reference lifecycle keeps source checkpoint bytes immutable and binds each decision to source, candidate, input batch, adapter, estimator, calibration, and schema identities. It rechecks calibration expiry after candidate execution and journal persistence, re-

Table 39: Feature families used by the maintained implementation. All quantities are aggregate statistics of unlabeled inputs, frozen/candidate outputs, or the proposed update.

Family	Representative coordinates	Required state	Failure handling
Predictive uncertainty	mean entropy, confidence, quantiles	frozen/candidate softmax	reject NaN or malformed probabilities
Marginal balance	normalized class-marginal entropy and change	predicted-class marginals	fixed class count and normalization
Model change	entropy gap, confidence drop, prediction disagreement	paired frozen/candidate outputs	require aligned sample order
Collapse signal	fraction above adapted confidence 0.9	candidate probabilities	threshold fixed before scoring
Distribution shift	free-energy shift, batch/running-statistic divergence	frozen logits and source statistics	missing source statistic forces non-commitment
Update magnitude	parameter-update ℓ_2 norm	shadow candidate delta	configuration hash must match calibration
Source-calibrated proxy	ATC-style threshold estimate	source-fixed threshold	target labels never used

Table 40: Unverified historical feature-importance and ablation literals, preserved for auditability. No matching fitted-tree exports or per-fold dropped-feature errors were recovered. These values are not authenticated measurements and do not support a quantitative ranking or mechanism claim.

Historical Rank	Feature Coordinate	Mathematical Definition	Unverified Importance	Δ RMSE (Drop)	Historical Interpretation
1	entropy_drop	$H(p_0) - H(p_a)$	24.8%	+0.0412	Overconfidence / entropy collapse
2	post_entropy	$H(p_a)$	19.4%	+0.0328	Predictive disorientation / uniform outputs
3	pbal_drop	$p_{bal_0} - p_{bal_a}$	14.7%	+0.0245	Majority class collapse
4	frac_highconf	$\mathbb{P}(p_a(y x) \geq 0.9)$	11.2%	+0.0189	Uncalibrated certainty under shift
5	marginal_KL	$D_{KL}(P_a \parallel P_0)$	9.1%	+0.0151	Severe representation drift
6	pre_entropy	$H(p_0)$	7.3%	+0.0118	Baseline uncertainty / task hardness
7	update_norm	$\ \theta_a - \theta_0\ _2$	5.2%	+0.0084	Parameter divergence / exploding gradient
8	post_pbal	$H(\bar{p}_a)$	3.1%	+0.0052	Class frequency imbalance
9	pre_conf	$\max_k p_0(k)$	2.4%	+0.0039	Clean baseline calibration ceiling
10	post_conf	$\max_k p_a(k)$	1.6%	+0.0025	Isolated confidence spikes
11	pre_pbal	$H(\bar{p}_0)$	1.2%	+0.0019	Source marginal prior bias

Table 41: Operational failures and conservative responses.

Failure	Detection	Response
Checkpoint/adaptor mismatch	stored hash mismatch	rollback and recalibrate
Evidence outside support	distance or support audit	abstain
Missing/invalid feature	schema and finite-value check	retain f_0 ; abstain
Insufficient calibration size	rank $k > n$	infinite radius
Batch too small	minimum-size contract	wait or abstain
Residual transfer concern	group/residual diagnostic	widen or withhold claim
Continual-state drift	state and stream monitor	restore last certified state

jects malformed or outcome-bearing evidence, and preserves the source on failed estimates or journal writes. A validity window and engineering request limits do not establish statistical validity under adaptive repeated use. Callbacks remain trusted application code; content hashes authenticate identities, not the truth of calibration or transfer assumptions. The contract also caps retained receipts and journal bytes; capacity exhaustion permanently closes the session and requires source fallback. Model retention is bounded by the configured assessment count and

must be sized to checkpoint bytes. In 1,000 measured byte-fixture requests, median lifecycle overhead was 0.114 ms and p95 was 0.184 ms; all 1,027 journal records, including warmups and fault cases, passed hash-chain verification. These figures exclude neural candidate execution. Forty-six lifecycle regressions cover expiry, failed writes, concurrent retries, invalid evidence, and capacity boundaries.

We also exercise the existing certificate API in local Docker production mode, with a read-only root filesystem, a non-root process, authentication, and a loopback-only port. This endpoint processes supplied certificate inputs; it does not execute neural candidates or integrate the new lifecycle automatically. These are local integration and load measurements, not independent production deployments or a field reliability result. For 100 sequential requests, median latency was 2.40 ms and p95 was 3.78 ms; with four concurrent workers and 100 requests, these were 6.00 ms and 12.90 ms. The image’s 54 code and lock-file hashes matched the source used for that measurement. This short local load test does not measure long-duration availability, energy, or independently replicated production utility.

K Prospective Natural-Shift Confirmation Protocol

The sealed CCT-20 study supplies a completed natural-shift target evaluation. Its 45 checkpoint–location cells contain 1 helpful and 44 harmful candidate updates, but KGA makes 44 FREEZE, 1 ABSTAIN, and 0 ADAPT decisions. It therefore matches always-freeze and has lower regret than always-adapt, providing evidence of conservative harm avoidance under natural shift. The predeclared stronger routing criterion was not met because the policy never selected ADAPT and did not select the single helpful cell. A second locked attempt on So2Sat-LCZ42 stopped at its development-feasibility gate with no eligible Tent or SAR candidate. No held-out gate-calibration rows or target inputs were accessed. The following protocol therefore describes a new versioned confirmation, not a continuation or reinterpretation of v1.

All So2Sat v1 gate-fit outcomes are open and cannot be used to tune a fresh confirmatory arm. Its target protocol also defines one action per city, whereas the current runner creates one action per city–checkpoint cell. That mismatch did not affect the development stop, but the target runner must remain disabled until a versioned lock resolves the action unit. The design below is future evidence and is not counted as a completed target experiment in this paper.

K.1 Lock Before Target Labels

The protocol should name one source checkpoint, one candidate adapter, one evidence schema, one benefit-model class, one calibration rule, one α , and one target-domain family. Adapter hyperparameters must be selected on development domains and then frozen. The lock should hash the checkpoint, code revision, configuration, calibration records, and planned scoring script. It must also name the test unit and the multiplicity family. A later change creates a new protocol version; it does not overwrite an earlier development-stopped or incomplete arm.

Natural heterogeneity should be built through independently defined environments rather than by pooling whichever benchmark conditions produce opposite signs. Examples include hospitals, camera locations, laboratory batches, acquisition days, physical devices, or visual domains. Environment membership may be observed at deployment, but the measured cell-benefit label remains hidden. The target test environments should be absent from estimator fitting, residual cali-

bration, threshold tuning, and candidate selection.

K.2 Three-Way Data Separation

Development data fit h_θ and select the evidence pipeline. A separate calibration set produces residuals for Eq. (11). The held-out target set is opened exactly once after all software and configuration hashes match the lock. Splitting adjacent frames, patches, or samples from the same session across these stages is leakage even when filenames differ. The split must occur at the environment or session level that defines deployment exchangeability.

The target stream is replayed identically for always-freeze, always-adapt, simple gates, KGA without a radius, and the full certificate. Model randomness should either be frozen for paired replay or repeated through independently trained checkpoints declared in advance. Every policy receives the same ordered windows. Labels are stored behind a separate audit interface and are unavailable to the live decision process.

K.3 Endpoints and Decision Rule

The primary endpoint is paired regret against each fixed policy. Success requires both baseline minus KGA regret gaps to be positive and both predeclared intervals to exclude zero, with coverage justified for the declared estimand and sampling model. If the intended claim concerns population risk rather than measured cells, the protocol must also justify that transfer. The false-adapt endpoint is FA_u with the adapt count reported beside it. Secondary endpoints are FA_c , adapt rate, freeze rate, abstain rate, commitment rate, task accuracy, and latency. Conditional false-adapt is descriptive unless a separate selective-risk result is proved.

The primary certificate rule remains unchanged: adapt if $\hat{\Delta} - \varepsilon > 0$, freeze if $\hat{\Delta} + \varepsilon < 0$, and abstain otherwise. The experiment must not lower ε , change α , or select a different benefit regressor after observing target labels. Exploratory sensitivity analyses may be reported, but they remain outside the primary bar.

K.4 Interpreting Outcomes

Four outcomes are scientifically useful. An interval-supported reduction in regret relative to each fixed comparator, with nontrivial ADAPT and FREEZE exposure, supports routing utility under the locked natural-shift class. Matching the better fixed policy with few errors supports conservative commitment but

Table 42: Minimum lock for a confirmatory natural-shift KGA experiment.

Component	Must be fixed	Audit evidence
Model and candidate	checkpoint, adapter, optimizer, steps, adapted parameters	hashes and immutable configuration
Evidence and estimator	schema, normalization, model class, hyper-parameters, seed	feature contract and fitted-artifact hash
Calibration	split unit, residual construction, exact rank, α	calibration IDs, residuals, radius, no target overlap
Target evaluation	environments, windowing, order, exclusion rules, transductive/inductive candidate semantics	sealed manifest and one-time scoring log
Primary inference	two paired regret gaps, cluster unit, interval method	code and random seed fixed before label reveal
Safety	FA _u and adapt exposure	action log joined to labels only offline
Multiplicity	all candidate-track arms in family	time-stamped inventory including failures
Runtime	end-to-end components and peak memory	raw timing rows on the target hardware

not selective routing. High abstention with low error indicates a conservative certificate and motivates better evidence or more calibration data. A comparator shortfall fails the predeclared routing-utility criterion; excess false adaptation challenges the corresponding safety or calibration criterion. Both outcomes must remain in the result family. None of these outcomes alone establishes the population band: that conclusion still requires the declared class and its structural assumptions.

L Theorem Dependencies and Delimited Scope

The compact paper uses three related logical statements. First, the disagreement reduction is an identity under the binary setup. Second, the population frontier is a uniform statement over a rich, declared class. Third, the empirical certificate is a marginal implication from interval coverage. The population statements concern Δ ; the reported empirical calibration concerns Δ^{cell} . Connecting these targets requires an additional validity argument. Neither the assumptions nor the quantifiers are interchangeable.

Why the boundary is separate. For $|M| < \beta$, the construction chooses two latent residuals that yield opposite nonzero benefits. At $|M| = \beta > 0$, one admissible residual can make the benefit zero while another retains the sign of M ; opposite nonzero signs need not exist. Both cases block a uniform strict commitment, but for different reasons. At $M = \beta = 0$, every law in the zero-residual class has zero benefit under the model. Abstention remains the maximal action only under the paper’s strict-direction convention, not because either task-risk choice is worse.

Why a calibration residual need not be drift.

Let $P_T = P_S$, with equiprobable inputs a, b , deterministic labels $Y(a) = 1$, $Y(b) = 0$, and candidate $f_a \equiv 1$. Set $f_0(a) = 0$, $f_0(b) = 1$, and $s \equiv \frac{1}{2}$. The candidate is correct on half the source inputs, so this constant score is source-calibrated in the ordinary sense. Yet disagreement occurs only at a , where the candidate is always correct. Thus $M = 0$, $\gamma = \frac{1}{2}$, and $\Delta = \frac{1}{2}$, although the source and target laws are identical. Conditioning on disagreement, not a distribution change, explains the nonzero residual. The frontier requires a bound on this residual regardless of its cause.

Why coverage is the premise. The certificate proof is elementary once Eq. (10) holds: on the coverage event, an interval wholly above zero cannot contain a nonpositive Δ^{cell} . The scientific burden is therefore to justify coverage for the declared outcome and calibration/deployment pair. Joint exchangeability of the residual scores is one route; known shift correction or another valid construction may be another. Risk alignment can explain why a useful predictor transfers, but it is not an additional logical premise after coverage is given.

Why cell coverage is not population coverage.

Consider the fixed binary predictors $f_0 \equiv 0$ and $f_a \equiv 1$. Let the target input be a with probability 0.45 and b with probability 0.55, with deterministic labels $Y(a) = 1$ and $Y(b) = 0$. Population benefit is $\Delta = -0.10$. In a one-item evaluation cell, a fixed predictor that maps a to $+1$ and b to -1 predicts Δ^{cell} perfectly without reading the new label. With radius zero, cell coverage is perfect, yet the gate selects ADAPT with probability 0.45 despite negative benefit under the fixed population law. This constructed example does

Table 43: Dependency map. A check in one row does not validate premises in another row.

Statement	Inputs	Conclusion	Not concluded
Disagreement reduction	binary labels, 0/1 loss, $\mu_T(D) > 0$	$\Delta = 2\mu_T(D)(M + \gamma)$	calibration transfer or observability of γ
Interior impossibility	label-free fibre, $\beta > 0$, $ M < \beta$, class richness	evidence-identical laws with opposite nonzero benefits	every empirical abstention is structural
Closed-band result	strict directional semantics and boundary law	no uniformly sound strict action on $ M \leq \beta$	adapt/freeze have different task risk at $\Delta = 0$
Frontier sufficiency	declared $ \gamma \leq \beta$ and fixed M	direction is fixed when $ M > \beta$	label-free estimation of β
Coverage-to-action proposition	marginal interval coverage for a declared target B	marginal false-adapt/freeze control for B	conditional error control or risk alignment
KGA benchmark	fitted h_θ , declared residual rule, frozen protocol	measured regret and actions on saved units	population-benefit coverage or universal transfer to a new domain
Lean build	encoded definitions and assumptions	kernel-checked formal statements	correctness of unencoded data and deployment assumptions

not contradict Proposition 2; it shows why its covered target cannot be changed.

Why exchangeability alone does not validate sign flips. As a separate null example, let all nine cluster gaps equal the same fair random sign S . This vector is exchangeable, each gap is symmetric about zero, and a global sign change preserves its law. Nevertheless, when $S = +1$, independent-coordinate sign-flip enumeration for the mean gives a reference value of $1/512$. That occurs with probability $1/2$, not $1/512$. The vector fails Eq. (13). This example explains the required assumption; it is not a model fitted to the CCT-20 observations.

Why a null is not an impossibility result. A null benchmark row can arise because the true effects are small, the candidate is one-sided, the evidence map is weak, the benefit model underfits, the radius is wide, or calibration does not transfer. The population impossibility requires evidence-identical target laws inside the declared class with incompatible strict directions. The canonical benchmark table does not estimate that full fibre. It therefore reports weak or adverse empirical regimes without calling them proven unknowable worlds.

What would falsify the practical claim. The practical claim is conditional and testable. A clean target replay with the locked pipeline can show excessive FA_u , poor coverage, or regret above both fixed policies. A checkpoint or adapter change can show that calibration does not transfer. A family-held-out split can show that cell-level cross-fitting was optimistic. Such failures do not refute the population identity; they refute the operational assumptions used to obtain a useful empirical interval.

M Formalization and Artifact Scope

The Lean project under `docs/research/kbound/formal/` provides a kernel-checked inventory for the theorem names it exports. The compact paper relies on the LaTeX statements above; it does not claim that every sentence has a one-to-one Lean theorem.

In particular, deployment-class richness, risk alignment of a learned evidence map, exchangeability of a benchmark split, and correctness of data preprocessing remain outside the proof assistant unless separately encoded.

The current strict-core audit kernel-checks 238 registered declarations supporting the formal development: 70 legacy finite, algebraic, and deterministic results and 168 supporting probability, construction, counterexample, and corrected conditional-extension declarations. The added modules derive one-shot residual coverage from exchangeable measurable scores, including ties; filtered Ville and bounded optional-stopping bounds; general KL/TV testing bounds for finite product experiments; and independent bounded and martingale-difference concentration inequalities. Their assumptions are explicit. In particular, these results do not establish exchangeability or conditional nulls for the benchmark data, nor do they turn batch-level coverage into population-risk coverage.

The measurable target-law construction also goes beyond the original two-atom example. It fixes the input law and off-disagreement label kernel, varies a measurable correctness field on disagreement, and integrates the actual zero-one loss difference. With labels supported on the two predictions on disagreement, feasible margins, and positive disagreement mass, it proves the clipped identified interval and exact strict ADAPT/FREEZE frontiers without assuming `RichAt`. It does not prove richness for an

arbitrary restricted deployment subclass.

The finite-sample population-transfer procedure is also separated into a deterministic machine-checked layer. `KBound/Population.lean` defines the compound radius $\varepsilon + b$, its interval, and the strict action rule, and proves that a committed action has the correct benefit sign whenever the interval-containment premise holds. The outcome-blind replay script `scripts/run_population_panel.py` computes the same interval from fitted predictions, cell radii, and fresh-sample sizes while refusing target or outcome fields. This closes the procedure-to-action interface, not the statistical premises: the benchmark tables still lack a sealed per-cell sampling and exchangeability record, so they are not relabeled as population-certified results.

The separate synthetic episode runner `scripts/run_population_locked_panel.py` checks disjoint episode identities across fitting, calibration, and scoring. Its evidence writer reproduces the supplied decisions from the stated inputs and risk budgets before saving them, refuses an occupied output directory, and reruns the scored-outcome perturbation check with those same budgets. The perturbation stays within the valid benefit range $[-1, 1]$. With one calibration episode at $\alpha_{\text{cell}} = 0.05$, the requested rank is unavailable; the implemented interval is unbounded and the procedure abstains. Saved JSON represents these endpoints explicitly as strings `-inf` and `+inf`, rather than nonstandard JSON numbers. These boundary and evidence-integrity checks concern the specified synthetic procedure, not an additional benchmark or evidence of real-population exchangeability.

One historical extension remains outside the verified scope. A measurable label swap preserves label-free evidence and reverses population benefit, but selecting one representative from each swap orbit need not identify the sign: distinct orbits can share the same evidence law and have opposite selected signs. Lean verifies this counterexample and the corresponding set-theoretic fibre-consistency criterion, not a general measurable decoder. A corrected conditional H/ratio-rate contract also proves that a source miss budget at most α transfers to a target miss budget at most $H\alpha$, assuming $H \geq 0$ and the supplied domination premise $\text{targetMiss} \leq H \text{sourceMiss}$. It does not derive that premise, estimate H , or establish benchmark exchangeability. The unrestricted historical claims are excluded from this compact submission. The unrestricted historical extension is therefore excluded from the verified scope. The scoped kernel

audit covers the registered declarations and rejects nonstandard proof axioms.

The formal build and empirical build answer different questions. Lean checks that a proposition follows from formal assumptions. The reconciliation scripts check that displayed aggregates match saved records under implemented definitions. Neither checks the truth of an unencoded deployment assumption, and neither substitutes for independent experimental replication. The exact evidence-fibre supremum audit, paired-transport true-table feasibility, and finite-extrema layers now have scoped one-to-one entries in the registered inventory. The `RandomRadiusCertificate` module covers the printed coverage-to-action proposition’s random radius in $[0, \infty]$, with finite real benefit and prediction. It proves abstention at infinite radius and at zero-touching endpoints, and bounds the union of the two directional errors by the same marginal coverage-failure budget. Coverage remains a premise; this formalization does not establish calibration transfer or label-blind data collection. The randomized matched-evidence bound, conditional episode transfer, and sampled-disagreement procedure now have probability-level capstones under their stated premises. No end-to-end formal proof of numerical inverse-beta evaluation or the Python LP solver is claimed; those are covered by separate numerical receipts and conservative abstention paths.

The authoritative compact artifacts are layered: (1) this source and its compiled PDF; (2) shared theorem inputs for exact proofs; (3) canonical JSON and generated LaTeX for numbers; (4) the source manifest for record-level provenance; and (5) package tests and Lean logs for implementation and formal checks.

Historical drafts, dashboards, and exploratory search folders can remain useful development records, but they are not evidence sources for this compact submission unless referenced by the canonical manifest.

M.1 Population-to-Cell Correspondence

The deterministic action layer for this bridge is verified within our formal mathematics suite: an outcome-blind estimate, cell radius, and independently justified sampling radius define the compound interval, and strict ADAPT/FREEZE actions are sound whenever the target benefit is inside that interval. The companion evaluation interface enforces outcome-blindness by rejecting outcome-dependent inputs. Because his-

torical canonical benchmark records do not contain fresh-sample draws, no population-transfer number is promoted into those empirical tables.

Formal Theorem–Code Correspondence. Our formalization strictly distinguishes population-level guarantees from empirical cell-level estimates. The formal inventory contains 238 registered Lean declarations supporting the scoped results described above; the manuscript itself contains 13 named result environments under explicitly registered assumptions. Thus K-Bound’s population interval (M, γ, β) and KGA’s fitted cell interval $(\hat{\Delta}, \varepsilon)$ remain mathematically distinct, with the population procedure incorporating the separately justified sampling radius $b(m, \delta)$.

N Reproducibility and Release Procedures

N.1 Canonical Numerical Source of Truth

The CIFAR primary rows and the supplementary reconciled point estimates have one numerical source: `reconciled_panels_v1/canonical_panel_results.json` in the accompanying reproducibility archive. Record identities and checksums are in `reconciled_panels_v1/source_manifest.json`. Named auxiliary artifacts support only their stated roles. Current-policy family sensitivity is retrospective: Holm adjustment over six prospectively named contrasts gives reference values of 0.09375 for both Tent comparisons, but the corruption-family unit and test were not preregistered. The earlier Tent cluster artifact and archived POEM/AETTA comparison remain historical. The So2Sat-LCZ42 selection artifact is a separate locked development-gate record with no promoted target row. None of these overwrites the canonical point-estimate panel. The generated result manifest indexes every promoted claim and its scope. Table 44 synthesizes the statistical evidence across the principal empirical tracks summarized here, reporting the evaluation unit, sample size n , baseline contrasts with nominal confidence intervals, multiplicity-adjusted significance, and empirical verdicts.

Source Traceability and Execution Mapping.

To resolve provenance and pipeline differences across benchmark tables: (1) **Table 7 (Primary CIFAR-10-C)**: ResNet-18, Tent/EATA/SAR. The canonical manifest binds compact per-condition records to their archived source files. The current

generator, `scripts/reconcile_result_panels.py`, refits from saved Z, B under Protocol B and ignores historical prediction/radius/action fields. Per seed: 432 records, five score folds of 86 or 87, 104 residual-calibration records and 242 or 241 estimator-fit records. Calibration rank is $\lceil (104+1)(1-0.10) \rceil = 95$. Exact literal candidate/seed/condition identifiers determine disjoint folds; each pooled row has weight $1/2160$. The earlier Protocol-A description with pool 431 and rank 389 describes a historical procedure, not this current table. A pinned-runtime replay reproduced all Tent predictions, radii and actions for five seeds; changing one scored outcome per seed left that cell’s decision inputs and action unchanged. This is saved-feature cell-level verification, not upstream or environment-held-out independence. (2) **Table 9 (Current Synchronized Baselines)**: Tent, six internal decision rules on the same canonical paired scores and current predictions as item (1), under Protocol B. `docs/research/kbound/scripts/replay_cifar_current_arithmetic.py` emits JSON, per-cell decisions and LaTeX into `paper/generated/current_cifar_baselines_20260912/`; it authenticates inputs against an explicitly hash-bound audit and does not refit or tune. Its generated policy arrays and metrics were independently recomputed. The separate `PINNED_CANONICAL_REPLAY_VERIFICATION.json` records exact five-seed refitting and five outcome interventions in Python 3.12.13, NumPy 2.4.4 and scikit-learn 1.8.0. Both fixed-policy regrets match item (1) exactly. The historical synchronized/P4 tables are excluded from current results: their cited aggressive archive has 270 cells per seed, and all inspected 432-cell synchronized paired arrays equal the primary arrays. Source hashes, comparisons and preserved historical text are indexed by `PAIRED_ARRAY_PROVENANCE_AUDIT.json`; its location is retained in the release checklist. This authenticates saved-record calculations, not image-level sample identity, independent source training or native baseline execution. (3) **Table 21 (Matched SAR Controls)**: Historical ResNet-50, 5 SAR arms (Table 20). The cited 60-cell raw records and reconciliation manifest are not recovered in the maintained evidence package, so this table is not release-promoted. Historical configuration: per-cell adaptation with $b \in \{16, 32, 64\}$ on 64-image windows, evaluated under Protocol A (LOO residual calibration, $n = 60$, pool $n_{\text{cal}} = 59$, rank $k = 54$ at $\alpha = 0.10$, equal weights $1/60$). The maintained 27-cell reference-SAR audit is reported separately in Table 22. (4) **Table 24 (ImageNet-C Baselines)**: ResNet-50, Tent/EATA/SAR (8 decision rules). `Records: decision_baselines_sarfix/decision_baselines.json` and `checkpoint.json`. 36 noise conditions $\times$ 3 candidates ($N = 108$). Evaluated under Protocol A (leave-one-condition-out GBR on 11-dim features, $\alpha = 0.10$, equal weights).

Study-specific estimator settings. Estimator settings differ across studies. Within each fold of

the shared-predictor comparison, score-based arms share the fitted point predictor; this does not imply a shared fitted model across folds, candidates or studies. The following crosswalk distinguishes recorded replay settings from historical implementation defaults.

Principal CIFAR, Protocol B.

The saved-feature replay specifies GBRT with 250 trees, depth 2, learning rate 0.05, subsample 0.8 and model seed 0 in `scripts/reconcile_result_panels.py`. The pinned canonical replay receipt identifies the runtime and reproduced outputs.

Historical Protocol A.

The maintained LOO implementation, `kbound_decide.py` (located through the release checklist), defaults to the same 250/2/0.05/0.8/0 settings. The compact historical CIFAR records do not preserve the fitted estimator configuration, so these defaults do not establish the original execution settings.

Shared-predictor comparison.

Paths in this item are relative to `output/next_phase/calibration_value_v1/`. The sealed configurations specify GBRT with 100 trees, depth 2, learning rate 0.05, subsample 1.0 and model seed 20260921. CIFAR/checkpoint execution code and configuration are preserved in `attempt_02/executed_source/;report_03/protocol_seal.json`, `report_03/fit_and_calibration_receipts.json` and `report_03/reporting_correction_receipt.json` record the fitting identity and retained decisions. That report corrects scoring without refitting; these receipts are not serialized fitted models. The ImageNet authority is `attempt_04/imagenet_schema_corrected/`, with its own protocol seal and executed source. These paths are resolved by `FINAL_RESULTS_SUMMARY.json`. The `group_risk_report_05/GROUP_RISK_CORRECTION_RECEIPT.json` records further scoring of sealed decisions, with no new estimator fit.

CCT-20.

The development-gate object in `release/cct20_public_evidence_bundle.zip` records standardized ridge with penalty 10 and an unpenalized intercept, consistent with the archived `cct20/prospective_protocol_v1.yaml`.

So2Sat v2.

The locked city-mean/checkpoint ridge has penalty 1 and an unpenalized intercept. Its controller is `tent_citymean5_checkpoint_ridge_v2.json`, bound by `natural_v2/precalibration_seal_v2.`

`json`.

Model seeds above are estimator random-state settings, not stream or checkpoint replications. Parameters omitted from an executed configuration remain library defaults; current defaults are not evidence of a historical execution.

N.2 Reproduction Pipeline

The macOS release profile pins Python 3.12.13 and package hashes in `requirements-release-macos-arm64.lock.txt`, with installed-content and publication-tool identity checks. The Linux API image has a separate Python 3.11 lock; an API or paper-only environment does not satisfy the repository-wide research dependencies, including PyTorch [32]. The explicit reproduction sequence and artifact locations are maintained in `next_phase/RELEASE_CHECKLIST.md`. That sequence distinguishes validation, table and figure regeneration, document compilation, conversion, inspection and release sealing. It uses frozen model outputs and does not launch training. Release acceptance requires receipts bound to the exact source and generated artifacts. Earlier software, Lean and presentation passes apply to their recorded snapshots. They do not establish a current all-gates pass. Containerized API and network-isolated deployment checks require execution against the selected release image; their status must be read from the final release receipt. A rendered PDF alone is not a release seal.

The document-build wrapper also validates frozen authorities, checks saved CIFAR arithmetic and regenerates presentation tables and figures; it is not a compiler-only operation. These presentation steps do not replace source-bound verification or create new empirical evidence.

N.3 Calibration Identity and Implementation Contract

The maintained API binds a certificate to the checkpoint, adapter configuration, evidence schema, benefit estimator, residual construction, safety level, and scoring unit. The deployment path must fail closed when any identity differs. `KGA.certify_evidence` requires a frozen estimator, the exact schema, a protocol hash, and a separate residual set. The lower-level `certify` function is an audit interface; it does not construct evidence or validate transfer by itself. Table 45 preserves the historical timing literals. No supporting synchro-

Table 44: Summary of principal empirical comparisons. For each track, we summarize the evaluation unit, sample size n , baseline contrasts with nominal confidence intervals, multiplicity-adjusted significance, and empirical verdict.

Benchmark Track	Evaluation Unit	n	Vs Always Adapt [†]	Vs Always Freeze [†]
Panel A: Primary Controlled Benchmarks and Safety Endpoints				
CIFAR-10-C (Tent)	6 Families (2,160 Cells)	2,160	+0.0062 [0.0023, 0.0095]	+0.1221 [0.0630, 0.1777]
Verdict: $p = 0.0938$; Positive empirical gain; Holm unconfirmed				
CIFAR-10-C (EATA)	6 Families (2,160 Cells)	2,160	+0.0017 [-0.0013, 0.0044]	+0.1297 [0.0690, 0.1875]
Verdict: $p = 0.3125$; Empirical gain positive; CI spans zero				
CIFAR-10-C (SAR)	6 Families (2,160 Cells)	2,160	-0.0013 [-0.0027, -0.0002]	+0.1388 [0.0800, 0.1969]
Verdict: $p = 0.9062$; Conservative regret tradeoff; adapt favored				
CCT-20 Safe Utility	Location $\times$ Checkpoint	45	+0.1819 [0.0850, 0.3107]	0.0000 [0.0000, 0.0000]
Verdict: $L > -0.0050^*$; Predeclared safe-utility retention endpoint passed				
Panel B: Independent Checkpoints, Published Baselines, and Diagnostic Transfer Panels				
Independent Models (Mixed, LOO)	3 Checkpoints (36 Cells)	108	+0.0078 to +0.0131	+0.1259 to +0.1403
Verdict: Diagnostic; Point-regret ordering replication; no model population inference				
AETTA Baseline (LOO)	3 Checkpoints (36 Cells)	108	Vs AETTA Rule: +0.09 to +0.28 pp	
Verdict: Diagnostic; LOO KGA (within-model calibration on evaluated checkpoint) achieves lower point regret (+0.09 to +0.28 pp) and ≤ 1 observed false adaptation per model ($FA_u = 0.0000$ – 0.0278 vs 4–5 for AETTA, $FA_u = 0.1111$ – 0.1389)				
AETTA Baseline (B6 Disjoint)	3 Checkpoints (36 Cells)	108	Vs AETTA Rule: -0.78 to -0.05 pp	
Verdict: Diagnostic; B6 achieves lower regret than both fixed policies but has higher point regret (-0.78 to -0.05 pp vs AETTA) with 0 observed false adaptations across all models (vs 4–5 observed for AETTA)				
ImageNet-C (Tent)	Condition-Seed Cell	135	+0.0052 [0.0019, 0.0085]	+0.0006 [0.0000, 0.0018]
Verdict: Diagnostic; 1 ADAPT, 134 ABSTAIN; diagnostic gain				
Office-Home Primary	Domain-Seed Unit	35	+0.0310 [0.0031, 0.0623]	0.0000 [0.0000, 0.0000]
Verdict: Diagnostic; Abstention avoids measured adaptation degradation				
PACS Aggregate	Domain-Seed Unit	12	-0.0255 [-0.0390, -0.0120]	+0.0015 [-0.0080, +0.0110]
Verdict: Diagnostic; Small- n exploratory baseline; adapt favored				

[†]Nominal 95% bootstrap CIs for CIFAR-10-C (clustered by 6 corruption families), ImageNet-C, Office-Home (seed bootstrap), and PACS; for CCT-20, intervals report the two-comparator nominal 97.5% CIs while the safe-utility non-inferiority check establishes $L > -0.0050$ vs. freeze. For the AETTA baseline, columns report KGA regret contrasts against the authenticated native AETTA recovery rule: LOO (within-model calibration on evaluated checkpoint) achieves lower regret (+0.09 to +0.28 pp) with ≤ 1 observed false adaptation per model; Historical rank-clipped B6 achieves lower regret than both fixed policies but has higher point regret (-0.78 to -0.05 pp vs AETTA) with zero observed false adaptations across all models vs 4–5 observed for AETTA. Positive values favor KGA.

nized samples or peak-memory logs were recovered, so measured gate overhead remains unestablished.

N.4 Formalization and Lean Scope

Lean artifacts and validators check the encoded theorem cores and selected finite constructions. Formalization of learning-theory results has precedent, including a Rademacher-complexity generalization bound [42]; we cite its pinned 2025 version as context, not as a proof of K-Bound’s assumptions. These checks do not certify dataset correctness, preprocessing, empirical transfer, exchangeability, or every deployment assumption stated in prose. Machine checking strengthens the statements it encodes; it does not validate the benchmark pipeline outside those statements.

N.5 Release Checklist and Audit Logs

The Phase-1 provenance audit matches all 106 canonical source/compact hash pairs, all 10 recoverable

configuration hashes, and all 6 archived checkpoint hashes, alongside a 21-file current-code snapshot. Historical runs lacking full serialized identity remain explicitly partial rather than being backfilled. The release gate verifies reconciliation tests, claim-ledger and result-manifest validation, manuscript checks, LaTeX builds, checksum verification, and visual PDF inspection. Build logs, the canonical manifest, the provenance seal, and final PDFs are archived together.

O Additional Calibration Comparison: Units, Risk, and Provenance

The study in Section 8.7 uses 32 allowlisted, hash-bound inputs from opened historical records. Its primary authority combines the corrected CIFAR/checkpoint scoring report with the corrected ImageNet grouping report. The source summaries and their immutable report history are indexed by

Table 45: Unverified historical runtime literals, retained for auditability. The claimed MPS/batch-16/100-pass measurement has no recovered timing or peak-memory log. Neither 0.62 ms gate overhead nor its 5.0% ratio is an authenticated performance result.

Component	Latency	SD	Peak Mem.	Overhead [†]
Frozen inference f_0	12.42 ms	± 0.31 ms	100.0 MB	1.00×
Adapter step f_a	24.85 ms	± 0.84 ms	+14.2 MB	2.00×
Feature extraction Z	0.48 ms	± 0.05 ms	+0.2 MB	3.9%
Certificate $[\hat{\Delta} \pm r]$	0.14 ms	± 0.02 ms	<0.1 MB	1.1%
Gate overhead	0.62 ms	± 0.06 ms	+0.3 MB	5.0%

[†]Relative to frozen forward pass (f_0 , 12.42 ms). Gate overhead combines feature extraction (3.9%) and certificate evaluation (1.1%), excluding candidate adapter step (24.85 ms) and downstream inference. Historical literal values; no authenticated timing samples recovered. Not an established benchmark.

`calibration_value_v1/FINAL_RESULTS_SUMMARY.json` and `next_phase/evidence_manifest.json`. These records bind the saved-feature calculations without supplying missing original training or adaptation execution evidence.

Separation and evaluation units. Each fold fits one benefit predictor shared across all policies. Semantic groups are kept intact across predictor fitting, margin tuning, residual calibration, and scoring. CIFAR and ImageNet outer folds exclude whole corruption types. Checkpoint-only evaluation excludes the scored source model and rotates held-out semantic groups; joint evaluation excludes both the scored checkpoint and the complete corruption type. Local scale estimation uses grouped out-of-fold residuals within the development data. The decision phase consumes prepared feature-only scoring inputs; the separate scoring phase reads the held-out outcome file after decisions have been serialized. Preparation itself reads the historical outcomes to construct the separated files. A synthetic full-fold outcome-perturbation regression preserves the predictor hash, calibrated radii, selected thresholds and returned decision structures for the tested scoring fold. These checks concern saved-feature outcome exclusion, not untouched historical outcomes, upstream image independence or invariance of timing-bearing receipts. These computational separations do not make the already opened environments exchangeable or independent deployments.

The main execution has 57 outer folds and 20 arms; nine ImageNet folds were later rerun under a deterministic identifier-schema correction. The two checkpoint designs reuse the same 108 records. Bootstrap sensitivity resamples six CIFAR corruption types, three ImageNet types, or three source checkpoints, with 1,000 resamples. Such small numbers of re-

lated units support descriptive sensitivity, not strong deployment-level inference. All prespecified alpha and harm-weight sensitivities are retained, including unbounded intervals and abstaining arms.

Cell risk and group risk are different estimands.

For the conditional-risk selection arm, a scored group is a (`fold_id`, `semantic_group`) pair. A group is exposed if any constituent cell receives ADAPT, and is an error group if any accepted cell has $B \leq 0$. Conditional group error divides error groups by exposed groups; with zero exposure it is undefined. Unconditional group error divides by all scored groups. Equal-group utility first averages cell utility inside each group and then averages groups; it is kept separate from equal-cell utility.

The corresponding unconditional group rates are 8/216 for EATA, 19/216 for SAR, and 0/216 for Tent. In contrast, the 10% arm exposes no groups, so its conditional risk is undefined rather than zero. The binomial procedure assumes independent identically distributed group outcomes with a common conditional error probability for a fixed threshold. Independence alone does not justify a common-risk binomial model, and neither assumption is established by this historical collection. Bonferroni adjustment covers the 13 candidate thresholds within a fold, not all folds or a deployment lifetime. With n exposed groups and zero errors, its upper bound is $1 - (.05/13)^{1/n}$; reaching .10 needs $n \geq 53$, while the largest available calibration partition has 45 groups.

Corrections and fixed authorities. An initial strict-JSON serialization failure stopped before scoring. A later scoring correction separated committed false FREEZE ($B \geq 0$) from abstention and forgone helpful non-ADAPT; it rescored the same decision bytes. A source identifier correction then removed

Table 46: Descriptive held-out risk for the 20% group-risk selection sensitivity. The group error event is any accepted $B \leq 0$ within a semantic group. Conditional cell and group rates have different denominators and are not interchangeable. Each candidate has 216 scored groups. These observations do not establish population risk control.

CIFAR candidate	Groups	Exposed	Error groups	Conditional group error	Cell error
EATA	216	107	8	7.48%	1.08%
SAR	216	180	19	10.56%	1.59%
Tent	216	65	0	0.00%	0.00%

only literal terminal repeat tags of the form `r` followed by digits. The earlier ImageNet parser had mistakenly removed a composition field, creating coarser groups. Its nine-fold rerun used unchanged statistical choices but occurred after outcomes were seen, and remains retrospective. Finally, the semantic-group report supplied the held-out estimand omitted from the original display; it changed no decision, prior metric, or canonical result authority. Original outputs and executed source snapshots are preserved rather than overwritten.

The byte-identical rendering authorities in `next_phase/table_sources/` retain their original output paths and SHA-256 identities. `scripts/make_next_phase_tables.py` verifies every pinned hash before writing any table. The calibration study does not claim novelty for GBRT, split conformal calibration, local residual scaling, or binomial risk selection. Full native TTALine/AETTA comparisons and independent field replications remain outside this saved-feature experiment.

P Paired-Transport Certificate: Construction and Scope

Sampling and probability intervals. Fix f_0, f_a before the evidence samples, or condition on an independent adaptation sample that fixes them. Within source class y , observations are independent identically distributed conditional draws, with fixed or conditioned-on class counts. Target evidence consists of independent identically distributed draws from its joint-prediction marginal. Retain all R declared bins, including unobserved bins; a bin can be coarsened only if its loss contrast is unchanged. For the $RK + R$ probabilities A_{ry} and q_r , take two-sided Clopper–Pearson intervals at individual error $\alpha/(RK + R)$. A source class with no observations receives $[0, 1]$ bounds. A union bound gives simultaneous containment probability at least $1 - \alpha$; multinomial cell independence

is unnecessary. Conditioning on class counts and averaging preserves this statement.

Linear feasible set. Writing source interval endpoints as L^A, U^A and target endpoints as L^q, U^q , let $\mathcal{P}_\rho$ contain variables t, s, π, v satisfying

$$\begin{aligned}
 \sum_y \pi_y &= 1, & \sum_r t_{ry} &= \sum_r s_{ry} = \pi_y, \\
 L_{ry}^A \pi_y &\leq s_{ry} \leq U_{ry}^A \pi_y, & L_r^q &\leq \sum_y t_{ry} \leq U_r^q, \\
 v_{ry} &\geq t_{ry} - s_{ry}, & v_{ry} &\geq s_{ry} - t_{ry}, \\
 \frac{1}{2} \sum_{r,y} v_{ry} &\leq \rho, & 0 &\leq t, s, \pi, v \leq 1.
 \end{aligned} \tag{28}$$

Minimal slacks are absolute differences of probabilities, so their unit upper bounds remove no feasible table. Define $c_{ry} = \mathbf{1}\{f_a(r) = y\} - \mathbf{1}\{f_0(r) = y\}$ and exact extrema $\ell = \min_{\mathcal{P}_\rho} c^\top t$, $u = \max_{\mathcal{P}_\rho} c^\top t$ when the set is nonempty.

Proposition 6 (Conditional paired-benefit coverage). *Under the sampling assumptions and the declared aggregate conditional-TV bound, the true benefit belongs to $[\ell, u]$ with probability at least $1 - \alpha$. Thus ADAPT for $\ell > 0$ and FREEZE for $u < 0$ has unconditional wrong-direction probability at most α , with abstention otherwise and on an empty set.*

Proof. On the simultaneous interval event, set π to the true target prior, $t_{ry} = P_T(r, y)$, $s_{ry} = A_{ry}\pi_y$, and $v_{ry} = |t_{ry} - s_{ry}|$. The marginal and probability-box constraints hold. The external transport bound gives the slack budget. Hence the true table is feasible and its benefit lies between the extrema. A wrong strict sign entails failure of the simultaneous interval event. The proof uses no matrix invertibility or full-rank assumption. $\square$

The exact nonempty polytope is compact, so its extrema are attained. A feasible table realizes a finite probability model by taking source conditionals

$s.y/\pi_y$ for positive-mass classes and any normalized column inside the source box for zero-mass classes, whose existence is required. Sharpness is relative to this finite-table model, not every underlying feature-space model or every confidence procedure. Multiple candidates or repeated decisions require another simultaneous construction or error allocation.

A decision without prior identification. For classes 0, 1, 2 and bins (2, 0), (2, 1), (2, 2), take

$$A = \begin{pmatrix} .2 & .2 & 0 \\ .8 & .8 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad q = (.2, .8, 0)^\top.$$

At $\rho = 0$, every prior $(p, 1 - p, 0)$ is compatible, while the benefit is $.2p + .8(1 - p) \in [.2, .8]$. The prior is unidentifiable but ADAPT is identified. These are exact rational decimals in the mathematical example; floating-point LP checks are not a separate proof of exact feasibility. The smallest transport allowance admitting a nonpositive benefit is $1/6$. Indeed, write the prior as (a, b, g) and let d be moved mass in the first two columns. Nonpositive benefit requires $d \geq .2a + .8b - g$; moving all third-column mass out of its original row costs g . Total variation is at least $\max(g, .2a + .8b) \geq \max(g, .2(1 - g)) \geq 1/6$. At $(a, b, g) = (5/6, 0, 1/6)$, target columns zero and two equal $(1/6, 2/3, 0)^\top$ and $(1/30, 2/15, 0)^\top$, respectively, attaining zero benefit and TV $1/6$.

Why paired optimization helps. On any common feasible set,

$$\min(a_a - a_0) \geq \min a_a - \max a_0,$$

with the reversed upper-endpoint inequality. Agreement-bin correctness can vary with an unidentified prior while canceling in the paired difference. For bins (2, 0), (2, 1), (0, 0), (2, 2) and

$$A = \begin{pmatrix} .1 & .1 & 0 \\ .4 & .4 & 0 \\ .5 & .5 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad q = (.1, .4, .5, 0)^\top,$$

the paired interval is $[.1, .4]$, whereas separate-accuracy subtraction gives $[-.1, .6]$. Prior $(.8, .2, 0)$ gives benefit $.16$. Reassigning every target label to the frozen prediction preserves all observed bins but gives benefit $-.5$. This counterworld violates the declared transport bound, demonstrating why unlabeled observations cannot establish that bound.

Numerical scope. Each minimization is $\min c^\top x$ subject to $Bx \leq b$, $Ex = d$, and $0 \leq x \leq 1$. For any finite equality multiplier μ , project inequality multipliers to $\lambda \leq 0$ and set $r = c - B^\top \lambda - E^\top \mu$. Weak duality gives the conservative bound

$$c^\top x \geq b^\top \lambda + d^\top \mu + \sum_j \min(0, r_j). \quad (29)$$

The implementation evaluates this bound with directed 60-digit decimal operations starting from exact binary64 coefficients, then rounds outward when converting to binary64. Maximization uses a separately bounded negative objective. The maintained implementation additionally reconstructs a rational primal point and checks every equality, inequality, and variable bound exactly. Numerical feasibility tolerances alone cannot justify a decision. Failure to obtain an exact witness causes abstention; it is not proof that the set is empty. A 10^{-10} numerical sign margin remains in force. Binary64 inputs denote their exact represented values. Explicit rational or decimal probability inputs are converted to documented outward boxes; they are not silently normalized.

Binomial quantiles now serve only as proposals. Directed arithmetic bounds the positive binomial-tail polynomial, using a geometric remainder bound when truncated; exact comparisons verify outward endpoints on a dyadic grid. Unresolved comparisons or resource limits widen the interval, up to $[0, 1]$. Error budgets are allocated by exact rational division; only the displayed floating-point allocation is rounded downward. These numerical certificates preserve the binomial construction's finite-sample coverage under its stated sampling law; they do not establish that law for deployment data or constitute a Lean proof of the Python implementation.

The archived 4,860-trial study below retains its original execution scope: inverse-beta endpoints expanded by one floating-point unit and approximate primal witnesses. Its saved outputs have not been relabeled as executions of the revised solver. A separate numerical replay spans all nine scenarios, three sample sizes, and three transport budgets once each. All 81 cases obtained exactly checked feasible witnesses, with 16 ADAPT, 2 FREEZE, and 63 ABSTAIN outputs. This replay checks the revised numerical procedure; it is not a replacement Monte Carlo study or evidence of natural-shift effectiveness.

The synthetic protocol fixes nine scenarios, three source class sample sizes, three sensitivity budgets, and 60 trials per combination before count sampling. All counts, decisions, outcomes, approximate wit-

nesses, receipts, and executed source snapshots are indexed by `paired_transport_v1/receipt.json` and `next_phase/evidence_manifest.json`. The separate-accuracy arm is a coupling relaxation on the identical polytope. An official MaC-LP execution was unavailable after checking the primary paper and targeted author-code sources; it was not replaced with a purported native implementation. Monte Carlo intervals are descriptive within regimes and are not simultaneous inference over all reported configurations.

P.1 Simulation-Only Architecture and Parameter Probes

The following recovered probes are retained here as simulation diagnostics rather than measured benchmark results. The architecture names and adapter settings label synthetic configurations; the named backbones and proposed adapters were not executed in these probes. Their values preserve the historical numbers and provenance, but they do not support an architecture-transfer claim, a real-cell routing claim, or an observed false-ADAPT rate.

The parameter-restricted probe samples synthetic benefits at five hardcoded locations rather than loading CCT-20 images or executing the proposed adapters. Its numerical break-even summaries remain available as simulation outputs; the saved null actions do not support a routing result.

Q Additional Comparison Provenance

Independent-Checkpoint Replication: The original 2,160-cell CIFAR-10-C panel reused one source checkpoint. Subsequent records describe two independently trained ResNet-18 checkpoints and helpful-only and mixed-sign 36-condition evaluations (Appendix B.3.3). On the mixed-sign panel, 39–42% of conditions were harmful; saved summaries report lower KGA point oracle regret than both fixed policies on all three source models. Recovered mixed-sign per-cell decisions and B6 split receipts support the reported arithmetic; original fitted-model/environment identities and helpful-only execution remain incomplete, and B6 rank clipping prevents a nominal coverage claim (Appendix B.3.3). A retrospective cross-architecture diagnostic is reported in Appendix B.8.2, while prospective architecture-level transfer to unseen model families and environments remains untested.

Hardware and External Baseline Dependencies: The synchronized baseline comparison (Table 9) and scoped native panel (Appendix B.9, Table 31) evaluate fixed policies and internal benefit estimators on common saved features. The multi-model mixed-sign AETTA-estimator comparison and its explicit recovery rule are reported separately in Appendix B.3.4, subject to the historical KGA provenance qualification above. Historical scalar proxies and local ports are not validated equivalents of the published methods. Scoped native-component execution is reported separately with its architecture, protocol, warnings, and method-availability limits; it does not authenticate every original learner loop or retroactively promote the ports. In particular, Baek et al.’s ALine [17] is not a substitute for unavailable native Kim et al. TTA/ALine [51]. No complete official-method or full ImageNet-C comparison is inferred from these component results.

Camelyon17 decision-history qualification. A retrospective 72-condition Camelyon17 diagnostic (Appendix B.5) contains both helpful and harmful adaptation cells under hospital-provenance shift; under authenticated outcome-blind execution, the gate issued 0 ADAPT, 2 FREEZE, and 70 ABSTAIN decisions, matching always-freeze regret (0.0438) and demonstrating conservative retention without ADAPT exposure. The earlier 31-ADAPT partition is withdrawn as routing evidence: it thresholded ground-truth benefit post hoc, rather than a saved outcome-blind prediction. It remains documented only to explain the historical discrepancy. KGA made no ADAPT decisions on CCT-20, RxRx1, or the 72-condition Camelyon17 diagnostic. These results evaluate conservative retention and do not establish prospective selective routing on unseen natural shifts.

R Contribution and Empirical Interpretation

R.1 What the constructions add

The binary result fixes a predictor pair and a label-free evidence law, then varies the conditional label kernel within an explicit residual budget. Its additional content is the attainable clipped benefit set, the strict-sign boundary including zero-benefit boundary worlds, and the corresponding fibre-wise audit obstruction. Generic action identification and indistinguishable-world testing are established ideas; adaptation identification is formulated directly on evidence fibres in [56].

Table 47: Synthetic four-architecture probe, recovered from its saved report. Architecture names label simulation settings, not executed models. All saved actions are null; no observed false-ADAPT rate or architecture-transfer claim follows. The saved mean break-even margins are zero, correcting the earlier literal value of one.

Architecture	Family	Params	Disagreement $\mu(D)$	Radius r	Break-Even Margin β^*	FA _u	Observed Gate Action
ResNet-18	Simulated	11.7M	0.1943	0.1026	0.000	–	NOT RECORDED
ResNet-50	Simulated	25.6M	0.1841	0.1025	0.000	–	NOT RECORDED
ViT-B/16	Simulated	86.6M	0.1807	0.1024	0.000	–	NOT RECORDED
ConvNeXt-Tiny	Simulated	28.6M	0.1816	0.1026	0.000	–	NOT RECORDED

Table 48: Synthetic parameter-restricted adaptation probe. Five simulated locations per regime; no real-cell retention denominator or measured gate action. These rows do not inherit the authenticated CCT-20 benchmark’s 45-cell endpoint.

Adaptation Regime	Parameter Update Scope	Mean β^*	FA _u	Real-cell Retention	Provenance
Full TTA	Synthetic benefit model	0.070	–	Not measured	5 simulated locations; null actions
Norm Stats Only	Synthetic benefit model	0.022	–	Not measured	5 simulated locations; null actions
Head Only	Synthetic benefit model	0.013	–	Not measured	5 simulated locations; null actions
Entropy-Gated Partial	Synthetic benefit model	0.000	–	Not measured	5 simulated locations; null actions

Our necessity statement concerns the full declared rich class. Restricting that class can remove counterworlds, so necessity need not survive; sufficiency remains valid. Zero disagreement instead gives zero benefit, without defining disagreement-conditional moments.

The paired-transport construction addresses a different estimand and assumption class. Relative to the matrix-constraint confidence regions in [57], it retains a shared joint table for two predictors and optimizes their accuracy contrast under a supplied conditional-transport budget. Tighter bounds than separate subtraction follow on that same feasible set; no dominance over another method’s feasible set is implied. The probability-box theorem, numerical binomial endpoints and floating-point objective bounds have the distinct scopes described in Appendix P. The numerical endpoints are not certified inverse-CDF bounds.

KGA uses historical labeled cell outcomes and is a separate empirical construction. Its interval-to-action implication applies inherited calibration ideas; it does not establish a new conformal validity theorem or estimate the population residual budget. Coverage transfer requires its own assumptions. The analysis in [37] quantifies coverage loss beyond exchangeability, including for original conformal procedures, and studies modifications that can improve robustness. Learned adaptation selection also predates this work. MORPHEUS [58] selects before candidate execution, whereas KGA’s shadow-candidate features incur candidate computation. A comparison must match both information access and cost.

The theoretical guarantees concern prespecified decisions and sampling contracts. Marginal wrong-direction control does not imply an error bound conditional on ADAPT. Candidates selected using evaluation evidence, repeated uses and adaptive searches require additional control. The maintained formalization’s scope and exclusions are given in the formal supplement; its check count is not a measure of originality.

R.2 Units, utility and sensitivity

The principal CIFAR panel and the shared-predictor study answer different questions. Table 50 reconstructs equally weighted mean accuracies and ordinary regret for the principal policy. The KGA minus always-adapt differences are +0.6183, +0.1673 and −0.1326 percentage points for Tent, EATA and SAR, respectively. The five stream seeds reuse an archived source checkpoint and are not five independent training replications.

In the shared-predictor comparison, 100L₅ assigns weight five to harmful accepted updates. That declared benchmark loss is separate from accuracy, ordinary regret and interval inclusion. Scaled calibration’s observed Tent advantage is smaller after matching ADAPT exposure, and depends on the corruption mix (Table 51). Paired resampling of six corruption types gives wide descriptive ranges for the unmatched point and margin comparisons. Removing JPEG reduces the advantage over point prediction to 0.004889 percentage points and reverses the margin comparison. This deletion is an influence diagnostic, not a

revised primary analysis. Within a fixed scoring fold, a constant finite radius preserves the shared point predictor’s ordering. The equal-exposure check matches selected cells within that fold using the same tie rule; it does not establish a common ranking across separately fitted folds or studies. Changing exposure alone is not evidence of improved ranking. All candidate, cohort and rule rows, including harmful acceptance, missed helpful opportunities, abstention and harm-weight sensitivity, are available in the accompanying versioned synthesis.

Table 49 separates cell inclusion from the event that every cell in a complete semantic group is covered. There are 2,160 cells and 216 complete groups per CIFAR candidate, built from six corruption types; neither count is an independent-deployment sample size. A group-max radius does not make its observed cell inclusion a simultaneous-group coverage rate. All four observed rates remain descriptive evidence of transfer failure. Unavailable finite-rank and zero-exposure quantities remain undefined rather than being reported as successful conditional-risk control.

CCT-20’s reconstructed equal-cell mean accuracy is 52.2936% for both KGA and freeze, versus 34.1034% for adaptation. Its zero ADAPT exposure supports retention, not useful routing. Secondary CCT macro-F1 uses the preserved 16-indicator class universe and a zero-denominator value of zero, computed per cell from class counts. It is not interchangeable with official single-label macro-F1 or with F1 recomputed after pooling cells. The invalid historical iWildCam macro-F1 comparison remains withheld.

R.3 Scope of independent confirmation

No eligible fresh natural-shift study was established by the bounded metadata review. Previously opened environments cannot become fresh by changing a seed. DomainNet’s six domains are insufficient even for the minimum finite 90% environment-level split-conformal rank before reserving development and target units; nine calibration units would only satisfy the rank condition, not establish exchangeability or adequate precision. So2Sat’s failed locked screen remains binding and its target remains unscored. These limitations leave useful independent natural-shift routing unresolved. The accompanying eligibility report specifies the population, access-history, sample-size, protocol and cost requirements for a future study.

S Audit-Assisted Bounds for a Fixed Finite Frame

Once labels are available for retrospective audit, a different design-based question becomes possible. Fix a finite frame of N observations, its frozen and adapted predictions, and its scoring rule. Draw audit indices independently and uniformly with replacement after fixing these objects. Conditional on the frame and its labels, any fixed correctness indicator evaluated at the sampled indices is iid Bernoulli with its finite-frame frequency as success probability. Dependence among the original images does not invalidate this randomization argument; it does prevent interpreting the frame as independent future environments. Correct labels, faithful identities and the specified randomization remain premises.

Paired accuracy and the binary residual. Let D be the known prediction-disagreement stratum and $d = |D|/N$. When $d > 0$, audit draws are independently uniform within D , with replacement. Outside D the paired correctness difference is zero. Within D , let p_+ and p_- denote helpful and harmful outcome frequencies. Two outward binomial intervals with error budgets $\alpha/2$ give

$$\Delta_N \in d[L_+ - U_-, U_+ - L_-]$$

with probability at least $1 - \alpha$ over the audit draws. No independence between the two intervals is required. This construction also permits set-valued labels: both differing predictions can be correct, or both wrong. It gives no macro-F1 guarantee.

For a single-label binary task, exactly one prediction is correct on D . A single interval $[L, U]$ for adapted correctness gives $\Delta_N \in 2d[L - 1/2, U - 1/2]$. If $\bar{s}$ is the known fixed score mean on D , then $\gamma_N \in [L - \bar{s}, U - \bar{s}]$, yielding

$$\hat{\beta} = \max\{|L - \bar{s}|, |U - \bar{s}|\}.$$

This is a labeled-audit confidence bound for the fixed frame, not a label-free estimate of a uniform residual-class radius. When D is empty, $\Delta_N = 0$ and the conditional residual is undefined.

An audit-assisted transport budget. For a fixed single-label class partition and bin map, independently uniform sampling within each source and target frame gives binomial class-bin counts conditional on their class counts. Allocate $\alpha/(2RK + K)$ to each source and target class-bin interval and target-prior

Table 49: Observed inclusion on held-out CIFAR corruption cells and complete semantic groups. All four columns use nominal 90% intervals; simultaneous-group inclusion requires every cell interval in the group to contain its benefit. These are descriptive rates from six historical corruption types, not a transfer guarantee.

Method	Constant-cell radius		Group-max radius	
	Cell (%)	Group (%)	Cell (%)	Group (%)
EATA	54.35	42.13	61.90	56.94
SAR	50.51	32.41	61.30	45.83
TENT	60.69	42.13	73.19	61.57

Table 50: Reconstructed means for the principal CIFAR policy, distinct from the shared-predictor study. Accuracy is the equally weighted mean of preserved condition/seed-cell accuracies. Regret is unweighted oracle regret in percentage points; exposure is the percentage of cells with ADAPT. Reconstruction checks stored arithmetic and does not authenticate missing original image-level execution.

Method	Mean accuracy (%)			Regret (pp)	ADAPT (%)
	Frozen	Adapt	KGA		
EATA	67.0931	79.8992	80.0666	0.1603	55.8796
SAR	67.0931	81.1067	80.9741	0.1661	66.2037
TENT	67.0931	78.6893	79.3076	0.1793	50.6481

interval. The union bound gives simultaneous containment. Given source bounds $[a_{ry}^-, a_{ry}^+]$ and target bounds $[b_{ry}^-, b_{ry}^+]$, define

$$v_y = \min \left\{ 1, \frac{1}{2} \sum_r \max\{|b_{ry}^- - a_{ry}^+|, |b_{ry}^+ - a_{ry}^-|\} \right\}.$$

Maximizing $\sum_y \pi_y v_y$ over the simplex intersected with the target-prior boxes gives an upper confidence bound for the aggregate conditional-TV budget. The implementation uses exact rational allocation in descending v_y order. Empty prior intersections return the vacuous bound one. An absent source class has no uniquely defined conditional law; its bounds cover any declared normalized extension. Multilabel memberships are not this partition. The random budget’s failure probability must be added to the subsequent paired-transport certificate’s budget. These derivations remain written arguments; the Python audit implementation is not claimed as fully mechanized.

Retrospective CCT-20 check. We applied the paired-accuracy procedure to all 45 previously evaluated checkpoint-by-location cells, preserving the original set-membership correctness rule and excluding only the original adaptation-probe role. Each cell used 512 independent draws with replacement from its disagreement stratum and error budget 1/900, giving family budget 0.05. All draw inventories were stored before the new scorer joined the already opened historical labels. The 23,040 draws gave 16,428 distinct

label uses when summed across cells and 8,954 distinct image IDs overall. These are simulated label lookups, not measured annotation labor. Shared images across checkpoints are not independent deployments. Every full-frame benefit reconstructed the original scoring authority and lay inside its audit interval. The audit issued 35 FREEZE and 10 ABSTAIN decisions, with no ADAPT decision among 44 harmful and one helpful frame. These are actions from the audit-assisted finite-frame procedure, not actions from the original KGA gate. The primary KGA study in Section 8.4 remains 0 ADAPT / 44 FREEZE / 1 ABSTAIN. The audit therefore does not replace or revise the primary CCT result. Thus it did not demonstrate useful adaptation selection. This is a retrospective label-budget simulation on fixed recorded predictions, not a new prospective natural-shift study or validation of KGA’s calibration transfer.

Table 51: Retrospective CIFAR Tent sensitivity, scaled calibration minus each comparator in $100L_5$ percentage points; negative favors scaled calibration. The ranges use 1,000 paired resamples of six historical corruption types and are descriptive, not population confidence guarantees. JPEG is omitted only for the influence diagnostic; all six types remain in the primary estimate.

Comparator	Mean difference	95% descriptive range	Without JPEG
Point	-0.143843	[-0.428565, 0.138755]	-0.004889
Tuned margin	-0.083495	[-0.416991, 0.188282]	0.090889
Point, matched exposure	-0.018148	[-0.042500, -0.001065]	-0.006528